Sources and attribution

Historical manuscript notes of the I. M. Gelfand Seminar. MCCME digital archive: Notes of the talks at the I. M. Gelfand Seminar.

MCCME identifies M. A. Shubin as the note-taker. The Clay Mathematics Institute funded the scanning. MCCME acknowledges M. A. Shubin, S. I. Gelfand, the Clay Mathematics Institute and Jim Carlson. MCCME archive notice.

Chapter 10. Manuscript frames 659–704. Catalogue snapshot: 5 October 2026.

Speakers named in the catalogue associations for the complete edition, using the catalogue's spelling:

I.M. Gelfand; R.L. Dobrushin; Irving Segal; Laurent Schwartz; S.G. Gindikin; A.Ya. Povzner; V.I. Arnold; A.A. Kirillov; I.I. Pyateski-Shapiro; M.I. Vishik; Yu.V. Linnik; E.A. Gorin; D.A. Kazhdan; V.P. Palamodov; A.S. Dynin; S.P. Novikov; Ya.G. Sinai; I.M. Gelfand, A.Ya. Povzner,Berezin; V.N. Latyshev; S.I. Gelfand; I.N. Bernstein; A.M. Stepin; R.A. Minlos; L.D. Faddeev; Paul Halmos; J.J. Kohn; Robert Finn; S.V. Smirnov; V.I. Matsaev; A.G. Kushnirenko; M.S. Agranovich; F.I. Karpelevich; V.A. Borovikov; V.B. Lidskii; G.E. Shilov; Lee Lorch; Louis Nirenberg; O.A. Oleinik; A.N. Parshin; D.B. Fuchs; S.G. Krein; Jean Leray; L.G. Makar-Limanov; A.Yu. Geronimus; Alan Weinstein ; M.L. Gromov; G.A. Margulis; Chandler Davis; V.M. Buchstaber; B.R. Vainberg; Gaetano Fikera; Dusa McDuff; Edwin Hewitt; B.V. Fedosov; A.M. Gabrielov; S. Rolevich; A.M. Perelomov; B.I. Rosenfeld; Vlastimil Ptak; Michael Atiyah; Pierre Deligne; A.M. Vinogradov; L.N. Vasserstein; Jack Morava; M.Sh. Birman; Yu.I. Manin; A.G. Kostyuchenko; V.I. Lomonosov; F.A. Berezin; G.M. Henkin; A.S. Mishchenko; A.N. Rudakov; A.I. Schnirelman; V.A. Ponomarev; Tony Springer; V.E. Zakharov; V.N. Tulovskii; A.B. Shabat; B.M. Gurevich; Yu. Koblenz; Enrico Guisti; Czeslaw Bessaga; B.A. Dubrovin; Graeme Segal; A.M. Polyakov; M.I. Graev; James King; Bert Kostant; E.S. Fradkin; I.M. Krichever; V.G. Drinfeld; M.V. Losik; A.V. Roiter; A.A. Migdal; V.N. Popov ; J.J. Duistermaat; A.M. Vershik; A.S. Tyurin; D.A. Kirzhnits; L.A. Dickey; S.M. Vishik; Bela Sekefalvi-Nagy; Kazuhiko Aomoto; Francesco Calogero; V.S. Kruskal; A.S. Schwarz; Alain Guichardet; A.A. Belavin; A.A. Beilinson; V.I. Ogievetskii; D.R. Lebedev; G.B. Shabat; Renata Kallosh; Raphael Hoegh-Krohn; I.A. Batalin; I.Yu. Kobzarev; A.V. Zelevinsky; Robert MacPherson; G.I. Olshanskii; A.B. Goncharov; Bernard Gaveau; A.G. Khovansky; I.V. Cherednik; B.Ya. Kazarnovskii; P.G. Grinevich; Jean-Luc Brylinski; Eckart Viehweg; Christian Kassel; A.V. Bobylev; B.V. Lidskii; Benedict Seifert; A.G. Elashvili; Werner Mueller; V.L. Ginzburg; John Milnor; Andre Haefliger; A.B. Givental; Mariusz Wodzicki; Al-dr.B. Zamolodchikov; Jurgen Frohlich; B.L. Feigin; Yu.A. Neretin; M.A. Semenov-Tyan-Shanski; George Wilson; Yu.L. Daletskii; V.A. Chulaevskii; P.B. Wiegman; F. Nikiforov; Yu.M. Berezanskii; D.A. Leites; Ivan Penkov; Jean-Pierre Serre; M.M. Kapranov; Marie-France Vigneras; Yu.S. Ryzhikova; V.G. Knizhnik; M.L. Kontsevich; A.Yu. Morozov; A.N. Varchenko; Yan I. Kogan; V.A. Kazakov; V.Yu. Ovsienko; I.S. Zakharevich; Don Zagier; D.V. Alekseevskii; A.P. Veselov; Klyachko; A.O. Radul; N.Yu. Reshetikhin; A.G. Usheridze; Gregory Moore; V.S. Dotsenko.

A catalogue association does not by itself establish an exact talk boundary within a manuscript page or identify the writer of every note.

The modern LaTeX text is AI-transcribed. The English edition includes AI-translated passages and AI-transcribed passages originally written in English. Diagram reconstructions are AI-created and typesetting is AI-produced. Historical lecture and manuscript authorship remains with its human authors.

Rights in the original notes, scans and other third-party material remain with their authors and respective rights holders, including heirs. To the extent that the project holds copyright or related rights in its own contributions, it dedicates those rights under CC0 1.0 Universal. The dedication applies only to rights held by the project.

These editions are freely available and are published solely for non-commercial educational purposes. If a rights holder or their representative objects to publication, please identify the affected material in GitHub Issues so we can take it down.

Exact image URLs, file digests and catalogue associations accompany the editable sources.

Chapter 10

GELFAND-P000659

6/IX-76. Gelfand Sem.

What should one know from physics? [At right] (Following Sinai?)



Uncertain reading. The note at right reads "Following Sinai?" with a question mark.

Original source details: GELFAND-P000659

1. Classical mechanics and Ham. formalism.
   Arnold (≤ 30 pp.), Landau and Lifshitz.
2. Quantum mechanics and scattering theory.
   Landau and Lifshitz, Quantum mechanics (minus the first two chapters).
   Fermi, Quantum mechanics.
3. Field theory and relativity.
   Landau and Lifshitz, Weinberg, Cosmology and the theory of gravitation (chapters 1–4).
4. Quantum field theory.

[crossed out]



Undeciphered passage. The line under item 4 is heavily crossed out.

Original source details: GELFAND-P000659

For next time, consult the physicists and type it up.

Quantum field theory

1. Second quantization (Bose and Fermi). [Inserted below: statistics] — Landau and Lifshitz.
2. Writing the equations [inserted in callout: of the field] using a functional integral,

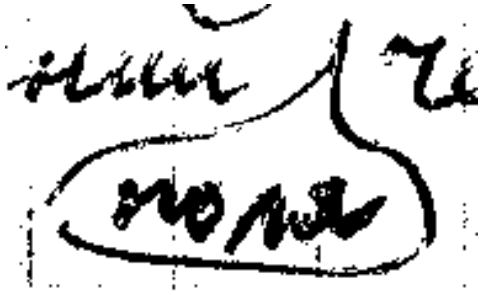


Diagram and typed description.

of the field

The callout after "equations" reads "поля" (of the field).

Original source details: GELFAND-P000659

perturbation series, Feynman diagrams.

3. Renormalizations. — Hepp.

4. Yang–Mills fields — there are surveys.

Gindikin (and Gelfand). All about $SL(2,\mathbb{R})$.

$$G = SL(2,\mathbb{R})$$

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \det(\,) = 1$$

(hyperboloid in $\mathbb{R}^4$)

$$\alpha\delta - \beta\gamma = 1$$

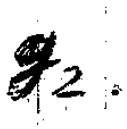
It corresponds to a form with 2 pluses and 2 minuses:

$$x^2 + y^2 - z^2 - t^2 = 1.$$

The group acts on the hyperboloid $G \times G$ by left and right translations

$$T_{(g_1,g_2)}g = g_1^{-1} g g_2 \,.$$

Uncertain reading. The final g_2 is heavily retraced.



Original source details: GELFAND-P000659

Any lin. transf. preserving the hyperboloid can be written in this form.

The stabilizer of the point $e = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ consists of transfs. [crossed out]

Undeciphered passage. A parenthesis is visible. Its contents are heavily crossed out.



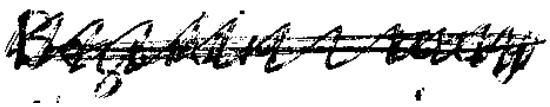
Original source details: GELFAND-P000659

of the form (g, g).

GELFAND-P000660

[Line crossed out at top]

Undeciphered passage. The line is heavily crossed out. Individual letter strokes remain visible.



Original source details: GELFAND-P000660

Exercise: compute the stabilizer subgrps. of other points.

Consider fns. on the hyperboloid.

For $\beta \neq 0$ convenient coordinates are [crossed out: α, β, δ] α, δ, β.



Uncertain reading. The first sign of the crossed-out triple is retraced and the reading is tentative.

Original source details: GELFAND-P000660

Fns. on the hyperboloid can be regarded as fns. $\psi(\alpha, \delta, \beta)$.

The functions are to be integrated against the invariant measure:

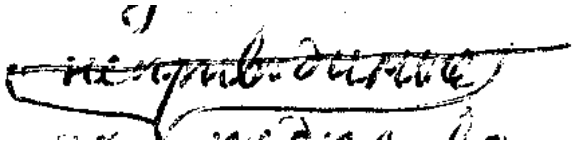
$$dg = \frac{d\alpha\, d\beta\, d\delta}{|\beta|}$$

(inv. under motions of the hyperboloid).

We will consider the space $\mathcal{H} = L^2(G)$ with respect to this measure.

$$\|\psi\|^2 = \int |\psi(g)|^2\, dg.$$

Problem: decompose the fns. into [circled and crossed out: irreducible] subspaces inv. under



Undeciphered passage. The enclosed insertion above the line is crossed out.

Original source details: GELFAND-P000660

$$T_{(g_1,g_2)}\psi(g) = \psi(g_1^{-1} g g_2).$$

In $\mathbb{R}^4$ we consider hyperplanes

$$\xi\alpha + \eta\delta - p\beta - q\gamma = \zeta.$$

G. Philosophy: divide into series (an algebraic problem)

(Bernstein, Gelfand, Gelfand)



Uncertain reading. The surname “Gelfand” is written twice in succession.

Original source details: GELFAND-P000660

Series into representations — a different (transcend.) problem.

This well-known example (Bargman, Gelfand and Graev, Molchanov, Strichartz, Vilenkin et al.) — everything about hyperboloids is already known.



Uncertain reading. The first surname appears to be written “Bargman” in Latin script.

Original source details: GELFAND-P000660

Into irreducibles — no need.

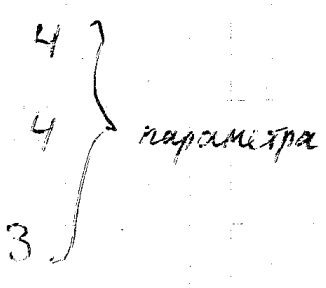
The following types of sections are possible.

$$\text{I.}\quad \Delta = \begin{vmatrix} \xi & p \\ q & \eta \end{vmatrix} < 0 \quad \text{one-sheeted hyperboloid} \qquad 4 \text{ parameters}$$

$$\text{II.}\quad \Delta = \begin{vmatrix} \xi & p \\ q & \eta \end{vmatrix} > 0$$

a)	$\lvert\zeta\rvert > 2\sqrt{\Delta}$	one-sheeted hyperboloid	4
b)	$\lvert\zeta\rvert < 2\sqrt{\Delta}$	two-sheeted hyperbol.	4
c)	$\lvert\zeta\rvert = 2\sqrt{\Delta}$	cone	3

} parameters



Uncertain reading. The numbers 4, 4 and 3 are written at right and grouped with a brace.

Original source details: GELFAND-P000660

III. $\Delta = 0$ — flat thing in hyperbol. cylinder 2 parameters.

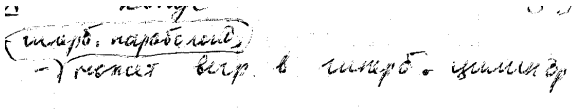


Diagram and typed description.

hyperbol. paraboloid

"Hyperbol. paraboloid" is enclosed above the line. The single sign before "hyperbol. cylinder" resembles "в" (in).

Original source details: GELFAND-P000660

GELFAND-P000661

A — one-sheeted hyperboloid (*cases* I, IIa), 4-param..

B — two-sheeted hyperboloid (—»— IIb), 4-param..

[Single letter crossed out: A?]



Uncertain reading. The crossed-out letter between B and D resembles A.

Original source details: GELFAND-P000661

D — cones, 3-param..

E — hyperbolic paraboloid (case III), 3-param..

$$\psi \in \mathcal{H}.$$

$$\widehat{\psi}(\xi, \eta, p, q, \zeta) = \int \psi(g)\delta(\xi\alpha + \eta\delta - p\beta - q\gamma - \zeta)\, dg.$$

(Radon transform of the fn. in the whole space)

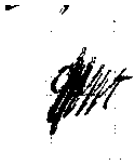
(G. The technique goes back to that of the 1947 paper by G. and Naimark.)

$\mathcal{H}_A, \mathcal{H}_B, \mathcal{H}_D, \mathcal{H}_E$ — fns. for which the integrals are 0 [sign crossed out: φ?] over the hyperplanes of the given family.



Original source details: GELFAND-P000661

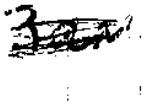
Uncertain reading. The third subscript is overwritten. Its outline reads D.



Original source details: GELFAND-P000661

Uncertain reading. A loop and a descending stroke in the crossed-out sign resemble φ.

[Crossed out: Problem [abbr.]?] How does one describe which fns. of 4 and 3 variables appear?

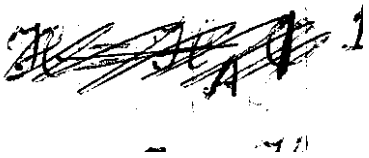


Original source details: GELFAND-P000661

Uncertain reading. The crossed-out opening tentatively reads “Зад.” (Problem).

Main theorem 1.

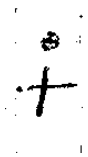
[Formula crossed out: $\mathcal{H} = \mathcal{H}_A \ldots$]



Original source details: GELFAND-P000661

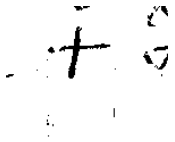
Uncertain reading. The crossed-out formula shows $\mathcal{H}$, an equals sign and $\mathcal{H}$ with subscript A. The final sign is unclear.

1. $\mathcal{H} = \mathcal{H}_A \overset{\circ}{+} \mathcal{H}_B.$



Original source details: GELFAND-P000661

Uncertain reading. There is a small retraced circle above the plus sign.



Uncertain reading. A separate dot is also visible to the left of the plus sign.

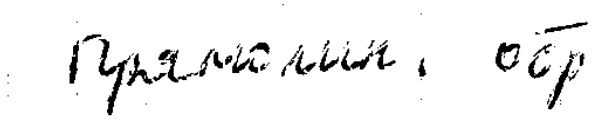
Original source details: GELFAND-P000661

$$2. \quad \mathcal{H}_A = \mathcal{H}_D = \mathcal{H}_E.$$

The fn. $\widehat{\psi}$ satisfies a second-order ultrahyperbolic eq. that can be written down.

The families A, D, E are minimal.

Straight-line rulings — a 3-param. family C.



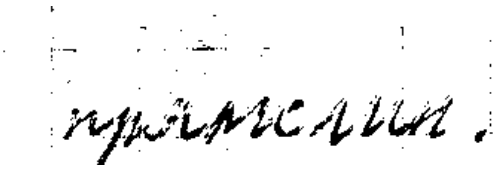
Uncertain reading. The abbreviation reads “Прямолин.” (straight-line).

Original source details: GELFAND-P000661

$\begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix}$ — an example of a straight line, $\begin{pmatrix} h & z \\ 0 & h^{-1} \end{pmatrix}$ (h — fixed)

or

$$g_1^{-1} \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} g_2 \quad \text{(general form of a straight-line ruling)}.$$



Uncertain reading. The same abbreviation for “straight-line” is repeated in parentheses.

Original source details: GELFAND-P000661

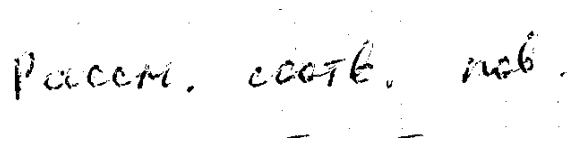
C — straight-line rulings.

$$2'. \quad \mathcal{H}_A = \mathcal{H}_D = \mathcal{H}_E = \mathcal{H}_C. \qquad \text{(discrete series)}$$

Explicit specification of projectors and description of these spaces.

$$\alpha\delta - \beta\gamma = 1.$$

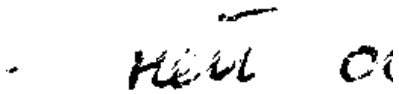
Consider the corresponding surface in $\mathbb{C}^4$ with the same eq..



Uncertain reading. The abbreviation before “in” reads “пов.” (surface).

Original source details: GELFAND-P000661

and on it, the domains:



Uncertain reading. The pronoun is “ней”, the feminine form used with “surface”.

Original source details: GELFAND-P000661

$$\underline{(\Im\alpha\bar{\beta})(\Im\delta\bar{\beta}) > (\Im\beta)^2} \qquad (1)$$

Consider it in $\mathbb{C}^3$.



Uncertain reading. The pronoun after "Consider" tentatively reads "eë" (it).

Original source details: GELFAND-P000662

It splits into 2 domains whose distinguished boundary is the real space $\mathbb{R}^3$.

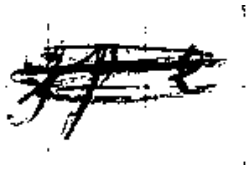
The inequality is inv. under fractional-linear transfs. (one of the Siegel domains).

Distinguished boundary $G = SL(2, \mathbb{R})$.

The domains are separated by the sign of the parentheses.

(The distinguished boundary of a domain D is the smallest set in ∂D such that a fn. holom. in D and cont. on $\bar{D}$ is uniquely recovered from its restriction to that set.)

Lemma. In $\mathbb{C}^3$ with coords. (α, β, δ) [text crossed out] inequality (1) singles out 2 domains, each with distinguished boundary $\mathbb{R}^3$.



Undeciphered passage. The short entry after the coordinate triple is heavily crossed out.

Original source details: GELFAND-P000662



Uncertain reading. Before the 2 the writing tentatively reads "выд." (singles out).

Original source details: GELFAND-P000662

Theorem. On the surface $\alpha\delta - \beta\gamma = 1$ in $\mathbb{C}^4$ there exist 2 domains inv. under real motions, with distinguished boundaries given by the hyperboloid [sign crossed out] $\alpha\delta - \beta\gamma = 1$ in $\mathbb{R}^4$.



Undeciphered passage. The initial sign to the left of the hyperboloid equation is crossed out.

Original source details: GELFAND-P000662

Denote these domains by D_+ and D_-.

Main thm. 2.

$$\psi \in \mathcal{H}_A \iff \psi = \psi_+ + \psi_-, \quad \text{where}$$

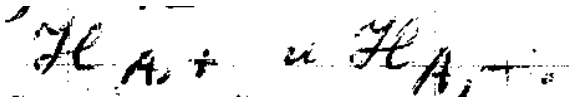
ψ_+ continues analyt. into D_+, ψ_- continues analyt. into D_-.

[Above the line: Denote [sign crossed out] the subspaces by $\mathcal{H}_{A,+}$ and $\mathcal{H}_{A,-}$.]



Original source details: GELFAND-P000662

Undeciphered passage. A sign with diverging strokes between “Denote” and “the subspaces” is crossed out.



Original source details: GELFAND-P000662

Uncertain reading. A thin vertical stroke crosses the long bar in the second subscript. The reading as a minus is tentative.

The projector onto $\mathcal{H}_{A,+}$ is given by the fn.

$$\int_G K(g_1, g_2)\psi(g_2)\, dg_2,$$

where

$$K(g_1, g_2) = c\,\frac{1}{[-\alpha_1\delta_2 - \delta_1\alpha_2 + \gamma_1\beta_2 + \beta_1\gamma_2 + 2]^{3/2}}.$$

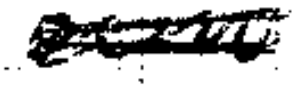
$$g_1 = \begin{pmatrix} \alpha_1 & \beta_1 \\ \gamma_1 & \delta_1 \end{pmatrix}, \qquad g_1 \in D_+.$$

The 2nd projector is analogous.

$$\mathcal{H}_{A,+} \cap \mathcal{H}_{A,-} = \{0\},$$

since [text crossed out] everything is in L^2.

Undeciphered passage. The word after “since” is heavily crossed out.



Original source details: GELFAND-P000662

The submerged part: Fourier-integral technique.

$K(g_1, g_2)$ — analogue of the Cauchy kernel.

Arnold’s conjecture. D_+ — those that map the disc into itself,

D_- — those that map the exterior of the disc [text crossed out] into itself,



Original source details: GELFAND-P000662

Undeciphered passage. A short entry after “disc” is heavily crossed out.

$SL(2, \mathbb{R})$ — preserves the circle.

GELFAND-P000663

13/IX-76 Gelfand Sem..

A. A. Kirillov. Superalgebras, supergroups and supermanifolds.

D. B. Fuchs. All about the Weil algebra.

$\mathfrak{g}$ — Lie algebra. Construct from it the assoc. Weil algebra:

$$W(\mathfrak{g}) = \underset{\text{exterior algebra}}{\Lambda^*(\mathfrak{g}')} \otimes_{\text{II}} \underset{\text{symmetric algebra}}{S^*(\mathfrak{g}')}.$$



Uncertain reading. Two vertical strokes with shared crossbars appear below the tensor product sign.

Original source details: GELFAND-P000663

If $\mathfrak{g}$ has basis $e_1, \ldots, e_n$, then the elements of $W(\mathfrak{g})$ — functions of $2n$ variables, super $\mathfrak{g}$.

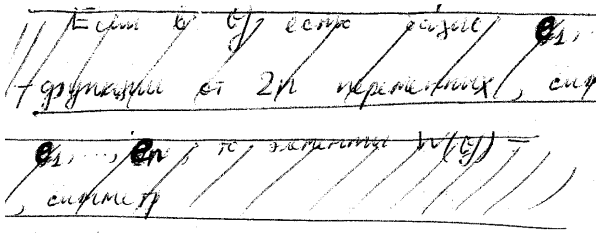
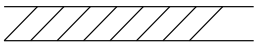


Diagram and typed description.



Diagonal strokes cross the two lines between the horizontal rules. The last entry tentatively reads “super” followed by a Fraktur g.

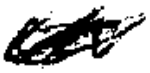
Original source details: GELFAND-P000663

Grading: $\Lambda^p(\mathfrak{g}') \otimes S^q(\mathfrak{g}')$ has degree $p + 2q$.

Differential: $d : W(\mathfrak{g}) \to W(\mathfrak{g})$ with properties

1) $d^2 = 0$, 2) acts in the usual way on [entry crossed out] the product.

Undeciphered passage. The short entry before “the product” is heavily crossed out.



Original source details: GELFAND-P000663

Let $h \in \underset{\Lambda^1(\mathfrak{g}')}{\mathfrak{g}'}$. Consider $h \otimes 1 \in W(\mathfrak{g})$.

$$d : \Lambda^1(\mathfrak{g}') \otimes S^0(\mathfrak{g}') \to \Lambda^2(\mathfrak{g}') \otimes S^0(\mathfrak{g}') \oplus \Lambda^0(\mathfrak{g}') \otimes S^1(\mathfrak{g}').$$

$$h \otimes 1 \text{ [}h\text{? crossed out]} \mapsto \{(g_1, g_2) \mapsto h([g_1, g_2])\} \otimes 1; \qquad 1 \otimes h.$$



Uncertain reading. The crossed-out sign before the arrow resembles h.

Original source details: GELFAND-P000663

This extends uniquely to a diff. with properties 1) and 2).

Property 2).

$$\xi \in W^k(\mathfrak{g}), \quad \eta \in W^\ell(\mathfrak{g}) \implies d(\xi \cdot \eta) = d\xi \cdot \eta + (-1)^k \xi \cdot d\eta.$$

On [entry crossed out] it is determined by the condition $d^2 = 0$.



Undeciphered passage. The entry after “On” is crossed out with diagonal strokes.

Original source details: GELFAND-P000663

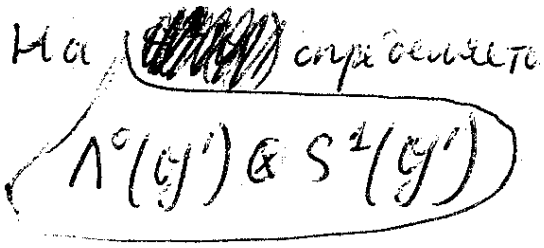


Diagram and typed description.

$\Lambda^0(\mathfrak{g}') \otimes S^1(\mathfrak{g}')$

The enclosed insertion is connected to the place after “On”.

Original source details: GELFAND-P000663

$$d : W^k(\mathfrak{g}) \to W^{k+1}(\mathfrak{g}).$$

$$d(h \otimes 1) = (dh \otimes 1) \oplus (1 \otimes h),$$

where $(dh)(g_1, g_2) \doteq h([g_1, g_2])$.

$\doteq$ means $\overset{\text{def}}{=}$.

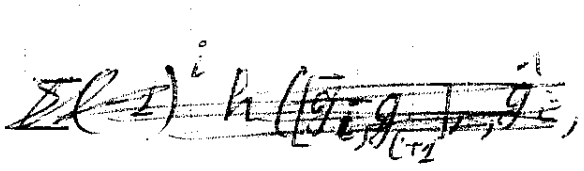
$$(dh)(g_1, \ldots, g_{\ell+1}) =$$



Uncertain reading. The last sign in the subscript resembles 1, as in the exponent $i + j - 1$. The reading $\ell + 1$ is tentative.

Original source details: GELFAND-P000663

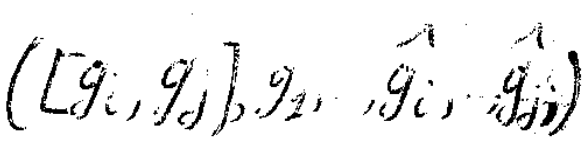
[First sum crossed out: $\sum(-1)^i h(\ldots)$.]



Undeciphered passage. The sum sign, the sign with superscript i, h, brackets and several instances of g are visible. The full entry cannot be read.

Original source details: GELFAND-P000663

$$\sum_{i<j} (-1)^{i+j-1} h([g_i, g_j], g_1, \ldots, \widehat{g_i}, \ldots, \widehat{g_j}).$$



Uncertain reading. The first subscript after the commutator tentatively reads 1. The last two gs carry hats. An extra stroke is visible at the end.

Original source details: GELFAND-P000663

$$dh = \sum a_i h_{i_1} \wedge h_{i_2} = \sum a_i (h_{i_1} \otimes 1)(h_{i_2} \otimes 1).$$

Differentiate by the Leibniz rule.

A nonzero element appears in $\Lambda^1 \otimes S^1$.

Let $e_1, \dots, e_n$ — a basis in $\mathfrak{g}$, also a basis of linear fns. on $\mathfrak{g}'$.

$h^1, \dots, h^n$ — the dual basis in $\mathfrak{g}'$.

Then

$$d(1 \otimes h^k) = c_{ij}^k (h^i \otimes h^j).$$

Main theorem: The homologies of the complex $\dots \to W^k \xrightarrow{d} W^{k+1} \to \dots$ are equal to 0.

Next time: how it works in char. classes.

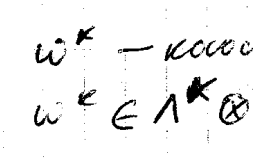
A. A. Kirillov. Devised to compute the homologies of simple compact Lie groups.

The easy part is the inv. forms.

Symmetric ones are easy to find. Skew-symmetric ones?

ω^k — a skew-symm. inv. diff. form.

$$\omega^k \in \Lambda^k \otimes S^0 + \dots; \qquad d\omega^k = 0.$$

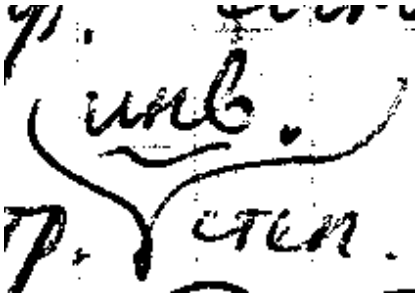


Original source details: GELFAND-P000664

Uncertain reading. The superscript on ω in the membership line is retraced. It tentatively reads k, as in the nearby entries.

Then ω^k — the differential of a symm. expr..

Conclusion: as many skew-symm. [inserted above: inv.] of deg. k as symm. of deg. $\frac{k+1}{2}$.



Original source details: GELFAND-P000664

Diagram and typed description.

inv.

The abbreviation “inv.” is inserted above the line between “skew-symm.” and “deg.”.

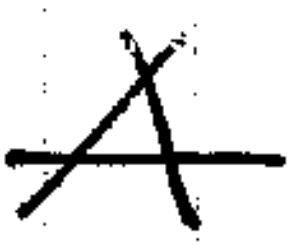
—”— 3 —”— 2.

Arnold. On works from the end of the last century on 2 questions.

1) $y'' = F(x, y, y')$ (e.g., $y'' = 0$).

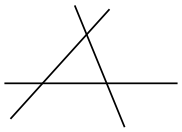
Tresse (a student of Sophus Lie) 1896, Acta Math. 1894.

$y'' = 0$ the graphs are straight lines.



Original source details: GELFAND-P000664

Diagram and typed description.

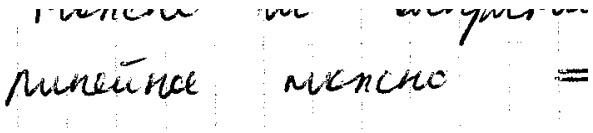


Three straight lines at right intersect pairwise to form a triangle.

There are configuration theorems (Pappus, Desargues, Pascal, etc.).

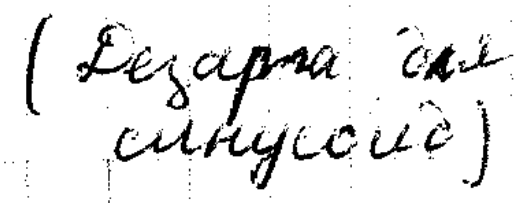
Can the solns. of the general eq. be straightened?

Linear menu $\Longrightarrow$ all the config. thms. hold (Desargues for sinusoids).



Original source details: GELFAND-P000664

Uncertain reading. The second word tentatively reads “меню” (menu).



Original source details: GELFAND-P000664

Uncertain reading. The word between “Desargues” and “sinusoids” tentatively reads “для” (for).

$$y'' + a(x)y' + b(x)y = 0$$

is reduced by a diffeom. of the plane (x, y), linear in y:

$$x = f(\tilde{x})$$
$$y = g(\tilde{x})\tilde{y}$$

Locally it is reduced to $y'' = 0$.

GELFAND-P000665

Consider an arbitrary family.

The infinitesimal theory is fairly well known.

E. Cartan in 1924–25, — projective connection —

— the solutions are geodesics with respect to this connection.

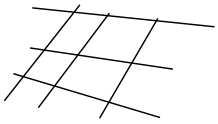
Projective connection — a connection in a G-bundle, where G — the projective group acting in the tangent space.

Space of line elements (x, y, y').

2) In $\mathbb{R}^3$ there are 2 direction fields.



Diagram and typed description.



Two families of three strokes each cross to form a slanted grid.

Original source details: GELFAND-P000665

The diffeom. group of $\mathbb{R}^3$ acts.

[retraced sign] To what form can one reduce it?

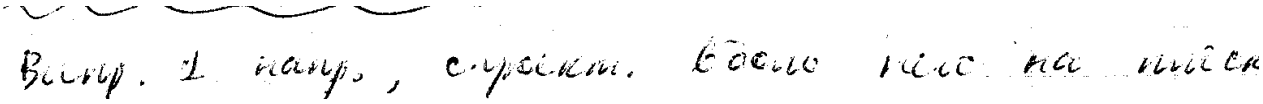


Uncertain reading. The isolated, heavily retraced sign before the question is undeciphered.

Original source details: GELFAND-P000665

These problems are equivalent.

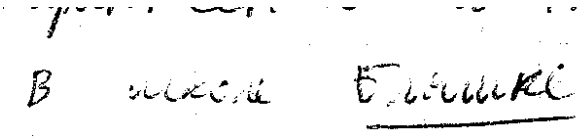
Choose 1 dir., project along it onto the plane. There is a 2-param. family there, etc.



Original source details: GELFAND-P000665

Uncertain reading. The abbreviation after “1 dir.” reads “спроект.” (project).

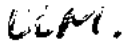
In his lifetime Blaschke worked on it.



Uncertain reading. The word before “Blaschke” tentatively reads “жизни” (life).

Original source details: GELFAND-P000665

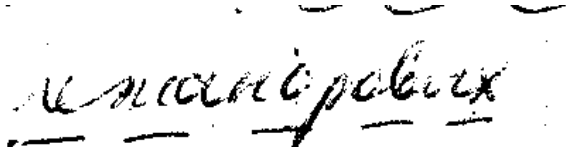
In $\mathbb{R}^3$ there is a contact structure. (Planes spanned by tangent vectors).
The curves have locally the structure of a projective line.



Uncertain reading. The abbreviation after “curves” tentatively reads “им.” (have).

Original source details: GELFAND-P000665

(from the general theory of multidimensional bundles)



Uncertain reading. The adjective before “bundles” tentatively reads “многомерных” (multidimensional).

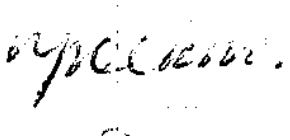
Original source details: GELFAND-P000665

On the circle there are many inequivalent structures of a locally projective space.

$$1)\quad \mathbb{RP}^1 \qquad 2)\quad S^1 \xrightarrow[\text{deg. } 2]{} \mathbb{RP}^1$$

2) inequivalent structure.

Proof of inequivalence: examine the vect. fields (infinitesim. automs. of this structure).

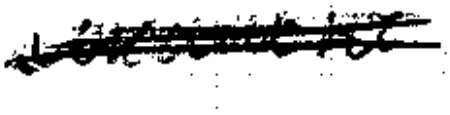


Uncertain reading. The abbreviation before “vect. fields” tentatively reads “просм.” (examine).

Original source details: GELFAND-P000665

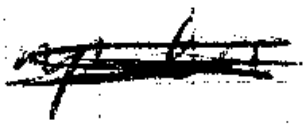
$$\text{on } \mathbb{RP}^1 \quad \frac{\partial}{\partial x}, \quad x\frac{\partial}{\partial x}, \quad x^2\frac{\partial}{\partial x}.$$

Problem: how [entry crossed out] the equation [entry crossed out] globally a locally projective space?



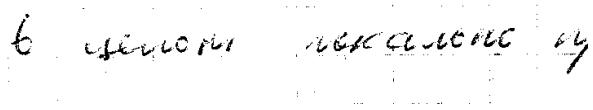
Undeciphered passage. The word between “how” and “the equation” is heavily crossed out.

Original source details: GELFAND-P000665



Undeciphered passage. The word after “the equation” is crossed out with several strokes.

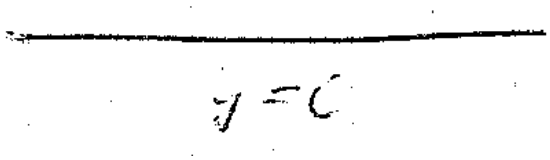
Original source details: GELFAND-P000665



Uncertain reading. After the crossed-out word the writing reads “в целом локально” (globally, locally).

Original source details: GELFAND-P000665

Take one soln. as a straight line.



Original source details: GELFAND-P000665

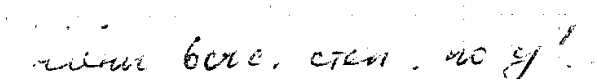
Diagram and typed description.

$y = 0$

The line at right is labelled $y = 0$.

$$y'' = a(x)y + b(x)p + \dots$$

[At right: terms of higher deg. in y'.]

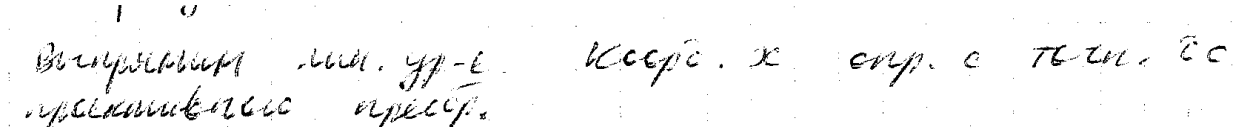


Original source details: GELFAND-P000665

Uncertain reading. The first words of the note read “члены выс. степ.” (terms of higher deg.).

$$p = y'.$$

Straighten the lin. eq.. The coord. x is associated with points of its projective space.



Original source details: GELFAND-P000665

Uncertain reading. The abbreviation after “with” tentatively reads “т-ми” (points).

$$\begin{cases} \tilde{x} = f(x), \\ \tilde{y} = g(x)y. \end{cases}$$

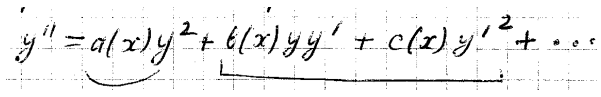
(f satisfies an eq. of Schwarz type).

GELFAND-P000666

Hence a projective structure arises on $\forall$ solns..

Now start with the quadr. part.

$$y'' = a(x)y^2 + b(x)yy' + c(x)y'^2 + \dots$$



Original source details: GELFAND-P000666

Diagram and typed description.

$\overbrace{a(x)y^2}\ \underbrace{+b(x)yy'+c(x)y'^2}$

An arc is drawn below the first term. A bracket runs under the next terms through $c(x)y'^2$.

Now diffeom. of the plane of the form:

$$\begin{cases} \tilde{x} = x + yf(x,y), \\ \tilde{y} = y(1 + yg(x,y)). \end{cases}$$

Kill b and c with these diffeom..

There remains a, determined up to a mult. constant.

$\underline{\underline{a(x)(dx)^{5/2}}}$ determined up to a mult. constant.

This is a 5/2-form along the solution, determined up to a mult. const. depending on the solution.

Theorem. If $a(x)(dx)^{5/2} \equiv 0$, then the eq. can be straightened.

(in formal series)

Further diffeoms. lead to $\underline{\underline{y'' = y'^2 A(x,y,y')}}$.

Consider an eq. of the form $\underline{y'' = P_3(y')}$ — a 3rd-degree poly. in y'.

This class is inv. under diffeomorphisms.

The occurrence of these eqs. was discovered by R. Liouville.



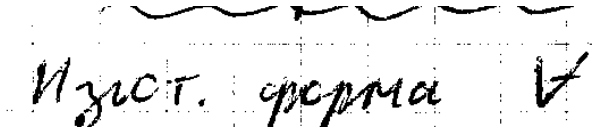
Original source details: GELFAND-P000666

Uncertain reading. The verb before the name tentatively reads “открыл” (discovered).

Scalar invariant.

From this, a form of $\forall$ other degs. (in partic. a vect. field).

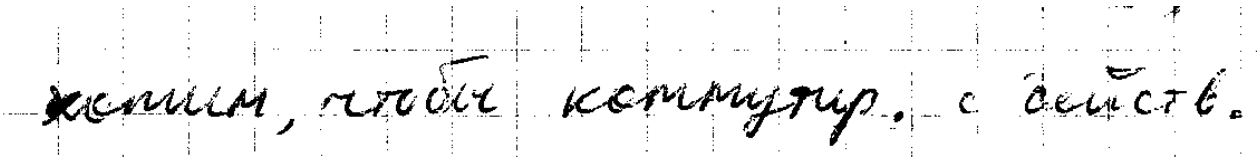
$$\underline{v(x) = (a(x))^{-2/5} \frac{\partial}{\partial x}.}$$



Original source details: GELFAND-P000666

Uncertain reading. The opening tentatively reads “Из эт.” (from this).

Now construct a scalar: we want it to commute with the action of the projective group.



Original source details: GELFAND-P000666

Uncertain reading. After “we want” the entry reads “чтобы коммутир.” (to commute).

We obtain invariants:

$I_2, I_3, \ldots$ — invariants.

Consider the 2-jet of the field. On 2-jets the projective group acts by the adjoint

representation.

$$I_2 = 2v''v - v'^2$$
$$I_3 = vI_2'$$
$$— \quad — \quad —$$
$$I_n = vI_{n-1}'$$

Gelfand conjecture: connected with Gelfand–Fuchs cohomology.

Pommaret worked on diff. invariants.

GELFAND-P000667

20/IX-76 Gelfand Sem..

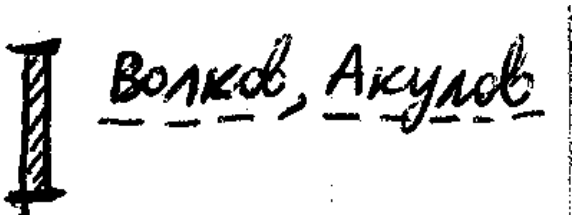
S. P. Novikov. Superalgebras in physics.

[Crossed out: Rev.?] Ogievetsky's review in UFN and Zumino's review.



Original source details: GELFAND-P000667

Uncertain reading. The crossed-out opening tentatively reads "Обз." (Rev.).



Original source details: GELFAND-P000667

Diagram and typed description.

Volkov, Akulov

The surnames are underlined with broken strokes in the right margin. A narrow hatched mark stands to their left.

Superalgebra [Crossed out: Lie?] — a direct sum $L = L_{\text{even}} \oplus L_{\text{odd}} = L_0 \oplus L_1$.



Original source details: GELFAND-P000667

Uncertain reading. The strokes of the short crossed-out word resemble "Ли" (Lie).

$[a, b]$ defines the struct. of a graded Lie algebra.

$$[a, b] = (-1)^{da \cdot db + 1}[b, a], \qquad da = \dim a.$$



Original source details: GELFAND-P000667

Uncertain reading. The definition of da reads "dim a".

The Jacobi id. also holds with signs.

Let L_0 and L_1 be embedded in an assoc. algebra (e.g., of matrices).

L — a subalgebra of the matrix alg., closed under the operation

$$1) \quad a, b \in L_0 \implies ab - ba = [a, b] \in L_0,$$
$$2) \quad a, b \in L_1 \implies ab + ba = [a, b] \in L_0.$$

$$3) \quad a \in L_0,\ b \in L_1 \implies ab - ba = [a, b] \subset L_1.$$

Or

$$[a, b] = (-1)^{\varepsilon_1 \varepsilon_2 + 1}[b, a],$$
$$\varepsilon_1 = \deg a, \qquad \varepsilon_2 = \deg b.$$

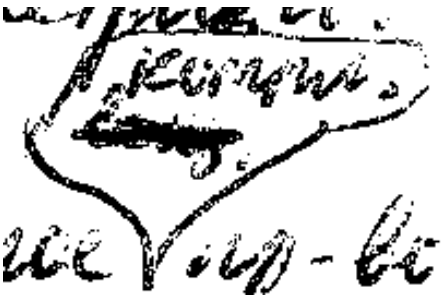
Concrete superalgebras were first considered in the work of Gelfand and Likhtman.

$$\mu = 0, 1, 2, 3.$$

Everything in $\mathbb{R}^4$ Minkowski.

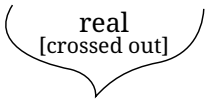
$$\eta_{\nu,\mu} = \begin{pmatrix} -1 & & & 0 \\ & 1 & & \\ & & 1 & \\ 0 & & & 1 \end{pmatrix} \quad \text{— metric.}$$

space of spinors — 4-dimensional space [inserted above: real. Word crossed out.] (with basis) Q_j, $j = 1, 2, 3, 4$.

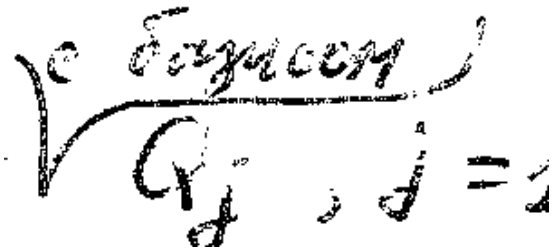


Original source details: GELFAND-P000668

Diagram and typed description.

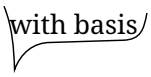


"Real" is written in the enclosed insertion. The lower word is crossed out.



Original source details: GELFAND-P000668

Diagram and typed description.



The second outlined insertion, "with basis", points to the start of the list.

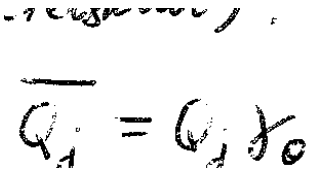
$$\alpha = \sum \alpha^j Q_j.$$

Matrix algebra generated by matrices γ_μ (real) — Majorana basis.

$$\gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2\eta_{\nu\mu}. \qquad (\approx \text{ quaternions})$$

Choose 4 matrices γ_μ, real and forming a Majorana basis
(see Akhiezer, Berestetskii).
Set

$$\bar{Q}_j = Q_j\gamma_0;$$
$$\bar{\alpha} = \alpha\gamma_0.$$



Original source details: GELFAND-P000668

Uncertain reading. The letter Q has a descending diagonal stroke.

L_0 — alg. of [crossed out: Poincaré] translations.



Original source details: GELFAND-P000668

Uncertain reading. “Poincaré” is crossed out before “translations”.

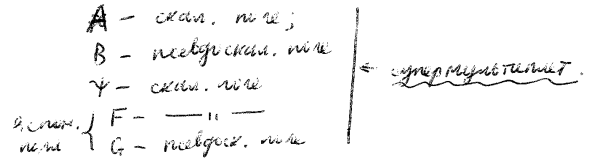
L_1 — space of spinors.

$$1) \quad [L_1, L_0] = 0,$$
$$2) \quad [L_1, L_1] \text{ — ?}$$

$$[\bar{Q}_i, \bar{Q}_j] = -2(\gamma^\mu)_{ij} P_\mu.$$

P_μ — Poincaré translation oper., μ — index.

G. Problem. extend the Poincaré alg. to a superalgebra.



Original source details: GELFAND-P000668

Diagram and typed description.

A — scalar field
B — pseudoscalar field
ψ — scalar field
aux. fields: F — —”—
aux. fields: G — pseudoscal. field

← supermultiplet

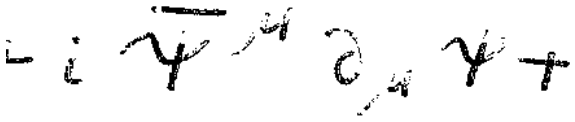
The brace at left groups F, G. The label at right points to the vertical line beside the list.



Original source details: GELFAND-P000668

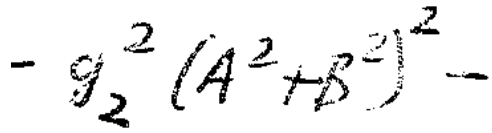
Uncertain reading. The label beside ψ also reads “скал. поле” (scalar field).

$$
\begin{aligned}
L = & -\tfrac{1}{2}\big((\partial_\nu A)^2 + (\partial_\mu B)^2 + i\bar\psi\gamma^\mu\partial_\mu\psi \\
& \quad + m^2A^2 + m^2B^2 + im\bar\psi\psi\big) \\
& + gmA(A^2+B^2) - g_2^2(A^2+B^2)^2 \\
& - ig\,[im\text{? crossed out}]\,\bar\psi(A-\gamma_5 B)\psi.
\end{aligned}
$$



Original source details: GELFAND-P000668

Uncertain reading. Faint strokes before the raised μ after $\bar\psi$ tentatively read γ.



Original source details: GELFAND-P000668

Uncertain reading. Two instances of 2 are visible, one above g and one below. No fraction bar is distinguishable.



Original source details: GELFAND-P000668

Uncertain reading. The crossed-out strokes after $-ig$ tentatively read im.

[crossed out: (supersymm. Lagrangian?)]



Original source details: GELFAND-P000668

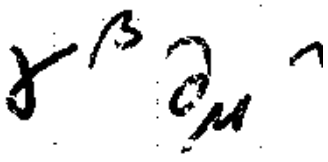
Uncertain reading. The final word in the crossed-out parenthesis tentatively reads “лагранжиан” (Lagrangian).

GELFAND-P000669

The opers. from L_0 and L_1 act on the Lagrangian L (a rep. is constructed) and it is invariant.

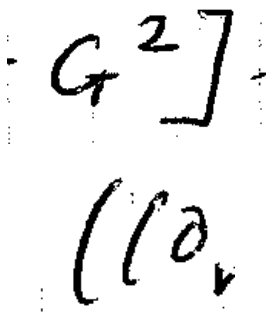
The Wess–Zumino supersymm. Lagrangian.

$$
\begin{aligned}
L = & -\tfrac{1}{2}\big((\partial_\nu A)^2 + (\partial_\mu B)^2 + i\bar\Psi\gamma^\beta\partial_\mu\Psi - F^2 - G^2\big] \\
& + m(FA + GB - i\bar\Psi\Psi) \\
& + g(FA^2 - FB^2 + 2GAB - i\bar\Psi\Psi A + i\bar\Psi\gamma_5\Psi B).
\end{aligned}
$$



Original source details: GELFAND-P000669

Uncertain reading. The upper index of γ is β. The lower index of ∂ is μ.



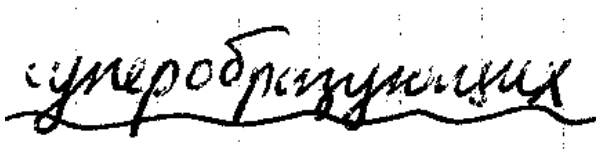
Original source details: GELFAND-P000669

Uncertain reading. The outer opening bracket is round and the closing bracket is square.

No derivs. in F and $G \Rightarrow$ they are eliminated by the Euler eqs. $\Rightarrow$

$\Rightarrow$ the preceding Lagrangian.

Action of the supergenerators.



Original source details: GELFAND-P000669

Uncertain reading. The heading reads “суперобразующих” (of the supergenerators).

$$L = L_0 + L_1.$$

$$\alpha \in L_1, \quad \alpha = \sum \alpha_j Q_j, \qquad \bar{\alpha} = \alpha\gamma_0.$$

1) $[\bar{\alpha}, A] = \delta A = i\bar{\alpha}\Psi$;
2) $\delta B =$ [crossed out] $i\bar{\alpha}\gamma_5\Psi$;
3) $\delta\Psi = \partial_\mu(A - \gamma_5 B)\gamma^\mu\alpha + (F + \gamma_5 G)\alpha$;
4) $\delta F = i\bar{\alpha}\gamma^\mu\partial_\mu\Psi$;
5) $\delta G = i\bar{\alpha}\gamma_5\gamma^\mu\partial_\mu\Psi$.



Original source details: GELFAND-P000669

Undeciphered passage. The entry after $\delta B =$ is crossed out. Individual signs are visible.

Migdal (Sasha). On gauge fields.

They arose from attempts to describe exp. data (on nucleons). These say that nucleons are made up of quarks.



Original source details: GELFAND-P000669

Diagram and typed description.

$p \sim$  (3 quarks)

After $p \sim$, an oval with three dots and the label “3 quarks”.

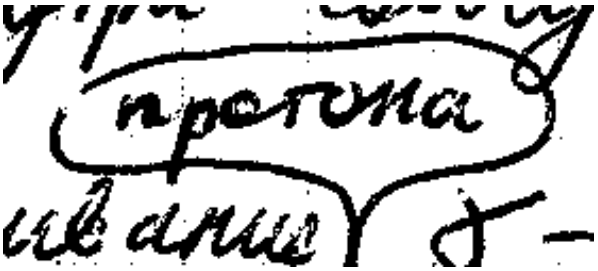
They are free inside, and it is impossible to take them out.

[crossed out: They.] Quarks are visible — probing [inserted above: the proton] γ-quanta (as in a cellophane bag).



Original source details: GELFAND-P000669

Uncertain reading. “They” is crossed out before “Quarks”.



Original source details: GELFAND-P000669

Diagram and typed description.



“The proton” is inserted before the quanta.

In the theory the forces must increase with distance!

[crossed out] By the uncertainty princ. they cannot merge.



Original source details: GELFAND-P000669

Undeciphered passage. The short entry before “By the principle” is crossed out.

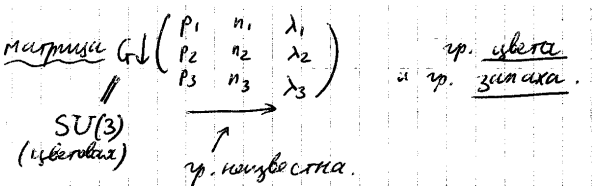
Relativ. inv. hinders specifying a law of forces (the interaction potential is relativ. noninvariant).

Locality is confirmed by probing — quarks are nearly pointlike.

one must construct Lagrangians.

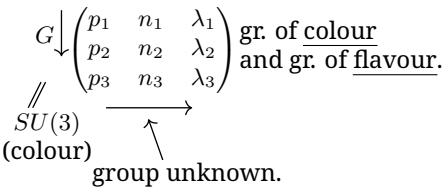
Gauge theory gives such a law of forces.

Quark — matrix.



Original source details: GELFAND-P000669

Diagram and typed description.



At left G and a downward arrow. Below the matrix a rightward arrow with an annotation. The labels at right name colour and flavour.

There are 9 quarks.

Forces that prevent breaking apart are connected with colour.

To the colour gr. corresponds [crossed out] a much stronger symm. — gauge.



Undeciphered passage. The isolated sign after "corresponds" is heavily crossed out.

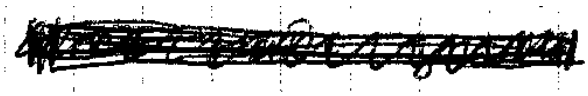
Original source details: GELFAND-P000669

Mesons are constructed from a quark and an antiquark.

GELFAND-P000670

18/X-76. Gelfand Sem..

[crossed out] Write down the basic notions and equations one needs to know. And the problems.



Undeciphered passage. The long entry before "Write down" is heavily crossed out.

Original source details: GELFAND-P000670

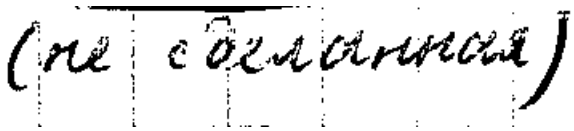
Problem: express II_1 factors using matrices of continuous order.



Uncertain reading. The subscript is heavily retraced. Its outline reads 1.

Original source details: GELFAND-P000670

(undone)



Uncertain reading. The note tentatively reads "не сделанная" (undone).

Original source details: GELFAND-P000670

I. Bernstein. (continued).

Superspace — informally, there are 2 sets of variables:

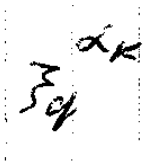
$$\left\| \begin{array}{l} x_1, \ldots, x_p, \\ \xi_1, \ldots, \xi_q \end{array} \right.$$

Formally

$$C^\infty(\mathbb{R}^p) \otimes \Lambda(\xi_1, \quad, \xi_q)$$

An element:

$$\sum f_{\alpha_1 \ldots \quad \alpha_q}(x) \xi_1^{\alpha_1} \cdots \xi_q^{\alpha_k}, \qquad \alpha_i = 0, 1.$$



Uncertain reading. The last ξ_q has exponent α_k. The coefficient ends in subscript α_q.

Original source details: GELFAND-P000670

All f commute with ξ, and $\xi_i\xi_j = -\xi_j\xi_i,\ \xi_i^2 = 0$.

$\mathbb{Z}_2$ — Grading. Degree — sum of α_i mod 2. ||

Another def. [crossed out] Ringed space.



Original source details: GELFAND-P000670

Undeciphered passage. The narrow sign before "Ringed" is blacked out.

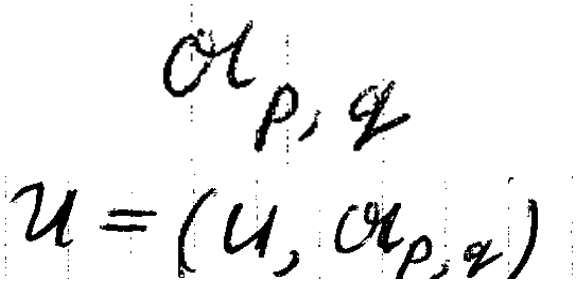
$\mathbb{R}^p$ with the sheaf $\mathcal{A}_{p,q}$.

$$U \subset \mathbb{R}^p \quad \longmapsto \quad \text{ring } C^\infty(U) \otimes \Lambda(\xi_1, \quad , \xi_q)$$

$U \subset V \quad \Rightarrow \quad$ there is a natural restriction map.

Superdomain. This is a topological space $U \subset \mathbb{R}^p$ with the same sheaf

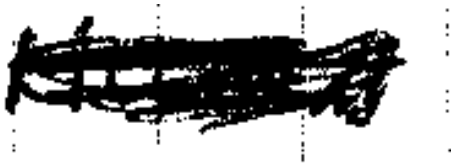
$$\mathcal{U} = (U, \mathcal{A}_{p,q}).$$



Original source details: GELFAND-P000670

Uncertain reading. The symbol is repeated in the ordered pair and reads as a script $\mathcal{A}$.

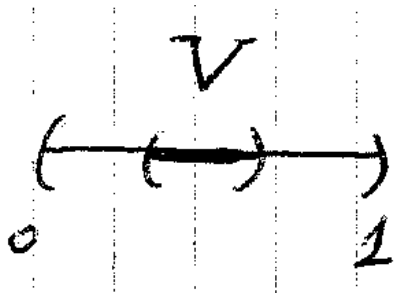
[Crossed out above the heading]



Original source details: GELFAND-P000670

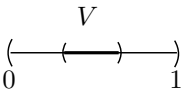
Undeciphered passage. The short word above "Example" is heavily crossed out.

Example. $u, \xi_1, \xi_2, \qquad 0 < u < 1.$



Original source details: GELFAND-P000670

Diagram and typed description.



The inner interval V is marked by a heavier line between parentheses.

functions on $V = C^\infty(V) \otimes \Lambda(\xi_1, \xi_2)$.

$$f_{00}(u) + [\text{crossed out}]\, f_{10}(u)\xi_1 + f_{01}\xi_2 + f_{11}(u)\xi_1\xi_2.$$



Original source details: GELFAND-P000670

Undeciphered passage. A group of signs after $f_{00}(u)+$ is crossed out.

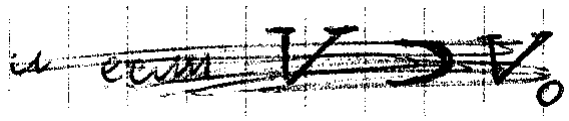
$C^\infty(\widetilde{U})$ — notation for the space of fns. on the superdomain $\widetilde{U}$.

$$\|$$
$$C^\infty(U) \otimes \Lambda(\xi_1, \ldots, \xi_q)$$

Map of superdomains $\varphi : \widetilde{U} \longrightarrow \widetilde{V}$
this is a collection of objects: a map $\varphi' : U \to V$
and a map

$$C^\infty(\widetilde{V}) \longrightarrow C^\infty(\widetilde{U})$$

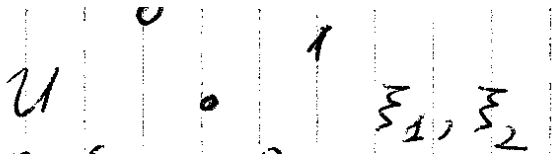
a ring hom. covering φ' (with unit) and preserving parity!!
[crossed out: and if $V \supset V_0$?]



Original source details: GELFAND-P000670

Uncertain reading. The sign between V and V_0 in the crossed-out line resembles $\supset$.

Example.



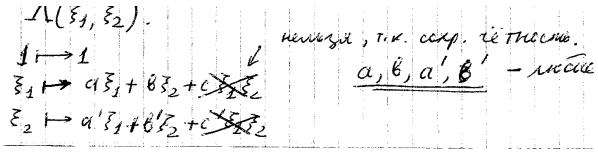
Original source details: GELFAND-P000670

Diagram and typed description.

$U \quad \circ \quad {}^{1}_{\xi_1, \xi_2}$

Beside the point are the labels U, 1 and ξ_1, ξ_2.

$p = 0,\ q = 2.$ Superdomain of dimension $(0, 2)$.
Geometrically, a point. a map of the superdomain to itself is an endomorphism.
Rings $\Lambda(\xi_1, \xi_2)$.



Original source details: GELFAND-P000670

Diagram and typed description.

$1 \mapsto 1$

$\xi_1 \mapsto a\xi_1 + b\xi_2 + c\xi_1\xi_2$

$\xi_2 \mapsto a'\xi_1 + b'\xi_2 + c\xi_1\xi_2$

The arrowed note reads “disallowed, since it preserves parity”. The letter c follows the final plus, before the crossed-out product. A prime is unclear.

a, b, a', b' — arbitrary.

GELFAND-P000671

The algebra $\Lambda(\xi_1, \xi_2)$ is commutative in the super sense.

$$xy = (-1)^{p(x)p(y)}yx.$$

Example 2.

$$\begin{matrix} u, \xi_1, \xi_2 \\ v, \eta_1, \eta_2 \end{matrix} \quad \left\| \begin{matrix} v = u + \xi_1\xi_2 \\ \eta_1 = \xi_1 \\ \eta_2 = \xi_2. \end{matrix} \right.$$

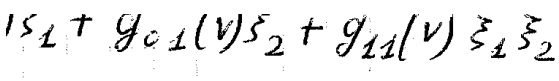
Problem: what is the image of $\sin v$?

An element of the left ring has the form

$$f_{00}(u) + f_{10}(u)\xi_1 + f_{01}(u)\xi_2 + f_{11}(u)\xi_1\xi_2$$

An elem. of the 2nd ring

$$g_{00}(v) + g_{10}(v)\xi_1 + g_{01}(v)\xi_2 + g_{11}(v)\xi_1\xi_2$$



Original source details: GELFAND-P000671

Uncertain reading. The 2nd ring is written with ξ_1, ξ_2, as in the 1st.

map

$$\widetilde{U} \longrightarrow \widetilde{V}$$
$$C^\infty(\widetilde{V}) \longrightarrow C^\infty(\widetilde{U})$$
$$g_{00}(v) \mapsto g_{00}(u + \xi_1\xi_2) = g_{00}(u) + \xi_1\xi_2 g'_{00}(u).$$

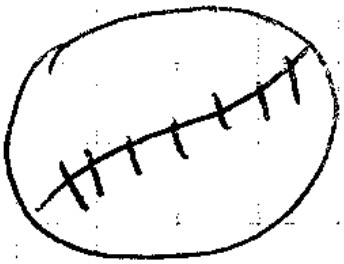
$x = (u_1, \ldots, u_p, \xi_1, \ldots, \xi_q)$ — a coordinate system.

$$\frac{\partial}{\partial u_i} f\xi_1^{\alpha_1} \ldots \xi_q^{\alpha_q} = \frac{\partial f}{\partial u_i}\xi_1^{\alpha_1} \ldots \xi_q^{\alpha_q}$$

Stokes' theorem — ?

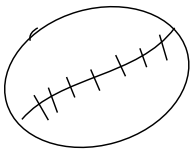
The integral of a superfunction over a superdomain.

Analogue in ordinary analysis



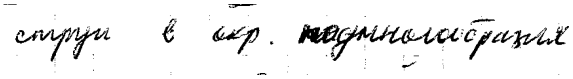
Original source details: GELFAND-P000671

Diagram and typed description.



A rising curve and seven visible cross-strokes lie inside the oval. A short mark meets the outline at the upper left.

jets in a nbhd. of a submanifold.



Original source details: GELFAND-P000671

Uncertain reading. The first words read "струи в окр." (jets in a nbhd.). The start of "submanifold" is retraced.

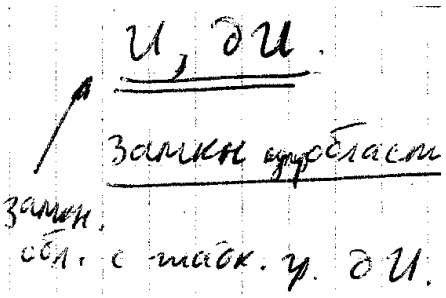
The integral for fns. on a manifold with boundary is noninv. under changes of variables.

Difference: here one cannot integrate forms. One can integrate densities (under change of coords. mult. by the Berezinian).

What does one integrate over the boundary?

$U, \partial U$. $f = 0$, where $f \in C^\infty(U)$ — a superfunction.

Closed superdomain. u, ξ_1, ξ_2



Original source details: GELFAND-P000671

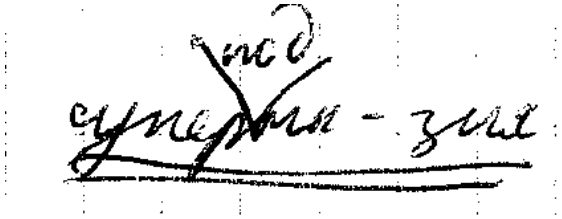
Diagram and typed description.

$U, \partial U$

closed domain with smooth bdry. ∂U

The arrow points from the note towards $U, \partial U$.

On ∂U there is the struct. of a [inserted above: sub] supermanifold.



Original source details: GELFAND-P000671

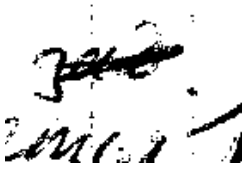
Diagram and typed description.

sub

supermanifold

The "sub" insertion points after "super". The ending of the abbreviation is tentative.

(by an even eq. [crossed out: this?] is cut out)



Original source details: GELFAND-P000671

Uncertain reading. The crossed-out word tentatively reads "это" (this).

$[0, \infty)$ u, ξ_1, ξ_2.

Different superdomains

1. $u = 0$
2. $u + c\xi_1\xi_2 = 0$



Original source details: GELFAND-P000671

Diagram and typed description.



even eq.

A narrow slanted quadrilateral with heavy lengthwise strokes. The words at right label the figure.

GELFAND-P000672

Submanifold: in the ring choose an ideal of fns..

Boundary — even.

$C^\infty(\widetilde{U})$ Boundary — eq. $f(\widetilde{U}) = 0$

$$\int \text{of a compactly supported fn.} = \int f_{11\ldots1}\, du_1 \ldots du_p$$

Different for closed domains.

Let the half-space be $u_1 \geq 0$

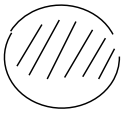
Then the integr. is given by the same formula.

It is preserved if it maps into itself as a supermanifold.

$u_1^2 + u_2^2 \leq 1$



Diagram and typed description.



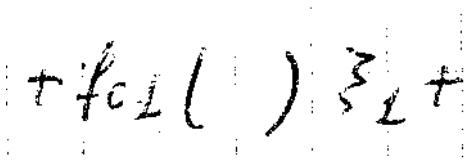
Slanted strokes fill the disc.

Original source details: GELFAND-P000672

$$u_1^2 + u_2^2 - 1 = 0 \quad (1) \quad \text{— 2 boundaries}$$
$$u_1^2 + u_2^2 - 1 - \xi_1\xi_2 = 0 \quad (2)$$

Integral over the domain with boundary (2).

$$\int [f_{00}(u) + f_{01}(\,)\xi_1 + \ldots]$$



Uncertain reading. After f_{01} the parentheses are empty, followed by ξ_1.

Original source details: GELFAND-P000672

Roughly

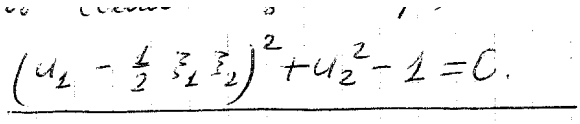
$$\int \xi_1 \xi_2 \, d\xi_1 \, d\xi_2 = 1$$

— For the ordinary boundary —

$$\int_{u_1^2+u_2^2<1} f_{11}(u) \, du_1 du_2$$

Check of invariance — the boundary eq. must stay the same.
If the boundary is $u_1^2 + u_2^2 - 1 - \xi_1\xi_2 = 0$
change variables so the boundary goes to the normal one

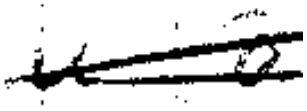
$$\left(u_1 - \tfrac{1}{2}\xi_1\xi_2\right)^2 + u_2^2 - 1 = 0.$$



Original source details: GELFAND-P000672

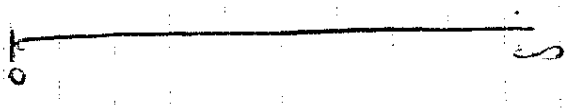
Uncertain reading. Inside the parentheses the term after u_1 is $-\frac{1}{2}\xi_1\xi_2$.

[crossed out: $u = 0$?]



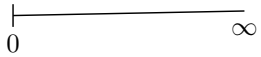
Original source details: GELFAND-P000672

Uncertain reading. The crossed-out formula above the line tentatively reads $u = 0$.



Original source details: GELFAND-P000672

Diagram and typed description.



A tick and 0 mark the left end. The right end is marked ∞.

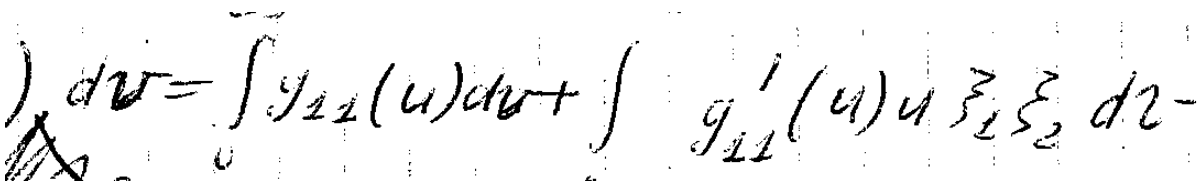
$$\int_0^\infty f_{11}(u) \, du \qquad u, \xi_1, \xi_2$$

bdry. $u = 0$.

$v = u + u\xi_1\xi_2$.

The boundary will be $v = 0$

$$\begin{aligned}\int_0^\infty g_{11}(v) \, dv &= \int_0^\infty g_{11}(u + u\xi_1\xi_2) \, dv \\ &= \int_0^\infty g_{11}(u) \, dv + \int_0^\infty g'_{11}(u) u \xi_1 \xi_2 \, dv\end{aligned}$$



Original source details: GELFAND-P000672

Uncertain reading. The differential in the third integral resembles dv.



Original source details: GELFAND-P000672

Diagram and typed description.

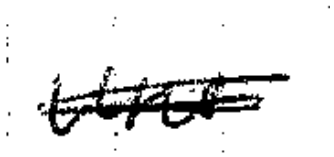


An open descending curve with diagonal and looping strokes. Some of their junctions are indistinct.

$$\frac{Dv}{Du}$$

Integral over a domain with boundary — — Bernstein and Leites

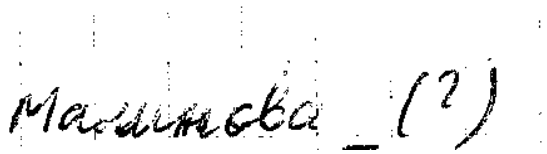
[crossed out]



Original source details: GELFAND-P000672

Undeciphered passage. The word before “There is work” is heavily crossed out.

There is work by Berezin and Marinov (?)



Original source details: GELFAND-P000672

Uncertain reading. The surname tentatively reads “Маринова” (Marinov). The question mark is in the manuscript.

class. mech. in Fermi and Bose variables

Problem: Stokes’ thm. for the circle $u_1^2 + u_2^2 + c\xi_1\xi_2 - 1 = 0$.

GELFAND-P000673

Notes on I. Bernstein’s previous super talks (from I. Dorfman’s notebook)



Original source details: GELFAND-P000673

Uncertain reading. The name in parentheses tentatively reads “И. Дорфман” (I. Dorfman).

Def. Superring

$$A = \underset{\text{even}}{A_0} \oplus \underset{\text{odd}}{A_1} \qquad \begin{aligned} A_0A_0 &\subset A_0 \\ A_0A_1 &\subset A_1 \\ A_1A_0 &\subset A_1 \\ A_1A_1 &\subset A_0. \end{aligned}$$

It is called commutative, if $ab = ba(-1)^{\deg a \cdot \deg b}$

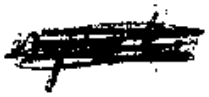
Linear algebra.



Undeciphered passage. Crossed-out mark before the bracket.

Original source details: GELFAND-P000673

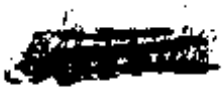
[Over a commut. superring consider



Undeciphered passage. Crossed-out word after "consider".

Original source details: GELFAND-P000673

any A-module A^n and its

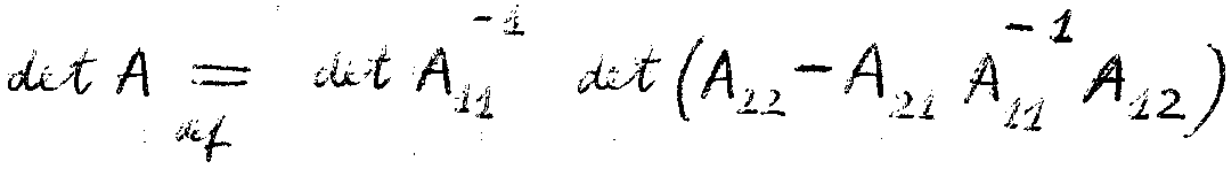


Undeciphered passage. Crossed-out word after "its".

Original source details: GELFAND-P000673

endomorphism given by a block matrix]

$$[A =] \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \qquad \det A \underset{\text{def}}{=} \det A_{11}^{-1} \det(A_{22} - A_{21}A_{11}^{-1}A_{12})$$



Original source details: GELFAND-P000673

Uncertain reading. The faint superscript on the first A_{11} reads -1.

$$\operatorname{str} A = \operatorname{tr} A_{11} - \operatorname{tr} A_{22}.$$
(supertrace)

Properties:

$$\det XY = \det X \cdot \det Y.$$
$$\det X = \exp \operatorname{str} \ln X$$



Uncertain reading. After X there is a dark mark below the line. It may be a comma or a crossed-out sign.

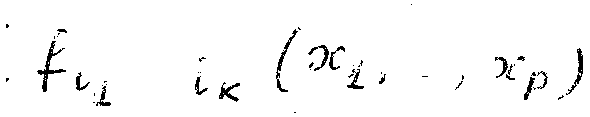
Original source details: GELFAND-P000673

$$\operatorname{str}(XY) = \operatorname{str}(YX)$$
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^* = \begin{pmatrix} A^* & C^* \\ -B^* & D^* \end{pmatrix}$$

Superanalysis.

Consider

$$\sum f_{i_1 \quad i_k}(x_1, -, x_p)\, \xi_1^{i_1} \dots \xi_k^{i_k}$$



Uncertain reading. A faint short stroke appears between the commas after x_1.

Original source details: GELFAND-P000673

Rules of differentiation

$$\frac{\partial}{\partial\xi_1}(\xi_1\xi_2)=\xi_2,\qquad \frac{\partial}{\partial\xi_2}(\xi_1\xi_2)=-\xi_1,$$
$$\frac{\partial}{\partial\xi_1}\xi_2=0,\qquad \frac{\partial}{\partial\xi_1}\xi_1=1,\qquad \frac{\partial}{\partial\xi_1}1=0.$$

_____ · _____

Consider block matrices $\begin{pmatrix}A & B\\ C & D\end{pmatrix}$

$$A=A_{\bar0}\oplus A_{\bar1}$$
$\mathbb{Z}/2\mathbb{Z}$ — grad. elem. $A,D\in A_{\bar0}$
$\bar0,\bar1$ — elem. elem. $B,C\in A_{\bar1}$
$p(a)$ — parity

$$\operatorname{Ber}J=(\det D)^{-1}\cdot\det(A-BD^{-1}C)\in A_{\bar0}$$
(Berezinian)

Assert

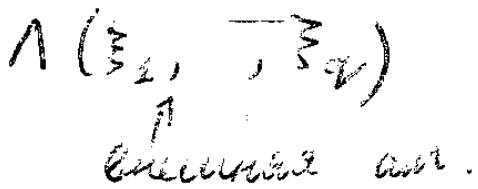
$$\operatorname{Ber}J_1J_2=\operatorname{Ber}J_1\cdot\operatorname{Ber}J_2.$$
$$\operatorname{str}J=\operatorname{tr}A-\operatorname{tr}D$$

Assert.

$$\operatorname{str}(J_1J_2)=\operatorname{str}(J_2J_1).$$

_____ · _____

In $\mathbb{R}^p$ consider $C^\infty(\mathbb{R}^p)\otimes\Lambda(\xi_1,—,\xi_q)$



Original source details: GELFAND-P000673

$p(\xi_i)=1,\qquad \xi_i\xi_j=-\xi_j\xi_i.$

Diagram and typed description.

$\Lambda(\xi_1,—,\xi_q)$
↑
exterior alg.

The arrow from the note points up towards the comma after ξ_1.

$$\Lambda(\xi_1, \dots, \xi_q) \ni \xi_1^{\alpha_1} \cdots \xi_q^{\alpha_q}, \qquad \alpha_i = 0, 1.$$
$$f = \sum f_{\alpha_1 \alpha_2 \dots \alpha_q} \xi_1^{\alpha_1} \cdots \xi_q^{\alpha_q}.$$

Def.

$$\mathbb{R}^{p,q} = (\mathbb{R}^p, C^\infty(\mathbb{R}^p) \otimes \Lambda(\xi_1, \dots, \xi_q))$$

— superspace of dim. (p, q).

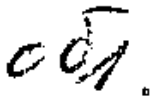
Def.

$$U = \left(u, \underbrace{C^\infty(u) \otimes \Lambda(\xi_1, \dots, \xi_q)}_{(\text{def})\, C^\infty(U)} \right)$$

superdomain.

(ringed sp.!)

— ring of fns. on the dom.



Uncertain reading. The abbreviation after "on" tentatively reads "обл." (dom.).

Original source details: GELFAND-P000674

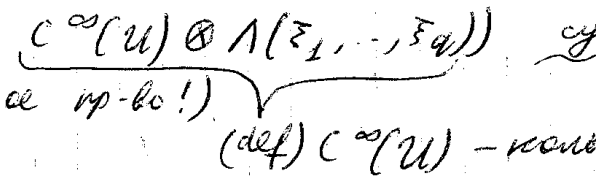


Diagram and typed description.
$\underbrace{C^\infty(u) \otimes \Lambda(\xi_1, \dots, \xi_q)}_{(\text{def})\, C^\infty(U)}$
The brace under the tensor product points to $(\text{def})\, C^\infty(U)$.

Original source details: GELFAND-P000674

map $\widetilde{\psi} : U \to V$



Uncertain reading. The isolated mark before the dash is undeciphered.

Original source details: GELFAND-P000674

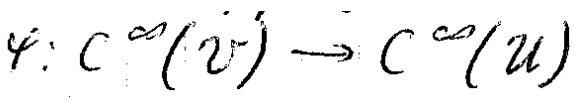
— morphism of ringed sp. — this is a map



Undeciphered passage. Crossed-out writing before ψ.

Original source details: GELFAND-P000674

$$\psi : C^\infty(V) \to C^\infty(U).$$



Uncertain reading. The arguments in the function rings are rendered as V and U. The manuscript barely distinguishes their case.

Original source details: GELFAND-P000674

Coord. notation. Let a tuple be chosen in the sp. U

$$(u_1, \dots, u_p, \xi_1, \dots, \xi_q)$$

— " — V — " — $(v_1, \dots, v_m, \eta_1, \dots, \eta_n)$.

It turns out that to specify the map $\psi : U \to V$



Undeciphered passage. Crossed-out word after $U \to V$.

Original source details: GELFAND-P000674

it suffices to specify

$$v_1^*, \dots, v_m^*, \eta_1^*, \dots, \eta_n^*.$$

Example.

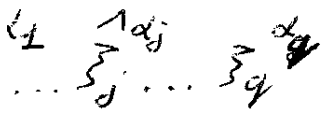
$$\begin{aligned} v &= u + \xi_1 \xi_2 \\ \eta_1 &= u \xi_1 \\ \eta_2 &= \xi_2. \end{aligned}$$

Problem. What is the form of $\sin v$?

Assert.

$$\frac{\partial}{\partial u_i}(f\xi^\alpha) = \frac{\partial f}{\partial u_i} \cdot \xi^\alpha,$$

$$\frac{\partial}{\partial \xi_j}(f\xi_1^{\alpha_1} \cdots \xi_q^{\alpha_q}) = \alpha_j (-1)^{\alpha_1 + \dots + \alpha_{j-1}} f \xi_1^{\alpha_1} \cdots \widehat{\xi_j^{\alpha_j}} \cdots \xi_q^{\alpha_q}.$$



Uncertain reading. The omission mark is over ξ_j, with α_j still written at upper right. The final q is retraced.

Original source details: GELFAND-P000674

Let U be a superdomain

$$\begin{aligned} x &= (u, \xi) \\ y &= (v, \eta) \end{aligned}$$

$$J = (J_{ij}), \qquad J_{ij} = \frac{\partial y_j}{\partial x_i}.$$

$$\operatorname{Ber} J \underset{\text{def}}{=} D(y)/D(x).$$

Def. U, $x = (u, \xi)$.

The expr. $\rho = fD(x)$ is called a volume form.

$$fD(x) = f\big(D(x)/D(y)\big) \cdot D(y).$$

Definition.

$$\int_u fD(x) \underset{\text{def}}{=} \int_u f_{11\dots1}(u)\, du_1 \dots du_p$$

(integral of a volume form)

Theorem (Berezin) This expr. is independent of the coordinate system.

Here one integrates a compactly supported fn.

GELFAND-P000675

25/X-76. Gelfand seminar.

Gelfand.

What is a connection? (for first-year students and Polyakov).

— see Nomizu, Sternberg, Bishop and Crittenden, Landau and Lifshitz.

Popov.

The founders' paper (Yang, Mills — translation in the coll. "Elem. particles and compensating fields").

Consider the Dirac eq.

without mass

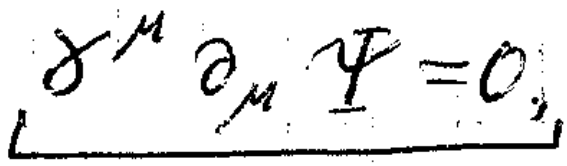


Diagram and typed description.

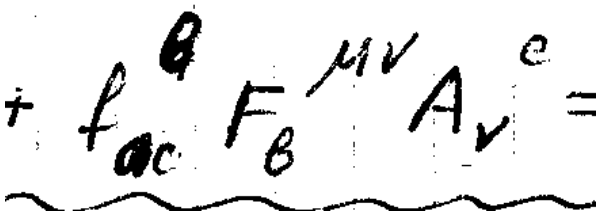
$\gamma^\mu \partial_\mu \Psi = 0,$

Original source details: GELFAND-P000675

$\Psi = (\Psi_\alpha)$ — ordinary 4-comp. spinor.

Yang–Mills equations:

$$\partial_\nu F_a^{\mu\nu} + f_{ac}^b F_b^{\mu\nu} A_\nu^c = 0,$$



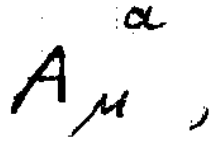
Uncertain reading. The retraced indices on the coefficient tentatively read f_{ac}^b.

Original source details: GELFAND-P000675

where

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + \tfrac{1}{2} f_{bc}^a (A_\mu^b A_\nu^c - A_\nu^b A_\mu^c)$$

on $\mathbb{R}^4$ consider the Minkowski metric $g_{\mu\nu}$, $\quad A_\mu^\alpha, \quad a = 1, \dots, \dim G$



Uncertain reading. The superscript on A_μ is α. The range immediately after it uses a.

Original source details: GELFAND-P000675

(G — some group, usually taken to be semisimple).
lowering of indices in the eq. with the aid of

$$K_{ab} = f^d_{ac} f^c_{bd},$$

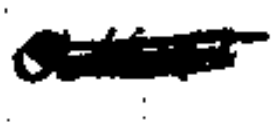
where f^d_{ac} — struct. const. of the Lie algebra of the group.

——— · ———

G. On quarks see Zel'dovich's article in UFN, "Quarks for pedestrians".

——— · ———

The Yang–Mills eqs. describe quarks.
$\mu, \nu = 1, \ldots, 4$;



Undeciphered passage. Crossed-out writing before $F^{\mu\nu}_a$.

Original source details: GELFAND-P000675

$F^{\mu\nu}_a$ — tensor with values in the Lie algebra.
A — analogue of the vect. potential

$$\partial_\nu = \partial/\partial x^\nu.$$

F — analogue of the field-strength tensor.
Example.
For $G = U(1)$ one obtains the eqs. of electrodynamics.
Equations:

$$\partial_\mu F^{\mu\nu} = 0$$
$$F^{\mu\nu} \underset{\text{def}}{=} \partial_\mu A_\nu - \partial_\nu A_\mu$$

We get Maxwell's eqs.

——— · ———

Gauge transformations.
In the el. and magn. field:

$$\begin{pmatrix} 0 & H_{12} & H_{13} & E_1 \\ -H_{12} & 0 & H_{23} & E_2 \\ -H_{13} & -H_{23} & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix} = F^{\mu\nu};$$

2-form



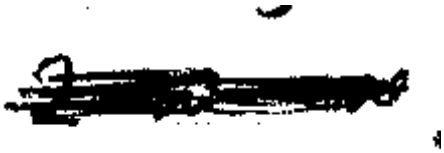
Uncertain reading. The third diagonal entry is retraced and reads as zero.

Original source details: GELFAND-P000675

Maxwell's equations:

$$F^{\mu\nu} = \frac{\partial}{\partial x^\mu} A_\nu - \frac{\partial}{\partial x^\nu} A_\mu.$$

— tensor



Undeciphered passage. Crossed-out writing after "tensor".

Original source details: GELFAND-P000675

of 2nd ord.

Transformations of A that leave F unchanged are called gradient transformations.

$$A_\mu \to A_\mu + \partial_\mu \alpha$$

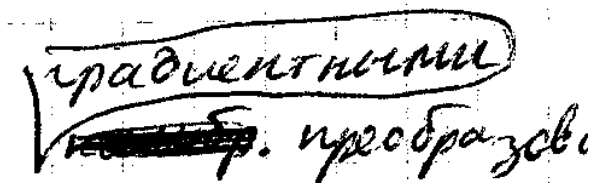


Diagram and typed description.

gradient

~~gauge~~

The outline of "gradient" points to the crossed-out abbreviation, tentatively "калибр." (gauge).

Original source details: GELFAND-P000675

GELFAND-P000676

Let (T_a) be a basis in the Lie algebra $\mathcal{L}(G)$.

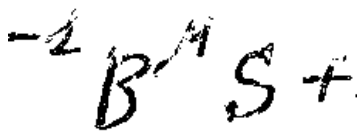
Consider

$$B_\mu = T_a A^a_\mu \quad \text{— field of matrices.}$$

$$F_{\mu\nu} = T_a F^a_{\mu\nu} \quad \text{—} \quad \text{— " —.}$$

Gauge transf. on fields of matrices:

$$B_\mu \to B'_\mu = S^{-1} B^\mu S + S^{-1} \partial_\mu S,$$



Uncertain reading. Between S^{-1} and S, μ is written as a superscript on B.

Original source details: GELFAND-P000676

$$F^{\text{[crossed out]}}_{\mu\nu} \to F'_{\mu\nu} = S^{-1} F_{\mu\nu} S,$$



Undeciphered passage. Crossed-out mark above the first F.

Original source details: GELFAND-P000676

where $S : \mathbb{R}^4 \to G$.

Direct calculation readily shows that the eqs. are inv. under these transformations.

Unknowns — A.

Nonlinearity due to the noncommut. group.

(F — curvature of the connection)

Yang–Mills equations.

Let (P, M, G) be a principal bundle

(M — manifold

G — group



Undeciphered passage. Crossed-out writing after “group”.

Original source details: GELFAND-P000676

— arbitrary)

on M there is a pseudo-Riemannian metric g.

In the principal bundle a connection is given by the form $\Omega \in \mathcal{L}(G) \otimes \Lambda^1(P)$.

The eq. will be for this form.

Fix an arbitrary (nondegenerate) quadratic form K on the Lie algebra $\mathcal{L}(G)$.

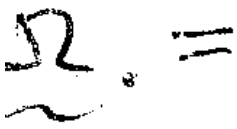
From these data a natural metric is constructed on P.

$$h_p(Y_p, Y'_p) = g_{\pi(p)}(d\pi Y_p, d\pi Y'_p) + K\big(\Omega(Y_p), \Omega(Y'_p)\big)$$

in general the metric is noninvariant.

Curvature of the connection.

$$\Phi = H\, d\Omega\,, \; = d\Omega + \tfrac{1}{2}[\Omega, \Omega]$$



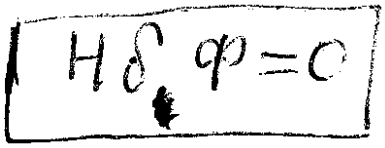
Uncertain reading. The low mark after $d\Omega$ before the second equality is rendered as a comma.

Original source details: GELFAND-P000676

H — operator taking the horiz. part of a form, constructed from the connection.

Yang–Mills equation for the free field

$$\boxed{H\delta_{\text{[crossed out]}}\varphi = 0}$$



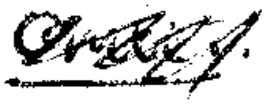
Undeciphered passage. Dark crossed-out mark below δ inside the box.

Original source details: GELFAND-P000676

δ — adjoint operator for the metric h.

This is the eq. on the principal bundle, and above is the eq. on the base. How do we identify them?

It turns out the eqs. are independent of the choice of metric K!



Uncertain reading. The retraced word tentatively reads “Оказ.” (it turns out).

Original source details: GELFAND-P000676

(on equiv. horiz. forms the operator $H \circ \delta$ is independent of the metric;

on arbitrary forms with values in a vector sp.

horiz. form — $\equiv 0$, if one of the vectors is vertical.

We must show that $H\delta\Phi$ — an equiv. horiz. form.

GELFAND-P000677

Consider a nbhd. $U \subset N$ and there take a trivialization ζ of the principal bundle (arbitrary section).



Uncertain reading. The letter after $\subset$ tentatively reads N.

Original source details: GELFAND-P000677



Uncertain reading. The section symbol is rendered as ζ. A mark to its lower right is crossed out.

Original source details: GELFAND-P000677

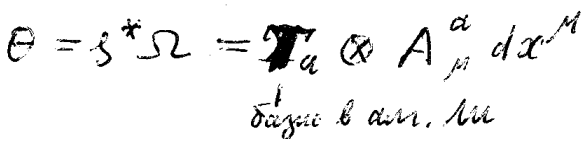


Diagram and typed description.

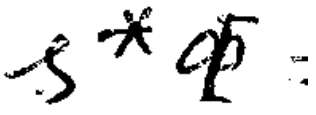
$\theta = \zeta^*\Omega = T_a \otimes A^a_\mu \, dx^\mu$

basis in the Lie alg.

The letter T is retraced. The stroke underneath marks the label.

Original source details: GELFAND-P000677

$$\zeta^*\Phi = T_a \otimes F^a_{\mu\nu} \, dx^\mu \wedge dx^\nu.$$

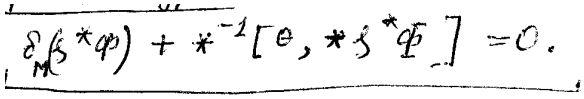


Uncertain reading. The symbol before the star is the same sign rendered as ζ.

Original source details: GELFAND-P000677

Projection of the eq.:

$$\delta_M(\zeta^*\varphi) + *^{-1}[\theta, *\zeta^*\Phi] = 0.$$



Original source details: GELFAND-P000677

Uncertain reading. The first section symbol is followed by φ, the second by Φ. Two small strokes lie at the lower edge of the bracket.

δ_M — the op. δ on the base.

Direct calc. shows this is the same eq.

It is the same eq. if the group has a 2-sided inv. metric.

The vacuum Einstein eqs. are here (Ricci tensor$= 0$)

$G =$ affine transformations of $\mathbb{R}^4$

(eq. in the bundle of aff. frames)

On the aff. frames there are 2 natural connections ind. by the base metric.

On the class of Cartan connections ind. by metrics.



Original source details: GELFAND-P000677

Uncertain reading. The retraced sign on the left resembles a 4.

There is a unique conn.



Original source details: GELFAND-P000677

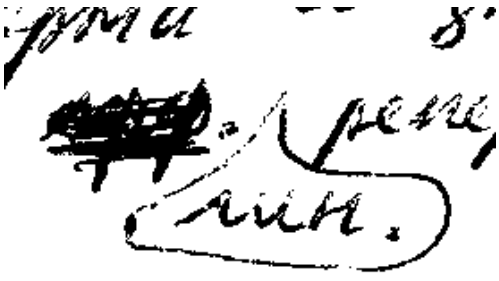
Undeciphered passage. Crossed-out mark before $\bar{\Omega}$.

$\bar{\Omega}$ in the bundle of aff. frames, such that

$$\text{if} \quad \underset{\text{lin.}}{L} \overset{i}{\hookrightarrow} \underset{\text{aff.}}{AL} \qquad i^*\bar{\Omega} = (\Omega^g, \chi);$$

χ — displacement form

(nat. form with values in $\mathbb{R}^4$ on the bundle of lin. frames)



Original source details: GELFAND-P000677

Diagram and typed description.

lin.

The outlined “lin.” below points to the position before “frames”. The earlier writing is crossed out.

Ω^g — metric connection

_____ · _____

def. of a Cartan connection — see Bishop and Crittenden.

on these connections $H\delta\varphi = 0$ — Einstein's equation. (1st order in the connection!! — unlike for general Yang–Mills fields)

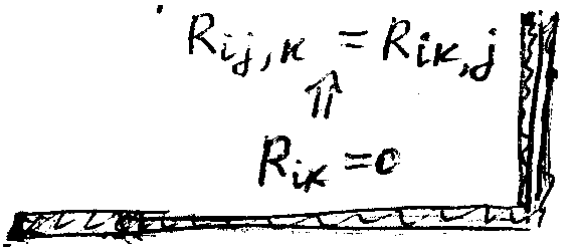
2nd order eqs. —



Original source details: GELFAND-P000677

Undeciphered passage. Crossed-out writing before "consequence".

a consequence of the 1st order eqs.



Original source details: GELFAND-P000677

Diagram and typed description.

$$R_{ij,k} = R_{ik,j}$$
$$\Uparrow$$
$$R_{ik} = 0$$

The double arrow points up. A hatched corner runs along the bottom and right.

GELFAND-P000678

Gelfand seminar.

I. N. Bernstein.

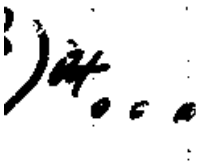
Kostant.

$$\varphi(x) = \prod_{n=1}^{\infty}(1-x^n) = (1-x)(1-x^2)(1-x^3)\times\dots$$
$$= \sum_{n=-\infty}^{\infty}(-1)^n x^{(3n^2+n)/2}.$$



Original source details: GELFAND-P000678

Uncertain reading. The retraced lower limit of the product reads $n = 1$.



Original source details: GELFAND-P000678

Uncertain reading. The dark mark between the third factor and the ellipsis resembles $\times$.

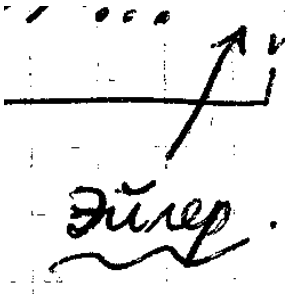


Diagram and typed description.

$= \sum$

Euler.

The arrow from the name points to the equality with the sum.

Original source details: GELFAND-P000678

Jacobi:

$$\varphi(x)^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) x^{n(n+1)/2}$$

(from the Θ-function).



Uncertain reading. The oval crossed by a horizontal stroke is rendered as uppercase Θ.

Original source details: GELFAND-P000678

Others have since been found (Dyson et al.)

Macdonald found a group-theoretic basis.

It is easy to check that

$$\varphi(x)^{-1} = \sum p(n) x^n,$$

$p(n)$ — number of partitions of n into a sum of nonnegative summands

$$x\varphi(x)^{24} = \sum \tau(n) x^n,$$

$\tau(n)$ — Ramanujan's function

this is its definition

$\tau(n)$ has a combinatorial definition



Undeciphered passage. Crossed-out writing above the line beginning "via".

Original source details: GELFAND-P000678

(via Dedekind's Θ-fn.)

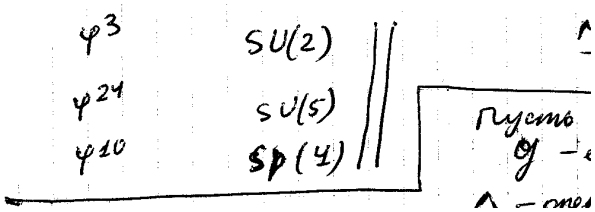


Uncertain reading. The glyph before "fn." is rendered as Θ, as in the line about Jacobi's formula.

Original source details: GELFAND-P000678

one can write

$$\varphi(x)^k, \quad \text{where } k \text{ — dimension of the Lie group (semisimple)}$$



Original source details: GELFAND-P000678

Diagram and typed description.

φ^3	$SU(2)$
φ^{24}	$SU(5)$
φ^{10}	$Sp(4)$

A double rule stands to the right of the three rows. The stepped line below rises to the next paragraph.

Macdonald: writes out the formula explicitly.
Let there be a compact semisimple group



Original source details: GELFAND-P000678

Undeciphered passage. Two crossed-out pieces of writing between “group” and its symbol.

G

g — its Lie algebra, on g there is the Killing form B

Δ — the Laplace operator on

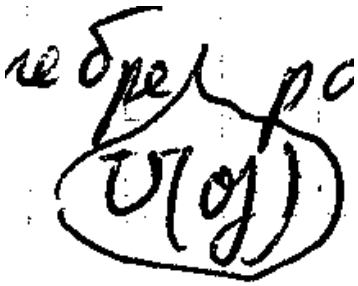


Original source details: GELFAND-P000678

Undeciphered passage. Two crossed-out pieces of writing after “on”.

G

$X_1, \ldots, X_k$ — orthonorm. basis in g.
In the enveloping algebra



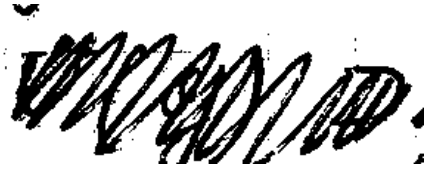
Original source details: GELFAND-P000678

Diagram and typed description.
algebra

$U(g)$

$U(g)$ is outlined below “algebra”.

consider the element



Original source details: GELFAND-P000678

Undeciphered passage. Crossed-out formula before the element.

$\Delta = \sum X_i^2$. it is 2-sided invariant.

$$\varphi(x)^k = \sum_{\text{irred. reps. } V} c_V \cdot \dim V \cdot x^{\Delta|_V}$$



Undeciphered passage. Crossed-out writing in the exponent before $\Delta|_V$.

Original source details: GELFAND-P000679

($\Delta|_V$ is a multiple of the identity; eigenvalue $\Delta|_V$.

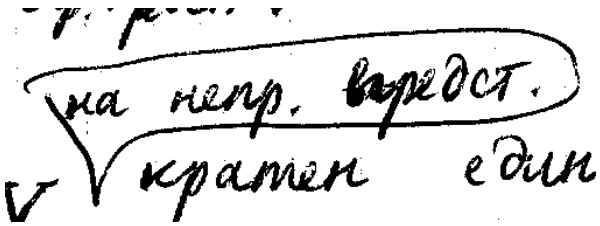


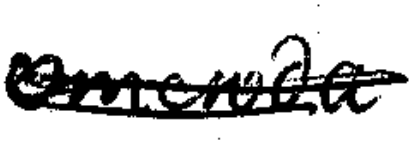
Diagram and typed description.

on an irred. rep.

The insertion above the line points after $\Delta|_V$.

Original source details: GELFAND-P000679

~~Hence~~ The asymptotics depend on the dimension.



Uncertain reading. The crossed-out word tentatively reads “Отсюда” (hence).

Original source details: GELFAND-P000679

Groups vary, and so do the formulas.

$$c_V = \chi_V(a), \qquad \text{where } \underset{\text{a certain element}}{a \in G}$$

~~For the series.~~ For $SU(n)$ there is a unique reg. element. (class of conjugate elements;) i.e., one whose eigenvalues are distinct and whose



Undeciphered passage. Crossed-out mark before the nth power.

Original source details: GELFAND-P000679

nth power is scalar.



Undeciphered passage. Crossed-out mark left of the matrix bracket.

Original source details: GELFAND-P000679

$$\begin{pmatrix} \xi & & & & 0 \\ & \xi^2 & & & \\ & & \xi^3 & & \\ & & & \ddots & \\ 0 & & & & \xi^n \end{pmatrix} c, \qquad c \quad \text{chosen so that } \det = 1.$$

Consider the ζ-function of the Laplace op.

V. Kac — proof via ∞-dimensional Lie algebras.

Consider $-\Delta + \frac{\dim g}{24}$ and set

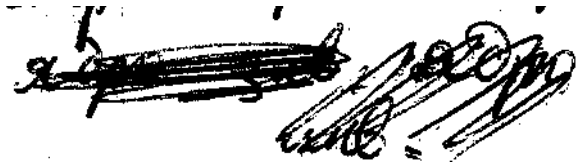
$$\zeta(s) \underset{\text{def}}{=} \left(-\Delta + \frac{\dim g}{24}\right)^{-s}\Big|_a$$



Undeciphered passage. Crossed-out writing after the defining equality.

Original source details: GELFAND-P000679

(op. of convolution with a fn.,



Uncertain reading. Two crossed-out passages. The second tentatively contains “ядро” (kernel).

Original source details: GELFAND-P000679

constant on classes of conj. elements, its value is taken at a)

$K(x,y)$ — kernel. In fact $K(xy^{-1})$

About ~~$\zeta(s)$~~ one can say something



Uncertain reading. The crossed-out expression tentatively reads $\zeta(s)$.

Original source details: GELFAND-P000679

Theorem.

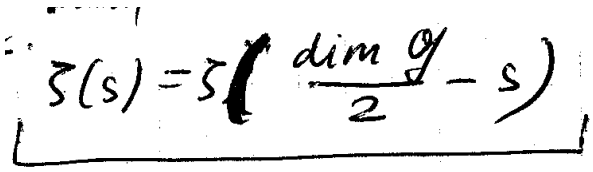


Diagram and typed description.

$$\underbrace{\zeta(s) = \zeta\left(\frac{\dim g}{2} - s\right)}$$

The line below the formula has upturned ends.

Original source details: GELFAND-P000679



Original source details: GELFAND-P000679

Uncertain reading. The letter after dim is rendered as g in all three formulas.

The theorem is equiv. to an



Original source details: GELFAND-P000679

Undeciphered passage. Crossed-out word after “equiv.”.

identity for $\varphi(x)^k$.

There are also φ^{39}, φ^{45} Van Esh



Original source details: GELFAND-P000679

Uncertain reading. The name is written in Latin script as “Van Esh”.

see Garland, Lepowsky, Inv. math., 34 (1976), 37–76

———— · ————

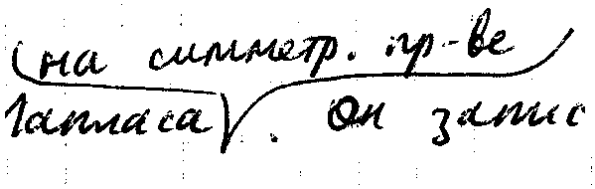


Original source details: GELFAND-P000679

Undeciphered passage. The underlined word before the surname is crossed out.

Olshanetsky.

Δ^0 — rad. part of the Laplace op.



Original source details: GELFAND-P000679

Diagram and typed description.

on the symmetric sp.

The insertion stands above the line after “Laplace”.

It is written in the form

$$\longrightarrow \frac{1}{2}\frac{\partial^2}{\partial t_k^2} + U(t_1, \ldots, t_n)$$

$$U = g \sum_{k>\ell} \frac{1}{\sin^2(t_k - t_\ell)}$$

And there is a complete system of observables in the quantum sense.

_____ · _____

I. Bernstein. on V. G. Kac's work.

Construction of a Lie algebra from a Cartan matrix.

$(a_{ij})_{n\times n}$ — matrix with integer entries, for which

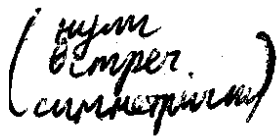
$$a_{ij} < 0 \quad \text{when } i \neq j \qquad \| \qquad \begin{array}{c} a_{ii} = 2, \\ a_{ij} \neq 0 \Rightarrow a_{ji} \neq 0 \end{array}$$

(zeros occur symmetrically)



Uncertain reading. The sign between a_{ij} and zero is written without a lower stroke.

Original source details: GELFAND-P000679



Uncertain reading. The ending of the last word runs into the bracket. It tentatively reads "симметрично" (symmetrically).

Original source details: GELFAND-P000679

GELFAND-P000680



Diagram and typed description. Vertical rule along the left margin.

Original source details: GELFAND-P000680

A Lie alg. is constructed from such a matrix.

Consider h_i, e_i, f_i, free generators (n of them), and write the relations

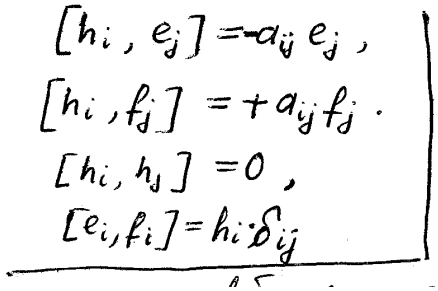
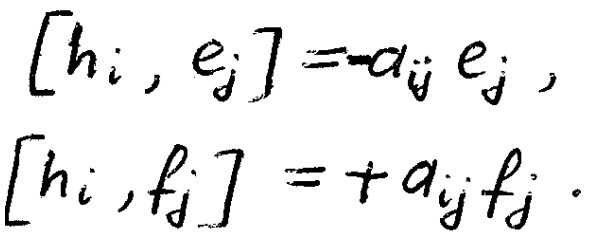


Diagram and typed description.

$$
\begin{aligned}
&[h_i, e_j] = -a_{ij} e_j, \\
&[h_i, f_j] = +a_{ij} f_j. \\
&[h_i, h_j] = 0, \\
&[e_i, f_i] = h_i \cdot \delta_{ij}
\end{aligned}
$$

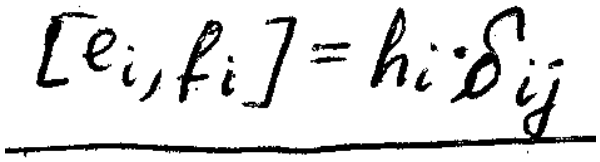
The outline runs below and to the right of the relations.

Original source details: GELFAND-P000680



Uncertain reading. The first line has a minus sign before a_{ij}, and the second has a plus sign.

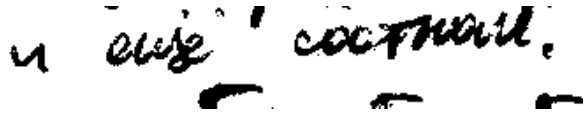
Original source details: GELFAND-P000680



Uncertain reading. The last line appears to give both e and f the same subscript i.

Original source details: GELFAND-P000680

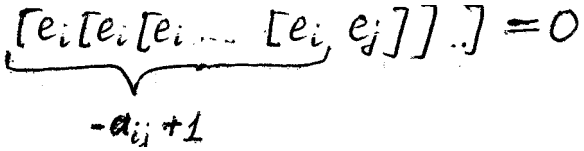
(take the quotient of the free alg. by these relations) and take the quotient of this alg. by the max normal subalgebra [Inserted at left above the formula: and further rels.]



Uncertain reading. The last word is abbreviated in the manuscript as “соотнош.” (relations).

Original source details: GELFAND-P000680

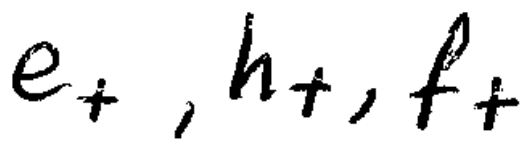
$$\underbrace{[e_i[e_i[e_i \dots [e_i}_{-a_{ij}+1}, e_j]]..] = 0$$



Uncertain reading. The underbrace ends before the comma. After e_j there are two closing brackets, two dots and a further closing bracket.

Original source details: GELFAND-P000680

For certain matrices, classical groups arise. $SU(2)$ e_+, h_+, f_+ matrix $1 \times 1 \| (2)$



Uncertain reading. All three letters have a plus subscript.

Original source details: GELFAND-P000680

$$[h,e]=2e, \qquad [h,f]=-2f, \qquad [e,f]=h,$$

$SU(3)$ is obtained for the matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

On roots: Serre "Lie Groups and Lie Algebras" — read only this section.

If $a_{ij}=a_{ji}$, then the following holds:

The group is finite-dimensional $\Leftrightarrow$ the matrix is positive definite.

$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$ — infinite-dimensional extension of $SU(3)$.

V. G. Kac. : the quotient of this alg. by its one-dimensional centre is

$$SU(3)\otimes \mathbb{C}[x,x^{-1}].$$

$$[x^k g, x^\ell g'] = x^{k+\ell}[g,g'].$$

all the non-strictly def. ones are obtained this way.



Original source details: GELFAND-P000680

Uncertain reading. The two abbreviated words read "нестрог. опред." (non-strictly definite).

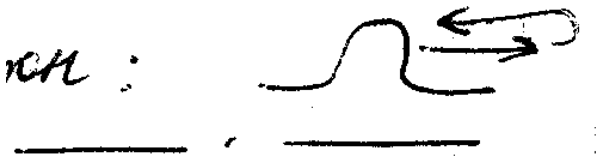
Happy New Year!

GELFAND-P000681

14/II-77. Gelfand Sem.

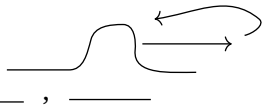
Yu. I. Manin.

Calogero: boomeron:

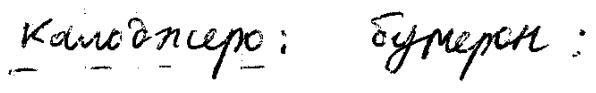


Original source details: GELFAND-P000681

Diagram and typed description.



The upper arrow hooks down at the right. A gap is visible between the hook and the point of the lower arrow.

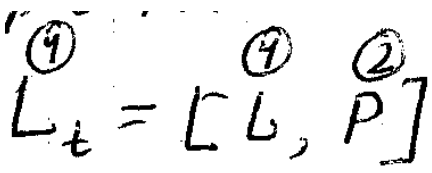


Original source details: GELFAND-P000681

Uncertain reading. The underline under "Calogero" is broken. The next word reads "бумерон" (boomeron).

Equation of Lax type

$$\overset{④}{L}_t = [\overset{④}{L}, \overset{②}{P}]$$



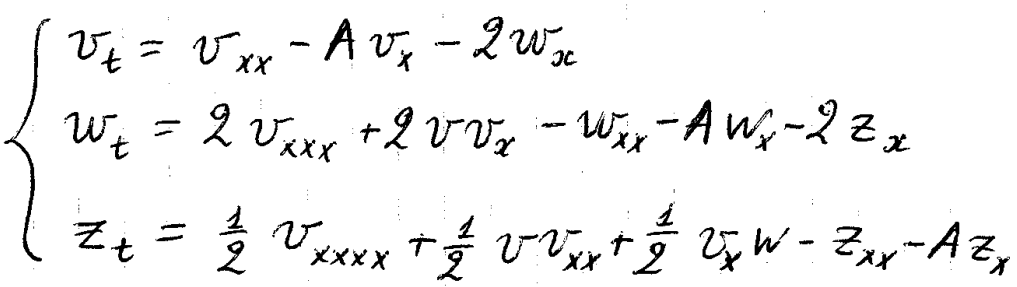
Original source details: GELFAND-P000681

Uncertain reading. A circled 4 stands over each L, and a circled 2 over P.

$$L = \partial^4 + v\partial^2 + w\partial + z, \qquad \partial = \partial/\partial x.$$
$$P = \partial^2 + A\partial + u$$

u is expressed in terms of v, w, z.

$$\begin{cases} v_t = v_{xx} - Av_x - 2w_x \\ w_t = 2v_{xxx} + 2vv_x - w_{xx} - Aw_x - 2z_x \\ z_t = \frac{1}{2}v_{xxxx} + \frac{1}{2}vv_{xx} + \frac{1}{2}v_x w - z_{xx} - Az_x \end{cases}$$

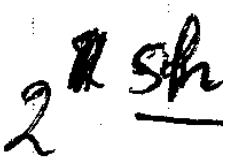


Original source details: GELFAND-P000681

Uncertain reading. The first line has no written coefficient before v_{xx}. The second has a 2 before vv_x.

1-soliton solution

$$\mu_1 = 2\frac{\operatorname{ch} 2\xi - \cos 2\xi + 2\sin \sigma t}{\operatorname{ch} 2\xi + \cos 2\xi - 2\cos \sigma t}$$
$$\mu_2 = 2\frac{\operatorname{sh} 2\xi - \sin 2\xi}{\operatorname{ch} 2\xi + \cos 2\xi - 2\cos \sigma t}$$

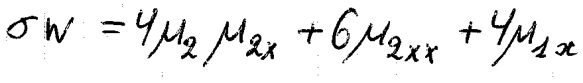


Uncertain reading. A thickened mark above and to the right of the factor 2, before sh, may be a crossed-out digit.

Original source details: GELFAND-P000681

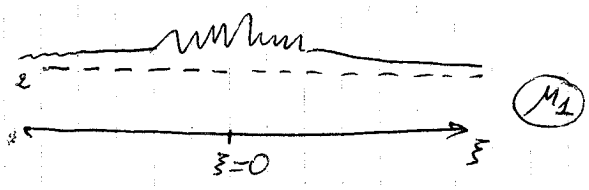
$$\xi = x + ct \qquad \begin{matrix} c \in \mathbb{R} \\ \sigma \in \mathbb{R} \\ \sigma \leftrightarrow A \quad \text{(is expressed).} \end{matrix}$$

$$\begin{cases} v = -4\mu_{2x} \\ \sigma w = 4\mu_2\mu_{2x} + 6\mu_{2xx} + 4\mu_{1x} \\ z = \dots \quad \text{(inessential)} \end{cases}$$



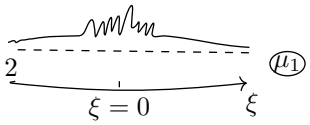
Uncertain reading. The left-hand side of the second line reads σw.

Original source details: GELFAND-P000681



Original source details: GELFAND-P000681

Diagram and typed description.



At the left end of the axis there is a hook and a set of small illegible marks.

[To the right of the graph:] every $\frac{2\pi}{\sigma}$ periodically



Original source details: GELFAND-P000681

Undeciphered passage. A sign or short passage is crossed out at the end of the line.

it spits (geyseron, spiton, ...) explosive instability.

Algebraic method.

a) construct an aux. stationary eq. $[L_1, P_1] = 0$ then introduce t.

b)

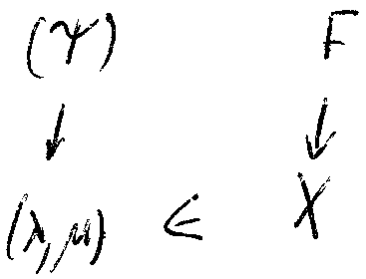
$$\begin{cases} L_1\psi = \lambda\psi \\ P_1\psi = \mu\psi \end{cases} \quad (\lambda, \mu) \in X \quad \text{— aff. alg. curve}$$

the case studied was: dim(space of ψ) $= 1$.

Krichever did not state it explicitly, but it followed from the constraints.

Drinfeld — did not specify the solution explicitly, but indicated the difficulties.

GELFAND-P000682



Original source details: GELFAND-P000682

Diagram and typed description.

$$\begin{array}{ccc} (\psi) & & F \\ \downarrow & & \downarrow \\ (\lambda, \mu) & \in & X \end{array}$$

Both arrows point down. The mark between the lower entries is rendered as $\in$.



Original source details: GELFAND-P000682

Uncertain reading. The sign in parentheses reads as ψ. There is an additional fine stroke in its centre.

The bundle extends to the completion.



Original source details: GELFAND-P000682

Uncertain reading. The middle word is abbreviated. It tentatively reads “продолж.” (extends).

We obtain a bundle on the curve

$$t \longrightarrow \text{path on the Jacobian.}$$

$$\underset{\text{Moduli of the system;}}{\underline{r > 1}} \quad \Rightarrow \quad \text{path in the moduli of bundles.}$$

Hyperell. curve $\longrightarrow$ intersection of 2 quadrics in $\mathbb{C}^5$.



Original source details: GELFAND-P000682

Uncertain reading. The space is written $\mathbb{C}^5$.

Analyt. paths?

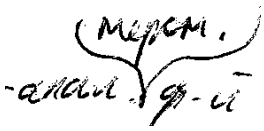
A soliton is a solution of the nonstationary eq. obtained by taking a curve with one double point with distinct tangents in the stationary eq.

Notation. k — field of char. 0.

$$\underset{\text{in the analytic setting,}}{(k = \mathbb{R} \text{ or } \mathbb{C})}$$

$\mathcal{B}$ — k-alg.

$$\begin{pmatrix} \mathcal{B} = \underset{\text{matrices}}{M_\ell}\ (\mathcal{B}_0), \\ \\ \mathcal{B}_0 = \text{field of rat. } k\text{-anal. } \overset{\text{merom.}}{\text{functions of}} \begin{matrix} x \\ x, t \\ x, t, z \end{matrix} \end{pmatrix}$$



Original source details: GELFAND-P000682

Uncertain reading. The abbreviation “мером.” (meromorphic) is inserted above the space before “ф-й” (functions) with a curved callout.

$$\underset{\text{derivations}}{\partial, \partial_1, \partial_2 : \mathcal{B} \longrightarrow \mathcal{B}}, \qquad [\partial, \partial_i] = [\partial_i, \partial_j] = 0$$

(in the anal. setting $\partial = \partial/\partial x$, $\partial_1 = \partial/\partial t$, $\partial_2 = \partial/\partial z$).

$\mathcal{B}[\partial]$

(L, P) — pair of ops. satisfying one of the 3 eqs.

$$\begin{cases} [L, P] = 0 & \text{stationary Lax} \\ \partial_1 L = [L, P] & \text{nonstationary Lax} \\ \partial_1 L + \partial_2 P = [L, P] & \text{Zakh.–Shabat} \end{cases}$$

① Krich.–Drinfeld bimodule $\mathcal{M}$.



Original source details: GELFAND-P000682

Uncertain reading. A circled 1 stands to the left of the heading.

Structures.

a) $\mathcal{M}$ — k-lin. space, left $\mathcal{B}$-module, right $\mathcal{O}$-module, where

$\mathcal{O}$ — ring of functions on an aff. curve over k; irreducible over $\bar{k}$.

b) distinguished element $\mathbf{1} \in \mathcal{M}$

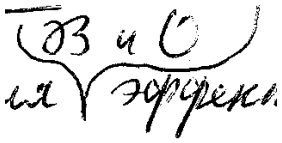
c) increasing filtration

$$\mathcal{M}_{-1} \subset \mathcal{M}_0 \subset \mathcal{M}_1 \subset \mathcal{M}_2 \subset \ldots, \qquad \bigcup_{i=0}^{\infty} \mathcal{M}_i = \mathcal{M}$$

d) $\nabla : \mathcal{M} \longrightarrow \mathcal{M}$ $\quad$ k-lin. operator.

Axioms.

e) the actions are $\mathcal{B}$ and $\mathcal{O}$ faithful and commute



Uncertain reading. "$\mathcal{B}$ and $\mathcal{O}$" is inserted above the space after "actions" with a curved callout.

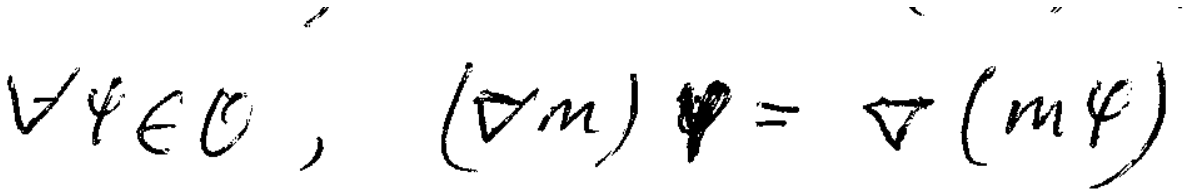
Original source details: GELFAND-P000682

f) $\mathcal{M}_{-1} = 0, \qquad \mathcal{B}\mathcal{M}_i \subset \mathcal{M}_i$

g) $\mathcal{M}_i/\mathcal{M}_{i-1}$ — free $\mathcal{B}$-module of rank 1; $\qquad \mathcal{M}_0 = \mathcal{B}\mathbf{1}$.

h)

$$\forall b \in \mathcal{B},\ m \in \mathcal{M}: \quad \nabla(bm) = (\partial b)m + b\nabla m,$$
$$\forall \varphi \in \mathcal{O}, \qquad (\nabla m)\varphi = \nabla(m\varphi)$$



Original source details: GELFAND-P000682

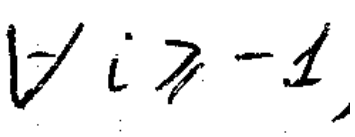
Uncertain reading. The letter in $(\nabla m)\varphi$ is retraced. The two nearby instances support reading it as φ.

GELFAND-P000683

("nabla" is the name of an old musical instrument of the lyre type).

i)

$$\forall i \geq -1, \qquad \nabla\mathcal{M}_i \subset \mathcal{M}_{i+1},$$
$$\nabla : \mathcal{M}_i/\mathcal{M}_{i-1} \xrightarrow{\sim} \mathcal{M}_{i+1}/\mathcal{M}_i$$



Uncertain reading. The start of the line reads $i \geq -1$.

Original source details: GELFAND-P000683

② Construction of the bimodule from a pair $L, P \in \mathcal{B}[\partial]$ with $[L,P]=0$ and $\bar{k}[L,P]$ without zero divisors.

Set

$$\mathcal{M} = \mathcal{B}[\partial]; \qquad \mathcal{O} = k[L,P].$$

— ring of polys. in L and P.

action of $\mathcal{B}$ — mult. on the left, action of $\mathcal{O}$ — mult. on the right

1 = identity operator

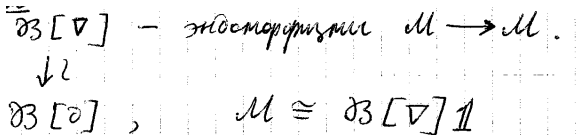
$$\mathcal{M}_i = \left\{ \sum_{k \le i} b_k \partial^k \mid b_k \in \mathcal{B} \right\}. \qquad \nabla m = \partial \circ m$$

The axioms are verified trivially.

Theorem 1. If there is a $(\mathcal{B},\mathcal{O})$-bimodule $\mathcal{M}$, then it gives an embedding $i : \mathcal{O} \longrightarrow \mathcal{B}[\partial]$, such that the reverse construction from i gives $\mathcal{M}$.

Construction.

a)



Original source details: GELFAND-P000683

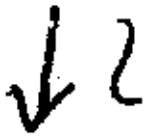
Diagram and typed description.

$\mathcal{B}[\nabla]$ — endomorphisms $\mathcal{M} \longrightarrow \mathcal{M}$.

$\downarrow$

$\mathcal{B}[\partial]$ $\quad \mathcal{M} \cong \mathcal{B}[\nabla]\mathbf{1}$

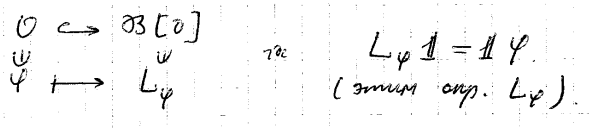
Uncertain reading. A separate vertical wavy stroke stands to the right of the down arrow.



Original source details: GELFAND-P000683

This step did not use multiplication by $\mathcal{O}$.

b)



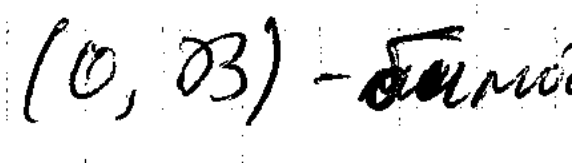
Original source details: GELFAND-P000683

$$\begin{array}{ccc} \mathcal{O} & \hookrightarrow & \mathcal{B}[\partial] \\ \Psi & & \Psi \\ \varphi & \longmapsto & L_\varphi \end{array}$$

Diagram and typed description.
such that $L_\varphi \mathbf{1} = \mathbf{1}\varphi$.
(this determines L_φ).

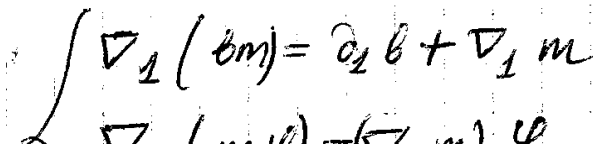
Theorem 2. Let there be an $(\mathcal{O},\mathcal{B})$-bimodule $\mathcal{M}$ and a deriv. $\partial_1 : \mathcal{B} \longrightarrow \mathcal{B}$ (in the anal. setting $\partial/\partial t$), and suppose there is also $\nabla_1 : \mathcal{M} \longrightarrow \mathcal{M}$

Uncertain reading. Theorem 2 writes $\mathcal{O}$ first and then $\mathcal{B}$.



Original source details: GELFAND-P000683

$$\begin{cases} \nabla_1(bm) = \partial_1 b + \nabla_1 m \\ \nabla_1(m\varphi) = (\nabla_1 m)\varphi \\ [\nabla_1, \nabla] = 0 \end{cases}$$



Uncertain reading. The right-hand side shows $\partial_1 b$ and $\nabla_1 m$, with a plus sign between them.

Original source details: GELFAND-P000683

$[\mathcal{M}, \nabla_1] \longrightarrow$ family of solns. of nonstationary Lax eqs.

$$\nabla_1 \mathbf{1} = P\mathbf{1}, \qquad P \in B[\partial].$$
$$\mathbf{1}\varphi = L_\varphi \mathbf{1}$$

Then

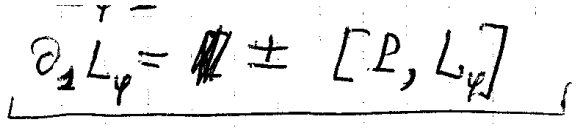


Diagram and typed description.
$\partial_1 L_\varphi =$ [crossed out] $\pm [P, L_\varphi]$

Original source details: GELFAND-P000683



Undeciphered passage. A sign is crossed out between the equality sign and $\pm$.

Original source details: GELFAND-P000683

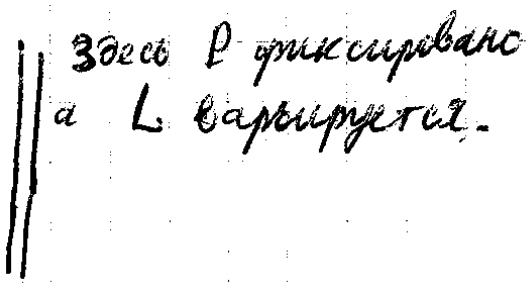


Diagram and typed description.
Here P is fixed
and L varies.

Original source details: GELFAND-P000683

GELFAND-P000684

Theorem 3. A module is given $\mathcal{M}$ with no mention of $\mathcal{O}$.

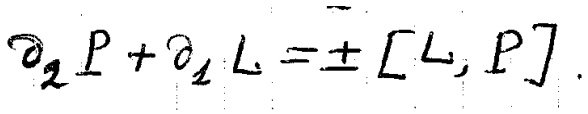


Uncertain reading. The wavy underline under "with no mention of" ends before $\mathcal{O}$.

Original source details: GELFAND-P000684

$\nabla, \nabla_1, \nabla_2$ commute pairwise. Then we obtain solns. of the Zakh.–Shabat eqs.:

$$\nabla_1 \mathbf{1} = P\mathbf{1}$$
$$\nabla_2 \mathbf{1} = L\mathbf{1}$$
$$\partial_2 P + \partial_1 L = \pm[L, P].$$



Original source details: GELFAND-P000684

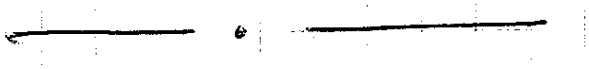
Uncertain reading. There is a plus sign between the two derivatives and a $\pm$ before the commutator.

Next time Duistermaat.



Original source details: GELFAND-P000684

Uncertain reading. The underlined surname is written “Дюйстермаат” (Duistermaat).



Original source details: GELFAND-P000684

Diagram and typed description.

There is a small hook at the left end of the shorter rule. A dot separates the two rules.

GELFAND-P000685

21/II-77 Gelfand Sem.

Duistermaat.

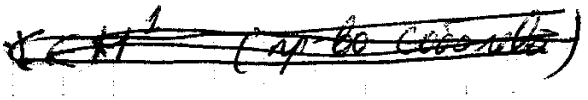


Original source details: GELFAND-P000685

Uncertain reading. The surname is written “Дейстермаат” here.

On the Morse index.

$I(\gamma)$ — functional, γ — curve in $\mathbb{R}^n$, ~~$\gamma \in H^1$ (Sobolev sp.)~~



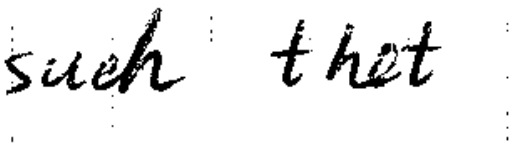
Original source details: GELFAND-P000685

Uncertain reading. The line crosses out $\gamma \in H^1$ and the parenthetical gloss.

$$I(\gamma) = \int_0^T f(\gamma(t), \dot{\gamma}(t))\, dt$$

Variational problem.

$$\gamma \in \text{space of curves such thet } (\gamma(0), \gamma(T)) \in R$$



Original source details: GELFAND-P000685

Uncertain reading. The line is in English in the manuscript. The word after “such” tentatively reads “thet”.

R — subset of $\mathbb{R}^n \times \mathbb{R}^n$.

Examples; a) $R = (0,0) \in \mathbb{R}^n \times \mathbb{R}^n$ — fixed endpoints

b) $R = \{(x,x)\}$ — $\gamma(0) = \gamma(T)$ — periodic condition



Uncertain reading. The thick horizontal mark between the formulas is rendered as a dash.

Original source details: GELFAND-P000685

Extremal — critical point in H^1.

$$\text{If } \frac{\partial^2 f}{\partial \dot{\gamma}^2} > 0, \qquad \text{then ind } \frac{\partial^2 I}{\partial \gamma^2} < \infty$$

(positive def.) (finitely many neg. squares)

GELFAND-P000686

$$\text{ind } \frac{\partial^2 I}{\partial \gamma^2} = \text{Morse index for the boundary cond.} = \text{ind}_R(\gamma).$$

There will be a formula for the Morse index in terms of the Arnold index of a curve in the Lagr. set.

Theorem.

$$\text{Morse index}_R(\gamma) = \text{ind}(\varphi) + i(I, \rho, \varphi(T)) + \dim \pi(\rho \cap I) - 2n.$$



Original source details: GELFAND-P000686

Uncertain reading. The right-hand side reads $i(I, \rho, \varphi(T))$ and $\dim \pi(\rho \cap I) - 2n$.

Eq. for stat. curves (Euler–Lagrange) $\underset{\text{Legendre tr.}}{\longrightarrow}$ Hamilton eq.

$$(\gamma, \dot{\gamma}) \underset{\text{Legendre transform}}{\mapsto}$$

The Hamilton eq. determines a flow of sympl. transformations.

Consider the eq. in variations along the extremal. We obtain a 1-param. family of lin. symplectic transformations.

$\Phi(t) =$ derivative of the Hamiltonian flow along the extremal curve

this is a lin. transformation:

$$\underset{\text{phase space}}{\mathbb{R}^{2n}_\sigma} \longrightarrow \mathbb{R}^{2n}_\sigma$$



Uncertain reading. The first $\mathbb{R}^{2n}$ has a heavily retraced subscript, tentatively read as σ.

Original source details: GELFAND-P000686



Original source details: GELFAND-P000686

Uncertain reading. The second $\mathbb{R}^{2n}$ also has a heavily retraced subscript, tentatively read as σ.

$\varphi(t) = \text{graph}\,\varphi(t) \subset \underline{\mathbb{R}^{2n}_{\sigma} \times \mathbb{R}^{2n}_{-\sigma}}$ (sympl. structure — difference)

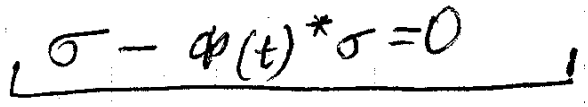


Original source details: GELFAND-P000686

Uncertain reading. The letter after "graph" looks lowercase and is transcribed as φ.

$\Phi(t)^*\sigma = \sigma.$ t fixed.

$\Updownarrow$



Original source details: GELFAND-P000686

Diagram and typed description.

$\underline{\sigma - \varphi(t)^*\sigma = 0}$

The formula has a ruled bracket below. The letter before (t) is transcribed as lowercase φ.

or

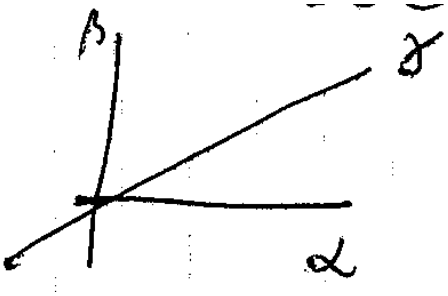
$\underline{\sigma \oplus (-\sigma) = 0 \text{ on } \varphi(t)}$



Original source details: GELFAND-P000686

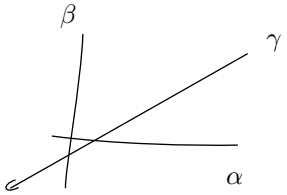
Undeciphered passage. The passage before the parenthesis is heavily crossed out.

(Lagrangian subspace)



Original source details: GELFAND-P000686

Diagram and typed description.



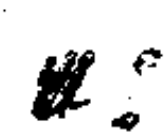
Three strokes labelled α, β, γ meet near a single point.

α, β, γ — 3 Lagr. planes

$$\alpha \cap \beta = 0$$
$$\gamma \cap \beta = 0$$

Then we can take

$$\gamma = \text{graph}\,\Gamma, \qquad \Gamma \text{ linear } : \alpha \to \beta$$



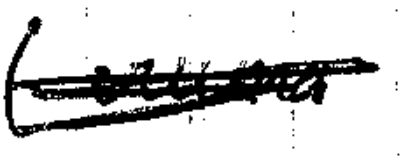
Undeciphered passage. A mark after “linear” and before the colon is overwritten.

Original source details: GELFAND-P000686

$$Q(u, v) = \sigma(\Gamma u, v), \qquad u, v \in \alpha$$

We obtain a bilin. form on α

$$\sigma = 0 \text{ on } \Gamma \quad \Leftrightarrow \quad Q \text{ is symmetric.}$$



Undeciphered passage. The passage before $Q = Q(\alpha, \beta, \gamma)$ is heavily crossed out.

Original source details: GELFAND-P000686

$$Q = Q(\alpha, \beta, \gamma)$$

If $\gamma = \gamma(t)$ (curve of Lagr. planes), then at some point γ may intersect α in a 1-dim. subspace.

At the moment of intersection take the difference of indices (it is independent of the choice of β).

$[\gamma : \alpha]$ — intersection index of the curve $\gamma(t)$ with the hypersurf.

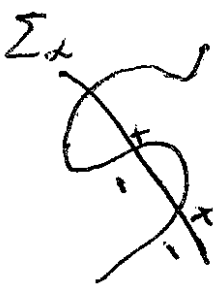
$$\Sigma_\alpha = \{\gamma \mid \gamma \cap \alpha \neq 0\}.$$



Uncertain reading. The overwritten abbreviation tentatively reads “гиперповерхн.” (hypersurface).

Original source details: GELFAND-P000686

GELFAND-P000687



Original source details: GELFAND-P000687

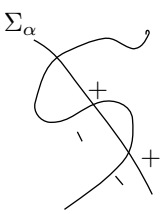


Diagram and typed description. The curved line crosses Σ_α three times. Two crossings have a plus sign and a short stroke each. A small loop marks the upper end of the curve.

$\Rightarrow$ this is a homotopy inv. with fixed endpoints.

Σ_α is embedded two-sidedly



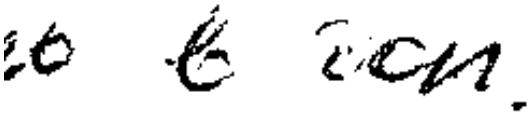
Undeciphered passage. A passage after an opening parenthesis is crossed out.

Original source details: GELFAND-P000687

Hence $\pi_1(\text{Lagr. planes}) = \mathbb{Z}$

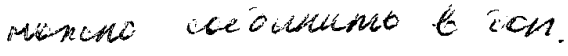
$$\text{since } (\text{Lagr. planes}) = \Sigma_\alpha \cup (\text{transverse planes})$$
$$\parallel$$
$$(\text{quadratic forms}) \quad \text{— homot. trivial}$$

Initial and final points outside Σ_α, can be joined in the compl. and the homot. type is independent of any choice.



Uncertain reading. The line ends “в доп.” (in the compl.).

Original source details: GELFAND-P000687



Uncertain reading. The words before “в доп.” read “можно соединить” (can be joined).

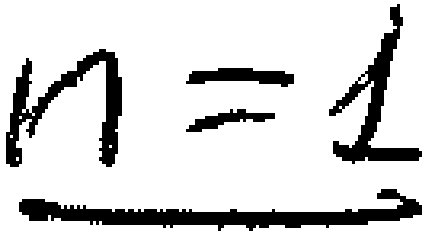
Original source details: GELFAND-P000687

For an open curve:

$$\text{ind}(\gamma) = [\gamma : \alpha] + \text{ind}\, Q\{\gamma(T), \alpha, \gamma(0)\}$$

(to make it independent of α).

Diagram and typed description. $\xrightarrow{n=1}$



Original source details: GELFAND-P000687

Lagr. planes form a circle

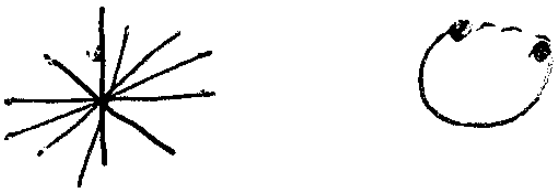
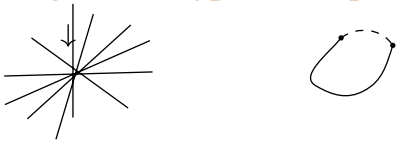


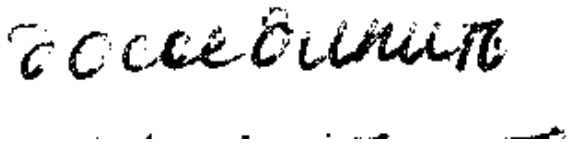
Diagram and typed description.



The pencil of lines has a downward arrow. The upper arc of the circle is dashed. Thick marks appear at its ends.

Original source details: GELFAND-P000687

[To the right of the sketches:] join up the open curve clockwise, from the initial point to the final one.



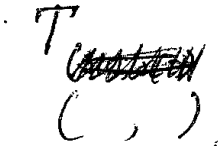
Original source details: GELFAND-P000687

Uncertain reading. The verb tentatively reads “досоединить” (join up).

ind(γ) — resulting number of turns.

R — set of bdy. conditions

$$R^\perp = \text{normal bundle}$$
$$= \left\{ \underset{\mathbb{R}^{2n}}{(x,\xi)}, \underset{\mathbb{R}^{2n}}{(y,-\eta)} \;\middle|\; (\xi,\eta) \perp T_{(x,y)}R \right\}.$$
$$\rho = T_{(\ ,\ )}(R^\perp)$$



Original source details: GELFAND-P000687

Undeciphered passage. The first subscript at T is crossed out. Parentheses with a comma and empty arguments appear below.

— fibre of the normal bundle over a pt. of the extremal

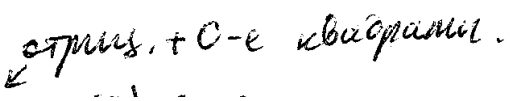


Original source details: GELFAND-P000687

Uncertain reading. The last word tentatively reads “экстремали” (of the extremal). Some strokes may cross it out.

this is a Lagrangian plane

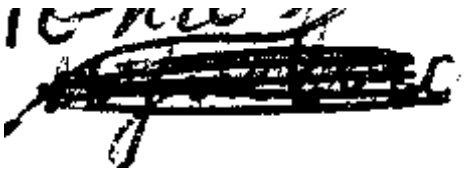
$$i(I,\rho,\varphi(T)) = (\text{ind} + \text{null})Q(I,\rho,\varphi(T))$$



Original source details: GELFAND-P000687

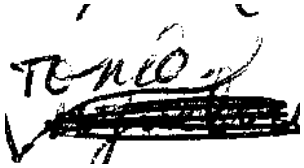
Diagram and typed description.
neg. $+0$ squares.

I — graph of the [inserted above: ident.] map.



Original source details: GELFAND-P000687

Undeciphered passage. The word between “graph” and “map” is crossed out.



Original source details: GELFAND-P000687

Uncertain reading. The insertion above the crossing-out tentatively reads “тожд.” (ident.).

π — projection to the base.

Selberg trace formula. in the simplest case:

G — semisimple connected real Lie group, K — max compact subgrp., $K\backslash G$

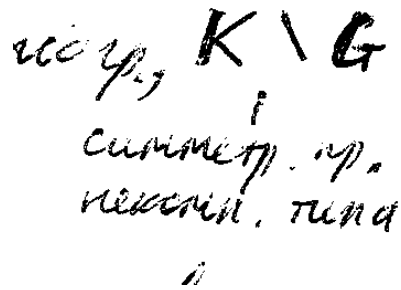


Diagram and typed description.

$K\backslash G$
symm. sp.
of noncompact type

Original source details: GELFAND-P000687

Γ — discrete subgrp. of G, acting on $M = K\backslash G$ on the right.

$M/\Gamma = X$ — compact set.

We will consider all ops. $C^\infty(M) \to C^\infty(M)$, commuting with the action of G.

$$f \longmapsto \varphi * f, \qquad (\varphi * f)(x) = \int_G \varphi(xy^{-1}) f(y)\, dy$$

$$\varphi \in \mathcal{E}'(K\backslash G)$$

GELFAND-P000688

f invar. $\Rightarrow$ we may take $\varphi \in \mathcal{E}'(K\backslash G/K)$ (two-sided invar.) — spherical fns.

The operator on X has a Schwartz kernel.

For φ supported at $e \in G$ we obtain a diff. operator.

The formula is for the spectrum of a diff. op. on M, descended to X.

Set

$$K_\varphi(x, y) = \sum_{\gamma \in \Gamma} \varphi(x\gamma y^{-1})$$

It follows from invariance that $K_\varphi \in \mathcal{E}'(X \times X)$.

Now consider an op. with smooth φ (compactly supported).

Γ — discr. $\Rightarrow$ [overwritten] the sum for fixed x, y is finite.



Undeciphered passage. A mark before “the sum” is heavily overwritten.

Original source details: GELFAND-P000688

$$\varphi \in C_c^\infty(K\backslash G/K) \quad \Rightarrow \quad K_\varphi \in C^\infty(X \times X).$$

We have:

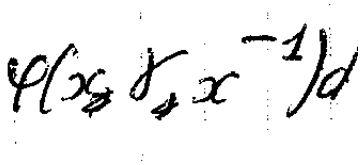
$$K_\varphi \circ K_\psi = K_{\varphi * \psi}$$
$$\varphi * \psi = \psi * \varphi \qquad \text{on spherical fns.}$$

$\Rightarrow$ we obtain a commut. algebra of ops. K_φ with involution. They are all compact $\Rightarrow$
$\Rightarrow$ $\exists$ orthon. basis of common eigenvectors $e_j \in L^2(X)$

$$K_\varphi(e_j) = \lambda_j(\varphi) e_j, \qquad \forall \varphi \in C_c^\infty(K\backslash G/K)$$

We have:

$$\sum_j \lambda_j(\varphi) = \operatorname{Sp} K_\varphi = \int_X K_\varphi(x,x)\,dx$$
$$= \sum_{\gamma \in \Gamma} \int_X \varphi\big(x_? \gamma_? x^{-1}\big)\,dx$$



Uncertain reading. Two similar small marks appear below the first x and below γ. Their readings remain uncertain.

Original source details: GELFAND-P000688

$$\big(C_c^\infty(K\backslash G/K), *\big) \xrightarrow{\sim} C_c^\infty(A)^W, \qquad W \text{ — Weyl group}$$



Uncertain reading. The delimiter after the final K and before the comma is heavily retraced.

Original source details: GELFAND-P000688

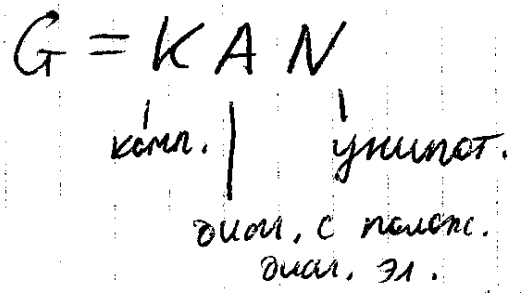


Diagram and typed description.

$G =$ K (compact) A (diag. with positive diag. entries) N (unipot.)

— Iwasawa decomposition.

Original source details: GELFAND-P000688



Uncertain reading. The manuscript spells the surname “Ивасавы”.

Original source details: GELFAND-P000688

$$f(a) = e^{\langle \rho, a \rangle} \int_N \varphi(an)\, dn.$$

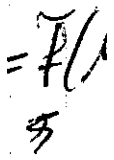
(horospherical map $\varphi \mapsto f$).

We have:

$$\lambda_j(\varphi * \psi) = \lambda_j(\varphi) \cdot \lambda_j(\psi).$$

Hence

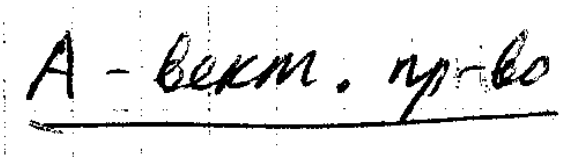
$$\lambda_j(\varphi) = \langle f, e^{i\mu_j} \rangle = \widetilde{f}_?(\mu_j), \qquad \mu_j \in A^*_{\mathbb{C}}.$$



Original source details: GELFAND-P000688

Uncertain reading. There is a separate undeciphered mark below $\widetilde{f}$.

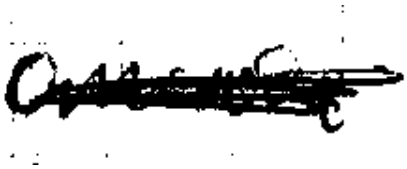
[Right margin:] A — vector sp.



Original source details: GELFAND-P000688

Uncertain reading. The note about A stands in the right margin and is underlined.

Here $\rho = \frac{1}{2} \sum$ pos. roots



Original source details: GELFAND-P000688

Undeciphered passage. A short word above “We have” is crossed out.

We have:

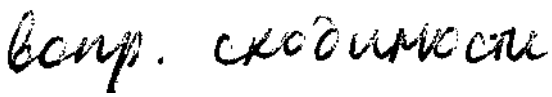
$$\sum_j \lambda_j(\varphi) = \sum_{\gamma \in \Gamma} \langle T_\gamma, f \rangle,$$

(we rewrote each summand)

T_γ — locally finite family of distributions.

GELFAND-P000689

All the brackets encountered mysteriously disappear



Original source details: GELFAND-P000689

Uncertain reading. The noun tentatively reads “скобочки” (brackets).

$$\sum_j \widetilde{f}(\mu_j) = \sum_{\gamma\in\Gamma} \langle T_\gamma, f\rangle, \qquad f \in C_c^\infty(A)^W$$

where $T_\gamma \in \mathcal{D}'(A)^W$

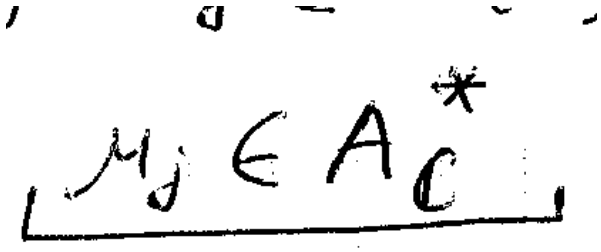


Diagram and typed description. $\mu_j \in A_{\mathbb{C}}^*$

Original source details: GELFAND-P000689

— a countable set of points called the spectrum [crossed out] Spec.

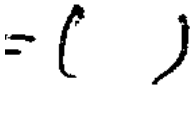


Undeciphered passage. Short marks before "Spec" are heavily crossed out.

Original source details: GELFAND-P000689

Next, information about μ_j from T_γ, given by explicit integrals.

$$T_\gamma(f) = \int_X \varphi(x\gamma x^{-1})\,dx, \qquad f(a) = (\ ) \int \varphi(an)\,dn$$



Uncertain reading. There are empty parentheses before the integral in the formula for $f(a)$.

Original source details: GELFAND-P000689

$$T_e(f) = \varphi(e) \cdot \mathrm{vol}(X)$$



Undeciphered passage. A short passage before the surname is crossed out.

Original source details: GELFAND-P000689

Harish-Chandra

$$T_e = \mathrm{vol}(X) \cdot \widetilde{\beta}, \qquad \beta \text{ — Plancherel measure.}$$

$\ell(\gamma)$ — length of the closed geodesic corresponding to a class in the fund. group.

Theorem.

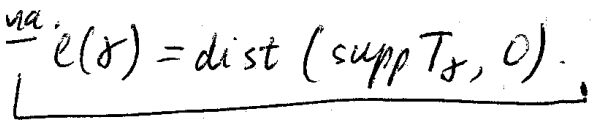


Diagram and typed description. $\ell(\gamma) = \mathrm{dist}(\mathrm{supp}\, T_\gamma, 0).$

Original source details: GELFAND-P000689

Thus,

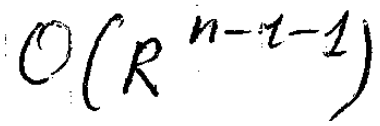
$$\sum_j \widetilde{f}(\mu_j) = \langle \widetilde{f}, \beta \rangle \operatorname{vol}(X) + \sum_{\gamma \neq e} \langle T_\gamma, f \rangle.$$



Diagram and typed description.

Original source details: GELFAND-P000689

$$\#\{j : |\mu_j| \leq R\} = \operatorname{vol}(X) \int_{\text{ball}(R)} \beta + O(R^{n-r-1})$$



Uncertain reading. The middle sign in the exponent tentatively reads r. It may also be τ.

Original source details: GELFAND-P000689

GELFAND-P000690

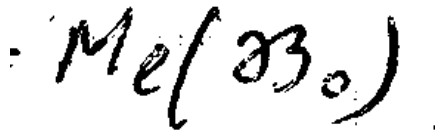
28/II-77 Gelfand Sem.

Manin (continued).

$k = \mathbb{R}$ or $\mathbb{C}$,

$\mathcal{B}_0$ — alg. of germs of k-merom. functions of x or x, t or x, t, y

$$\mathcal{B} = M_\ell(\mathcal{B}_0), \qquad \partial_x, \partial_y, \partial_t : \mathcal{B} \to \mathcal{B}$$



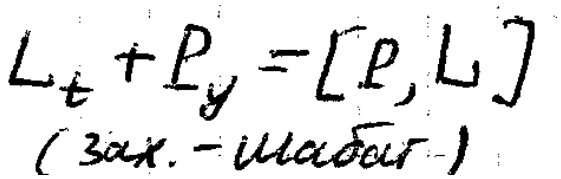
Uncertain reading. The subscript of M in the matrix algebra reads ℓ.

Original source details: GELFAND-P000690

$$\mathcal{B}[\partial_x]$$

Aim: find pairs $L, P \in \mathcal{B}[\partial_x]$, satisfying one of the eqs.

$[L, P] = 0$ (stationary Lax) or $L_t = [P, L]$ (nonstationary Lax) or $L_t + P_y = [P, L]$ (Zakh.–Shabat)



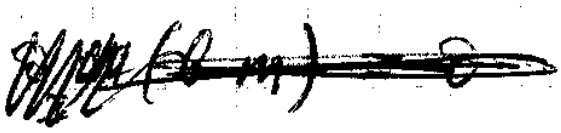
Uncertain reading. The third equation writes the commutator in the order $[P, L]$.

Original source details: GELFAND-P000690

$\mathcal{O}$ — ring of functions on an affine k-curve. Suppose we have constructed a k-space M, $(\mathcal{B}, \mathcal{O})$-bimodule,

$$0 \subset M_0 \subset M_1 \subset \cdots \quad \text{— filtration of } \mathcal{B}\text{-submodules}$$

$$\mathbf{1} \in M, \qquad \nabla : M \to M$$



Original source details: GELFAND-P000690

Undeciphered passage. The formula below $\nabla : M \to M$ is crossed out. Parentheses and bm are visible.

$$\nabla(bm) = (\partial_x b) \cdot m + b \cdot (\nabla m)$$

Further axioms [crossed out] see above.

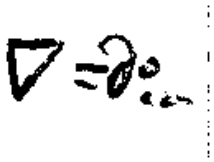


Original source details: GELFAND-P000690

Undeciphered passage. A short word before “see” is crossed out.

Example (basic)

$$\mathcal{O} = k[L, P], \quad M = \mathcal{B}[\partial_x], \quad \mathbf{1} = \mathrm{id}, \quad \nabla = \partial\,[?]$$



Original source details: GELFAND-P000690

Uncertain reading. There is a loop to the right of ∂ and several strokes below. The lower mark cannot be confirmed as x.

L, P — solve the stationary problem.

~~Definition.~~



Original source details: GELFAND-P000690

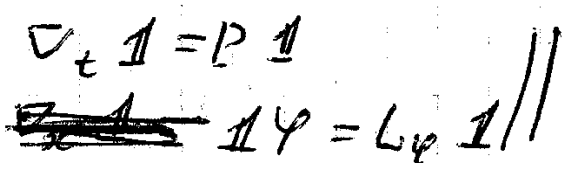
Uncertain reading. The crossed-out heading tentatively reads “Определение” (Definition).

Lemma. M — $\mathcal{B}[\partial_x]$-free module of rank 1 with gen. $\mathbf{1}$
(∂_x acts through ∇)
I. stationary Lax:

$$\mathcal{O} \ni \varphi \mapsto L_\varphi \in \mathcal{B}[\partial_x], \quad \mathbf{1}\varphi = L_\varphi \mathbf{1} \quad \Rightarrow \quad [L_\varphi, L_\psi] = 0$$

II. nonstationary Lax:

$$\nabla_t : M \to M, \qquad [\nabla_x, \nabla_t] = 0,$$



Original source details: GELFAND-P000690

Diagram and typed description.
$\nabla_t \mathbf{1} = P\mathbf{1}$
[crossed out] $\mathbf{1}\varphi = L_\varphi \mathbf{1}$ //



Original source details: GELFAND-P000690

Undeciphered passage. The short formula to the left of $\mathbf{1}\varphi = L_\varphi \mathbf{1}$ is crossed out.

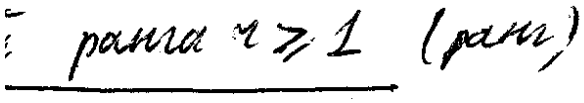
$$\Rightarrow L_t = [P, L]$$

GELFAND-P000691

III. $\nabla_y : M \to M$

$$\nabla_t, \nabla_y, \nabla_x \quad \Rightarrow \quad \nabla_t \mathbf{1} = P\mathbf{1}, \qquad \nabla_y \mathbf{1} = L\mathbf{1}$$

Explicit construction of a class of bimodules of rank $r \geq 1$ (rank)



Original source details: GELFAND-P000691

Uncertain reading. "(rank)" follows the underlined $r \geq 1$.

$r = 1 \Rightarrow N$-soliton solns. of KdV and higher KdV.
Fix $N \geq 1$ (number of solitons).

$$\alpha_1, \ldots, \alpha_N \in k, \qquad \alpha_i \neq \alpha_j$$
$$\beta_1, \ldots, \beta_N$$

$\alpha_i = \overline{\beta_i}$ when $k =$ [crossed out] $\mathbb{R}$. (meaning — the curve is real)



Original source details: GELFAND-P000691

Undeciphered passage. A mark between $k =$ and $\mathbb{R}$ is crossed out.

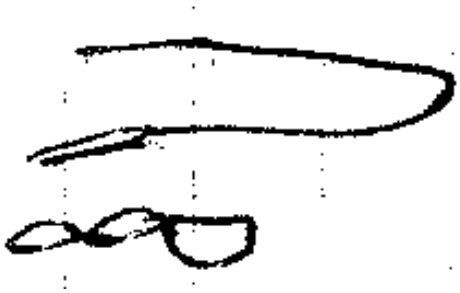


Original source details: GELFAND-P000691

Uncertain reading. The first word in the parentheses tentatively reads "смысл" (meaning).

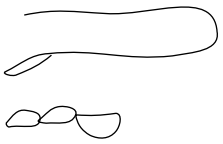
$$\mathcal{O} = \{f \in k[\lambda] \mid f(\alpha_i) = f(\beta_i), \quad \forall i = 1, \dots, N\}.$$

The module consists of sections of a bundle, trivial when lifted.



Original source details: GELFAND-P000691

Diagram and typed description.



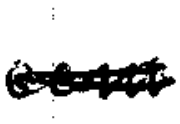
Two unlabelled sketches. The long upper line is open on the left. The lower line forms three loops.

Set

$$N = \mathcal{B}[\lambda]^r \quad \text{and} \quad N \supset M$$

All structures on M are induced by those on N. Introduce on N

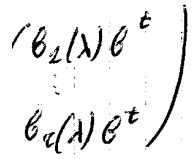
$$\mathcal{B} \times N \to N, \quad \text{[crossed out]} \quad N \times k[\lambda] \to N$$



Original source details: GELFAND-P000691

Undeciphered passage. A short word between the two actions is crossed out.

$$b \begin{pmatrix} b_1(\lambda) \\ \vdots \\ b_r(\lambda) \end{pmatrix} = \begin{pmatrix} b_1(\lambda)b^t \\ \vdots \\ b_r(\lambda)b^t \end{pmatrix}$$



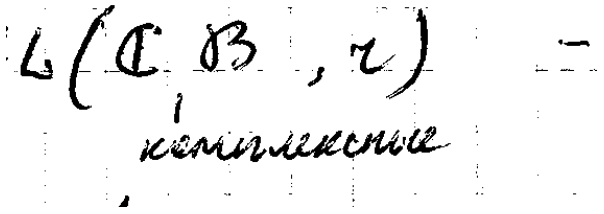
Original source details: GELFAND-P000691

Uncertain reading. Both visible components of the right-hand column have b^t.

$$\mathrm{GL}(\mathcal{B}[\lambda], r) \times N \to N$$

$$M = \{m(\lambda) \in N \mid m(\alpha_i) = g_i m(\beta_i), \quad \forall i = 1, \dots, N\}$$

(nontrivial gluing cond., as distinct from functions)



Original source details: GELFAND-P000691

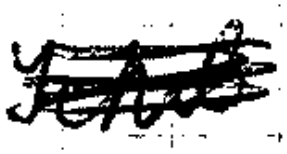
Diagram and typed description.

$g_i \in \mathrm{GL}(\mathbb{C}\,\mathcal{B}, r)$

complex

There is no clear operation sign between $\mathbb{C}$ and $\mathcal{B}$.

— parameters (gluing matrices, giving rise to nontrivial variants).



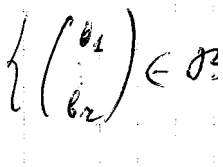
Undeciphered passage. A short passage before the conjugation condition is crossed out.

Original source details: GELFAND-P000691

$$\overline{g_i} = g_i^{-1} \quad \text{when } k = \mathbb{R}$$

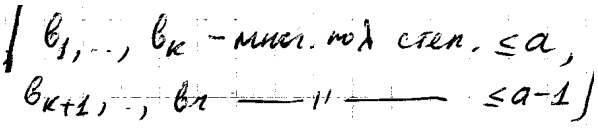
Filtration on N:

$$N_{\substack{ar+k \\ 0\le k\le r-1}} = \left\{ \begin{pmatrix} b_1 \\ \vdots \\ b_r \end{pmatrix} \in \mathcal{B}[\lambda]^r \;\middle|\; \begin{array}{r} b_1, \dots, b_k \text{ — polys. in } \lambda \text{ of deg. } \le a, \\ b_{k+1}, \dots, b_r \text{ — ” — } \le a-1 \end{array} \right\}$$



Uncertain reading. The column in the braces starts with b_1 and ends with b_r.

Original source details: GELFAND-P000691



Uncertain reading. The second line uses two rules and two short slanted strokes as a ditto mark.

Original source details: GELFAND-P000691

$$\mathcal{B}N_i \subset N_i,$$

M will start at the wrong place (there is a system of lin. eqs.).

$$M_{(i)} = M \cap N_i \quad \text{— still not what we want}$$

Lemma. Consider the matrix (block)

$$(\alpha_i^j \operatorname{Id}_r - \beta_i^j g_i)_{\substack{i=1,\dots,N \\ j=0,\dots,N-1}}$$

Suppose

$$\det(\alpha_i^j \operatorname{Id}_r - \beta_i^j g_i) \neq 0.$$

GELFAND-P000692

Then $M_{(rN)} = \{0\}$. For every $i > rN$ there is a canonical inclusion

$$M_{(i)}/M_{(i-1)} \xrightarrow{\sim} N_i/N_{i-1}$$

which is an isomorphism.

Def.: $M_i = M_{(i-rN)}$.



Uncertain reading. The index in parentheses reads $i - rN$.

Original source details: GELFAND-P000692

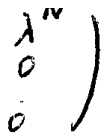
Find the explicit form of **1** from a system of linear equations. We have:

$$\mathbf{1} = \sum_{j=0}^{N-1} m_j \lambda^j + \begin{pmatrix} \lambda^N \\ 0 \\ \cdot \\ 0 \end{pmatrix} \qquad \text{(by def. of } M_1\text{)}.$$



Undeciphered passage. A short mark below the equals sign is crossed out.

Original source details: GELFAND-P000692



Uncertain reading. A single dot is visible between the two zeros in the column.

Original source details: GELFAND-P000692

The coeff. of $\lambda^N = 1$ [crossed out] — normalization condition.



Undeciphered passage. A mark after the one is crossed out.

Original source details: GELFAND-P000692

m_j satisfy a system of linear equations

$$\sum_{j=0}^{N-1} (\alpha_i^j E_r - \beta_i^j g_i) m_i + (\alpha_i^N - \beta_i^N g_i) \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = 0,$$

$$i = 1, \ldots, N.$$



Uncertain reading. The first parenthesis is followed by m_i.

Original source details: GELFAND-P000692

This uniquely determines m_i.

Variant: $\nabla_x \stackrel{?}{=} \partial_x$. The gluing conditions fail.

Add an endomorphism of the module M.

$$\nabla_x = \partial_x + \underbrace{d_x(\lambda)}_{\text{matrix}},$$

where

$$d_x(\lambda) \in M_r(\mathcal{B}[\lambda]) = \mathrm{End}_{\mathcal{B}}(N = \mathcal{B}[\lambda]^r).$$



Undeciphered passage. A mark before M_r is crossed out.

Original source details: GELFAND-P000692

Lemma. $\nabla_x M \subset M$, if g_i satisfy the system of diff. eqs.

$$\partial_x g_i = g_i d_x(\beta_i) - d_x(\alpha_i) g_i, \qquad i = 1, \ldots, N.$$

d_x — connection matrix. (include in the geom. structure in place of g_i).

d_x — arbitrary $\qquad$ g_i — to be found.

Simplest solutions for g_i:

Let $d_x(\lambda) \in M_r(M_\ell(k))[\lambda]$. Then (with fixed $\lambda = \mathrm{const}$)

$$g_i(x) = \exp(-x d_x(\alpha_i)) g_i(0) \exp(x d_x(\beta_i)).$$

Additional params.: the matrices $d_x(\lambda)$ with constant coefficients.

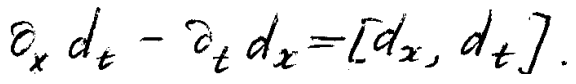
$r = 1 \Rightarrow$ ordinary exp.

Similarly $\to$ write equations for ∇_t.

$$\nabla_t = \partial_t + d_t(\lambda)$$
$$\nabla_y = \partial_y + d_y(\lambda).$$

Commutation ?

$$\partial_x d_t - \partial_t d_x = [d_x, d_t].$$



Uncertain reading. The commutator on the right has no minus sign.

Original source details: GELFAND-P000692

we need to be able to solve matrix equations. When $d_x = d_t =$ const we seek

$$[d_x, d_t] = 0.$$

Uncertain reading. The last word on the line tentatively reads “ищем” (we seek).

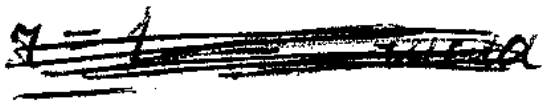


Original source details: GELFAND-P000692

The simplest choice is to take $d_x \in k[d_t]$.

[crossed out: $r = 1$ ...]

Undeciphered passage. The rest of the line after $r = 1$ is obscured by the crossing-out.



Original source details: GELFAND-P000692

Undeciphered passage. A separate mark to the right of the line is crossed out.



Original source details: GELFAND-P000692

We obtain solns. of uncontrolled order. Reduce the orders where possible.

Consider the nonstationary Lax eq. for def.

Diagram and typed description.





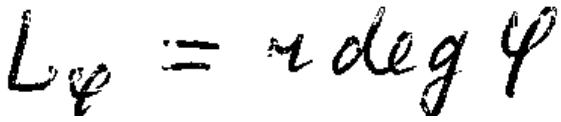
An upright flourish at lower right. No label is discernible.

Original source details: GELFAND-P000692

GELFAND-P000693

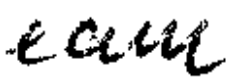
Lemma. ord $L_\varphi = r \deg \varphi$, if

Uncertain reading. The subscript of L and the symbol after deg are read as φ.



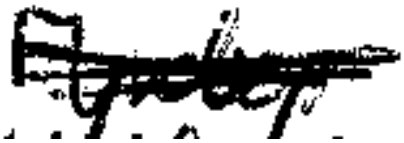
Original source details: GELFAND-P000693

Undeciphered passage. The line ends with “if.”



Original source details: GELFAND-P000693

[crossed out]



Original source details: GELFAND-P000693

Uncertain reading. The crossed-out word tentatively reads “Пример” (Example).

On the preceding discussion:

From

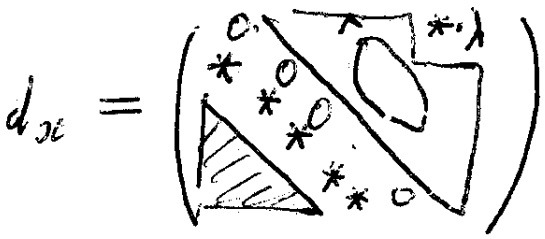
$$\nabla : M_i/M_{i-1} \xrightarrow{\sim} M_{i+1}/M_i$$

we obtain an add. condition on d_x:



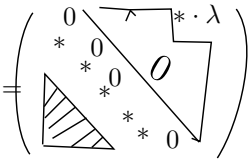
Original source details: GELFAND-P000693

Uncertain reading. The abbreviation before “condition” tentatively reads “доп.” (additional).



Original source details: GELFAND-P000693

Diagram and typed description.

$d_x =$ 

Schematic with five asterisks below the diagonal, a hatched triangle and a stepped outline around the zero region.

$*$ — invertible elements.

Example:

$$\begin{pmatrix} 0 & \lambda \\ b & 0 \end{pmatrix}.$$



Original source details: GELFAND-P000693

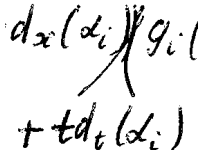
Uncertain reading. The lower left entry tentatively reads b. Another possible reading is β.

Remark. $\exists \psi$, $\deg \psi = 2$ in $\mathcal{O}$, if $\alpha_i + \beta_i$ is independent of i. (degenerate hyperelliptic curve)

$$\Rightarrow \quad \exists \psi : \operatorname{ord} L_\psi = 2r \qquad \text{(in particular one can take } 4.)$$

$$d_t = cE_2 + \dots$$

$$g_i(x) = \exp(-x d_x(\alpha_i) + t d_t(\alpha_i))\, g_i(0,0) \exp(x d_x(\beta_i) + t d_t(\beta_i)).$$



Original source details: GELFAND-P000693

Uncertain reading. The term $+td_t(\alpha_i)$ is inserted from below. The left side reads $g_i(x)$.

Question: behavior of the N-soliton solution?

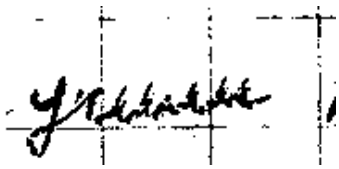
GELFAND-P000694

14/III-77. Gelfand Sem.

Gabrielov. "Combinatorial formulas for Pontryagin classes"

Gelfand:

MacPherson, Sullivan, Deligne learned from the work of G.–G.–L.



Original source details: GELFAND-P000694

Uncertain reading. The verb after "Deligne" tentatively reads "учились" (learned).



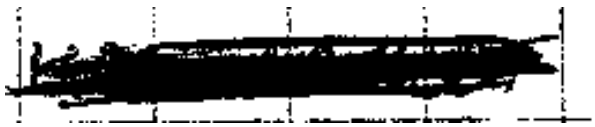
Original source details: GELFAND-P000694

Uncertain reading. The short word before "work" is overwritten. It tentatively reads "на," with "по" another possible reading.

Elementary geometry of higher-dimensional polyhedra has yet to be developed.

An n-dimensional polyhedron is a complex (a collection of simplices with boundaries specified).

Each $(n-1)$-dimensional simplex [crossed out] — is the boundary of exactly two n-dimensional ones.

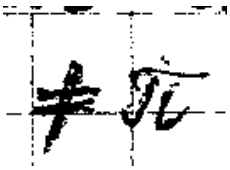


Original source details: GELFAND-P000694

Undeciphered passage. A short passage is obscured by dense horizontal strokes.

How do we assign a number to each vertex to compute the Euler characteristic?

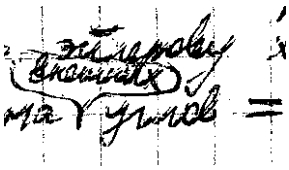
On going around a vertex, the sum of exterior angles is $\neq \pi$.



Original source details: GELFAND-P000694

Uncertain reading. Two horizontal strokes and a slanting stroke with a loop appear before π. The sign is read as $\neq$.

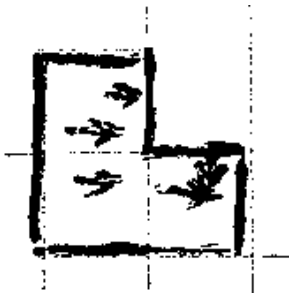
Sum of [inserted: exterior] angles $= \sum$ vertex defects.



Original source details: GELFAND-P000694

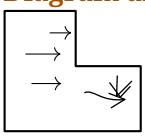
Uncertain reading. "Exterior" is circled between the lines, with a pointer before "angles."

In three dimensions it is much more complicated. Going around a vertex of a square gives a turn of $90°$.



Original source details: GELFAND-P000694

Diagram and typed description.



Stepped outline with three rightward arrows at left. Strokes overlap in the lower right and their individual directions are uncertain.

$SO(3)$ is already noncommutative. The geometry of three-dimensional polyhedra is unclear — there is no substantive theory.

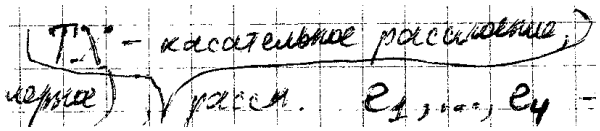
In two dimensions, commutativity reduces a contour to going around vertices.

For four-dimensional polyhedra there are two invariants (the Euler characteristic and one more). For smooth ones this has long been known.

MacPherson's exposition is in the Bourbaki Seminar (February 1977)

Gabrielov.

X — a manifold (four-dimensional). [above the line: TX — tangent bundle.]



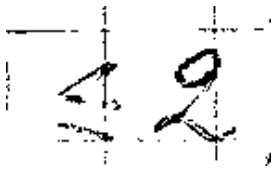
Original source details: GELFAND-P000694

Uncertain reading. The insertion is above the line inside a curved bracket pointing to the space before "consider."

consider $e_1, \dots, e_4$ — vector fields on X

$$Z = \{x : \mathrm{rk}(e_1, \dots, e_4) \leq 2\}, \qquad \mathrm{Codim}\, Z = 4$$

for fields in general position.



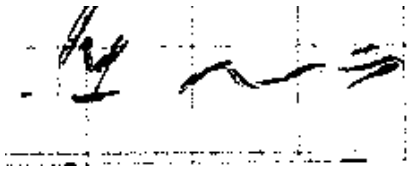
Original source details: GELFAND-P000694

Uncertain reading. The lower stroke of $\leq$ is separate from the angle above it.

Z — a set of points.

A cooriented subset — an orientation of the normal bundle is specified

(orient. $Y \rightsquigarrow$ orient. X)



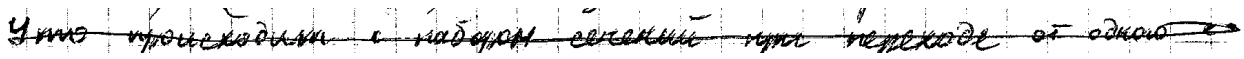
Original source details: GELFAND-P000694

Uncertain reading. Between the two orientations there is a wavy stroke and an arrowhead pointing right.

GELFAND-P000695

Let $L_{m,n} = \mathrm{Hom}(\mathbb{R}^m, \mathbb{R}^n)$. A point is specified by a tuple $(e_1, \dots, e_m)$, $e_j \in \mathbb{R}^n$. The set Z has a canonical coorientation (coorient. of the point x — orient. of $T_x X$). We obtain a set of points with signs $\pm$. The corresponding homology class is called the first Pontryagin class. (Number of plus signs) $-$ (number of minus signs) is independent of the choice of fields.

[crossed out with a rightward arrow: What happens to the collection of sections when passing from one]



Original source details: GELFAND-P000695

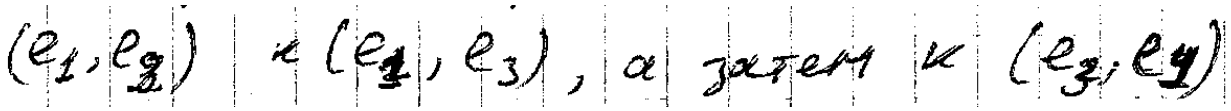
Uncertain reading. A horizontal stroke through the line ends in an arrowhead at right.

How does the cycle change when we replace the collection?

Let the bundle E be two-dimensional, $e_1(x), e_2(x)$ — sections.

$$Z = \{x \in X \mid e_1(x) = e_2(x) = 0\}$$

— a set of points. There is a canonical coorientation. This is the first Pontryagin class of the two-dimensional bundle. Pass from (e_1, e_2) to (e_1, e_3), and then to (e_3, e_4).



Original source details: GELFAND-P000695

Uncertain reading. Several subscripts are overwritten. The readings of the pairs are tentative.

[crossed out: also?]



Original source details: GELFAND-P000695

Uncertain reading. The short crossed-out word tentatively reads “ещё” (also).

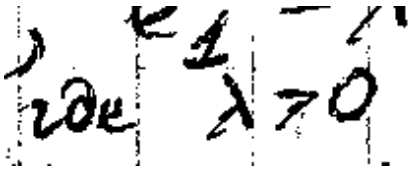
we can pass by elementary operations (changing one vector at a time). We have:

$$Z(e_2, e_3) - Z(e_1, e_3) = \partial Z^1(e_1, e_2; e_3), \tag{1}$$

where

$$Z^1 = \{x \mid e_3 = 0,\ e_1 = \lambda e_2\}, \qquad \mathrm{Codim}\, Z^1 = 3.$$

[inserted below $e_1 = \lambda e_2$: where $\lambda > 0$.]



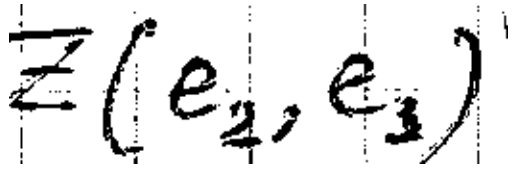
Original source details: GELFAND-P000695

Uncertain reading. The inequality is inserted below the equation.

The sign in the formula is easy to check. The boundary occurs at $\lambda = 0$ and $\lambda = \infty$. There is also $Z^1(e_1, e_3; e_2)$.

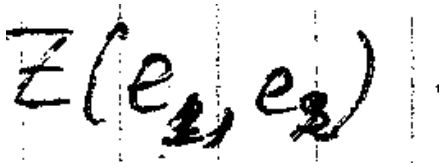
$$Z(e_1, e_2) - Z(e_2, e_3) = \partial Z^1(e_1, e_3; e_2) \tag{2}$$

$$Z(e_1, e_3) - Z(e_1, e_2) = \partial Z^1(e_2, e_3; e_1) \tag{3}$$



Original source details: GELFAND-P000695

Uncertain reading. The first subscript of the second term in (2) is read as 2.

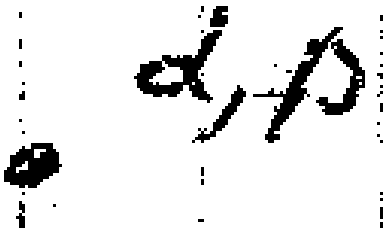


Original source details: GELFAND-P000695

Uncertain reading. Both subscripts of the second term in (3) are overwritten. The tentative reading is $1, 2$.

Add (1), (2) and (3). We obtain a closed curve. It is the boundary of the surface $e_1 = \lambda e_2 = \mu e_3,\ \lambda > 0,\ \mu > 0$.

0-dimensional hypersimplex



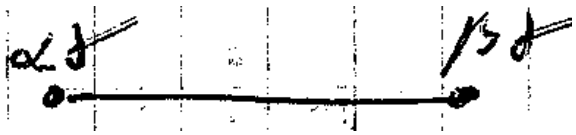
Original source details: GELFAND-P000695

Diagram and typed description.

α, β

(a point with two indices)

1-dimensional hypersimplex

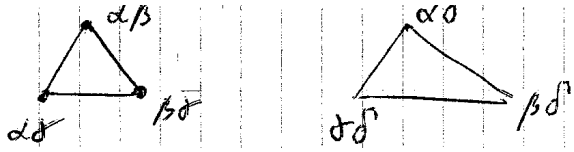


Original source details: GELFAND-P000695

Diagram and typed description.

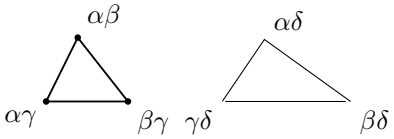


2 — || —



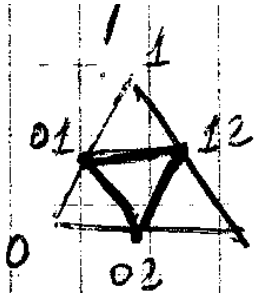
Original source details: GELFAND-P000695

Diagram and typed description.



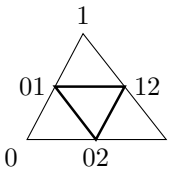
there are two types.

An n-dimensional hypersimplex is a polytope whose vertices are 0-dimensional hypersimplices, whose edges are 1-dimensional hypersimplices, and so on.

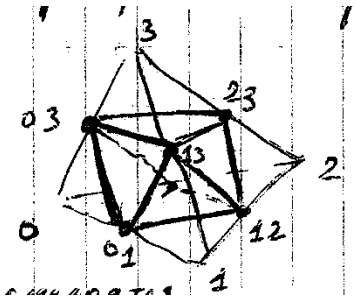


Original source details: GELFAND-P000695

Diagram and typed description.

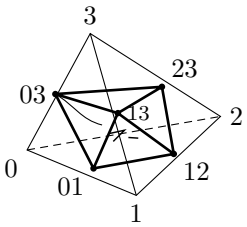


There is no discernible numeral at the lower right outer vertex.



Original source details: GELFAND-P000695

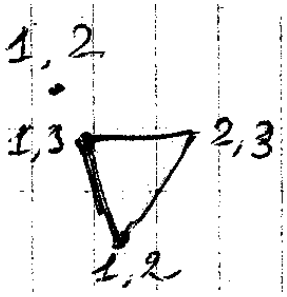
Diagram and typed description.



The heavy edges run through vertices labelled 03, 23, 13, 01, 12. A further mark and several overlapping strokes in the interior are unclear.

more complicated strata

[to the right of the drawings, beyond the two rules:] There are two types of triangles in the octahedron. Convex hull of the set of centres of the faces of the simplex. At each vertex are indices corresponding to the indices of the vertices of the corresponding face. One can also append the same index everywhere.

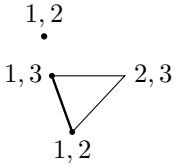


Original source details: GELFAND-P000695

Diagram and typed description.

$Z(e_1, e_2)$ — corresponds

$Z^1(e_1, e_2, e_3)$. corresponds

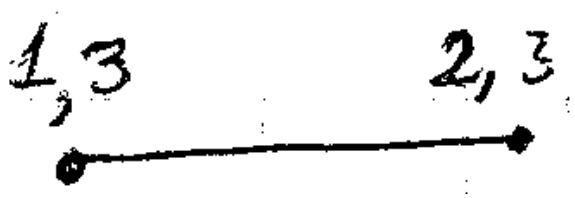


Problem: construct chains for each hypersimplex with the required properties.

GELFAND-P000696

A chain in the space of tuples of vectors in the plane. To the point 12 — in the space of pairs of vectors $e_1 = e_2 = 0$

To the segment

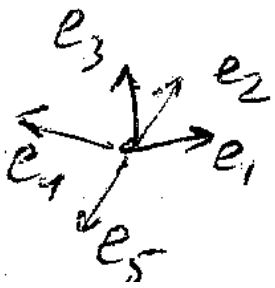


Original source details: GELFAND-P000696

Diagram and typed description.

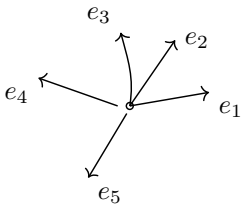


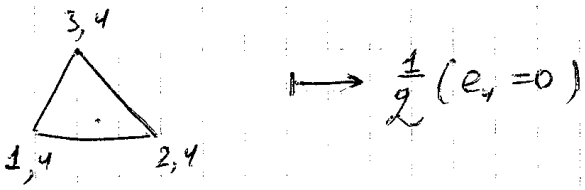
there corresponds $(e_1 = \lambda e_2,\ e_3 = 0)$. (3 — superfluous!)



Original source details: GELFAND-P000696

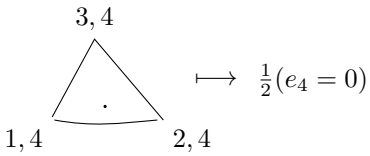
Diagram and typed description.

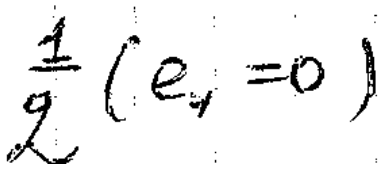




Original source details: GELFAND-P000696

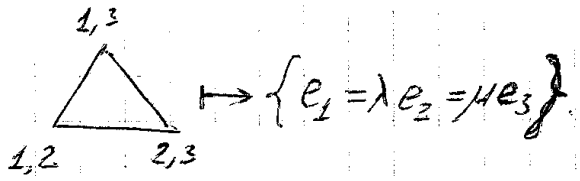
Diagram and typed description.





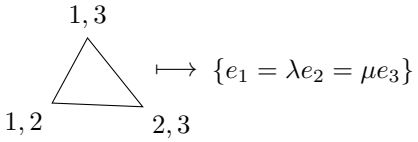
Original source details: GELFAND-P000696

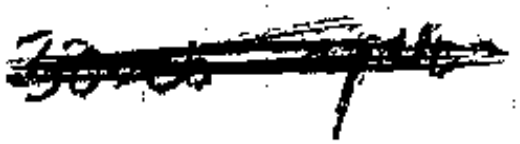
Uncertain reading. The coefficient reads $\frac{1}{2}$.



Original source details: GELFAND-P000696

Diagram and typed description.





Original source details: GELFAND-P000696

Undeciphered passage. Several strokes cross out a short passage under the left triangle.



Original source details: GELFAND-P000696

Diagram and typed description.

Two long strokes with a separate dot between them. The forms of their left ends are uncertain.

GELFAND-P000697

21/III — 77 Gelfand Sem.

M. I. Graev. Consider $\mathbb{C}^3$ and a fn. on it $f(x) = f(x_1, x_2, x_3)$. Consider the line

$$\begin{cases} x_1 = \alpha_1 x_3 + \beta_1 \\ x_2 = \alpha_2 x_3 + \beta_2 \end{cases}$$

$$\int f(\alpha_1 x_3 + \beta_1 \text{ [crossed out]}, \alpha_2 x_3 + \beta_2, x_3)\, dx_3\, d\bar{x}_3$$
$$= \varphi(\alpha_1, \alpha_2; \beta_1, \beta_2)$$



Undeciphered passage. A mark after β_1 in the first argument is crossed out.

Original source details: GELFAND-P000697

How can f be recovered from φ? The problem is overdetermined and trivial.

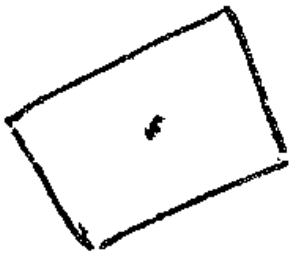
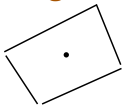


Diagram and typed description.



Original source details: GELFAND-P000697

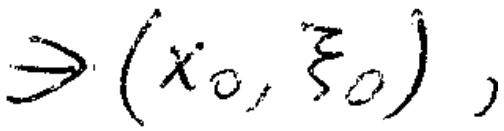
Draw a plane. This gives the classical Radon problem. Its soln. is well known. Solution. Write a $(1,1)$-form on the set of lines in the plane. Its integral gives the answer.

In the real case the equation is

$$\frac{\partial^2 \varphi}{\partial \alpha_1 \partial \beta_2} = \frac{\partial^2 \varphi}{\partial \alpha_2 \partial \beta_1}.$$

One can prove that it is sufficient.

Idea $WF(u) \ni (x_0, \xi_0)$, if the integral over the system of parallel planes with direction ξ_0 has a singularity.



Uncertain reading. The sign after $WF(u)$ is read as $\ni$.

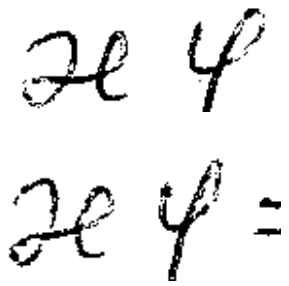
Original source details: GELFAND-P000697

Construct a differential form

$$\mathcal{H}\varphi = \sum_{i,j=1}^{2} \frac{\partial^2 \varphi}{\partial\beta_i \, \partial\bar{\beta}_j} \, d\alpha_i \wedge d\bar{\alpha}_j \quad (\mathbb{C}^3)$$

$$\mathcal{H}\varphi = \frac{\partial\varphi}{\partial\beta_1} d\alpha_1 + \frac{\partial\varphi}{\partial\beta_2} d\alpha_2 \qquad (\mathbb{R}^3)$$

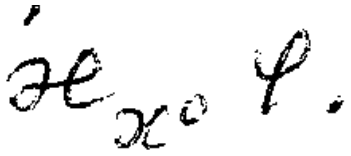
Uncertain reading. The handwritten sign before φ is rendered as $\mathcal{H}$.



Original source details: GELFAND-P000697

Let there be a point (x_1^0, x_2^0, x_3^0). Substitute $\beta_i = x_i^0 - \alpha_i x_3^0$. We obtain the restriction of the form to the lines through this point. Denote this by $\mathcal{H}_{x^0}\varphi$.

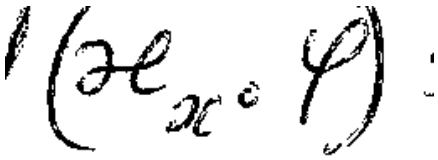
Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.



Original source details: GELFAND-P000697

Theorem. φ is representable in terms of $f \iff d(\mathcal{H}_{x^0}\varphi) = 0, \ \forall x_0$.

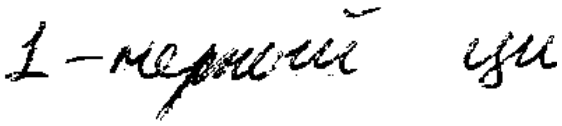
Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.



Original source details: GELFAND-P000697

Inversion formula in $\mathbb{C}^3$ (in $\mathbb{R}^3$ it is already nonlocal). Among the lines through x_0 choose a one-dimensional cycle γ and then

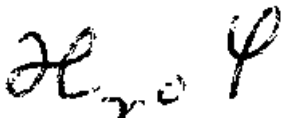
Uncertain reading. The manuscript reads “one-dimensional.”



Original source details: GELFAND-P000697

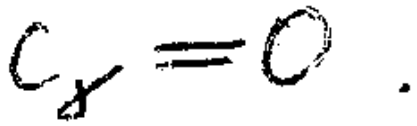
$$\int_\gamma \mathcal{H}_{x^0}\varphi = c_\gamma f(x^0)$$

Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.



Original source details: GELFAND-P000697

In the real case $C_\gamma = 0$.



Uncertain reading. The letter in this line is rendered as C, and in the preceding formula as c.

Original source details: GELFAND-P000697

A complex K — a three-dimensional set of lines.

When, given $\mathcal{H}\varphi$ on K, can we recover φ?



Uncertain reading. The handwritten sign rendered as $\mathcal{H}$ recurs here.

Original source details: GELFAND-P000697

All complexes K, with this property are of the following kinds:

a) all lines tangent to a surface

b) — ‖ —, meeting a curve

Linear complex — a complex of lines in $\mathbb{C}P^3$ ($\mathbb{R}P^3$). all its lines through a given point — a plane.

GELFAND-P000698

$$\begin{matrix} x \\ y \end{matrix} \quad \left\| \begin{matrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{matrix} \right\|$$

Plücker coordinates $p_{ij} = x_i y_j - x_j y_i$ — coordinates of the plane.



Uncertain reading. The surname is written “Плюккеровы” (Plücker).

Original source details: GELFAND-P000698

Lin. complex: $\sum c_{ij} p_{ij} = 0$.

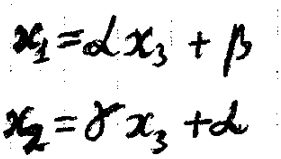
Lin. complex. a plane is associated with $\forall$ points of the space (it contains all the lines through that point)

a line is associated with $\forall$ lines, meeting all lines that pass through the given line.

There is a Legendre transform.

Example of a complex.

$$x_1 = \alpha x_3 + \beta$$
$$x_2 = \gamma x_3 + \alpha$$



Uncertain reading. α is the coefficient of x_3 in the first equation and the constant term in the second.

Original source details: GELFAND-P000698

Any other is isomorphic to it.



Uncertain reading. The final letter of “Любой” (any) is written over earlier strokes.

Original source details: GELFAND-P000698

References: Chapters 1 and 2 of vol. 5 of Generalized Functions.

F. A., vol. 1, $N = 1$, paper by Gelfand, Graev, Shapiro.



Uncertain reading. The notation after the volume number reads $N = 1$.

Original source details: GELFAND-P000698

GELFAND-P000699

28/III-77 Gelfand Sem.

I. M. Gelfand. Problem (from Griffiths’s paper [crossed out]: Variations on a theorem of Abel.)



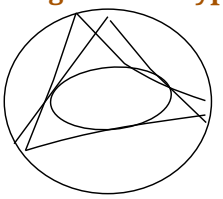
Undeciphered passage. A small mark between the surname and the colon is crossed out.

Original source details: GELFAND-P000699

An ellipse inside a circle.



Diagram and typed description.



A circle, an elongated oval, and overlapping sides. Some of the joins are uncertain.

Original source details: GELFAND-P000699

A polygon, inscribed in one curve of the second degree and circumscribed about another. When does this hold? (When can one inscribe it this way?)

Poncelet's thm.: if it closes from one point, then from $\forall$ other point it closes after the same number of steps.

Jacobi: a relation in elliptic integrals

From Poncelet's theorem $\Rightarrow \exists$ an invariant measure and an isomorphism to a rotation of the circle.

Latest issue of Compositia Math. — the simplest proof of the theorem of Dvoretzky.



Uncertain reading. The manuscript title reads "Compositia Math."

Original source details: GELFAND-P000699

(Figiel)

A. M. Vershik. Probability theory and the symm. group

S_n — group of permutations of a set of n elements.

S_∞ — group of finitary bijective permutations of the natural numbers, affecting only [crossed out] finitely many elements



Undeciphered passage. Several crossed-out strokes precede "finitely."

Original source details: GELFAND-P000699

Asymptotic problems related to S_n — asymptotics of Plancherel measures.

Young tableau. A collection of rows beginning at a vertical line and decreasing in length

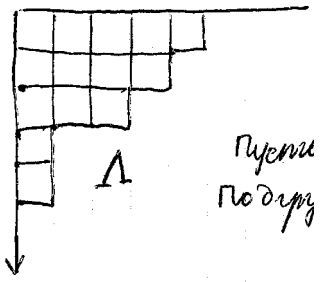
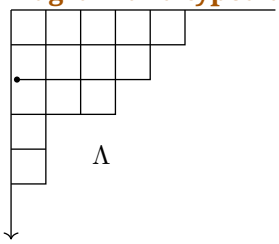


Diagram and typed description.



Original source details: GELFAND-P000699

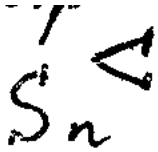
(see the end of van der Waerden's book
Curtis, Reiner,
Kirillov

Let there be n cells;

Subgroup

$$\underset{\substack{\text{preserves partition}\\ \text{into rows}\\ S_n^<}}{G_1,} \qquad \underset{\substack{\text{preserves partition}\\ \text{into columns}\\ S_n^\Lambda}}{G_2 \subset S_n}$$

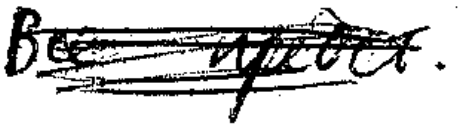
— permutes elements within a column.



Original source details: GELFAND-P000699

Uncertain reading. The superscript on the first S_n is an angular sign, rendered as $<$.

[crossed out: All reprs.]



Original source details: GELFAND-P000699

Uncertain reading. The crossed-out words tentatively read "Все предст." (All reprs.).

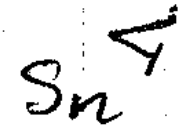
GELFAND-P000700

Consider the trivial representation of $S_n^<$ and a one-dimensional one of S_n^Λ.



Original source details: GELFAND-P000700

Uncertain reading. The adjective after "Consider" tentatively reads "единичное" (trivial). The ending is written together.



Original source details: GELFAND-P000700

Uncertain reading. The first S_n has an angular superscript, rendered as $<$.

$$\begin{pmatrix} \text{even} — \text{identity} & 1 \\ \text{odd} — & (-1) \end{pmatrix} \longleftarrow \text{sign representation}$$

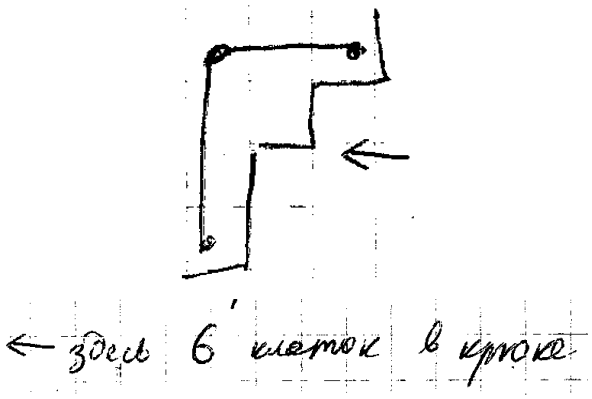
From each such representation induce a representation of S_n.

They are reducible, but have a unique common irreducible (theorem!) — this is the one assigned.

1955. hook formula (Robinson et al.)

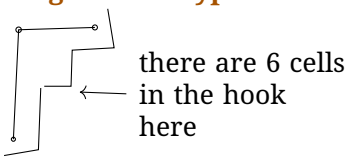
hook.

A hook is the total number of cells to the right of and below the given one —



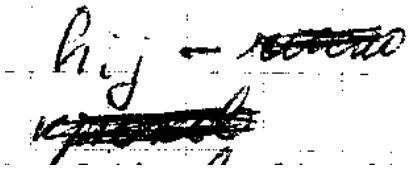
Original source details: GELFAND-P000700

Diagram and typed description.



$$\frac{n!}{\prod h_{ij}} = \dim \Lambda$$

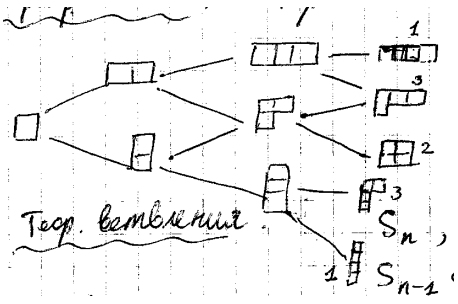
h_{ij} — [crossed out] (— number of elements in the hook



Original source details: GELFAND-P000700

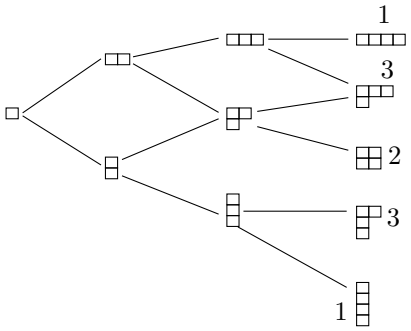
Uncertain reading. Two words after h_{ij} are crossed out. The first looks like “число” (number).

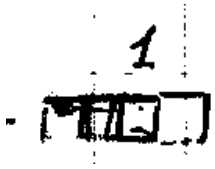
[overwritten] Young graph. The vertices are Young tableaux. In column n there are tableaux of n cells. There is an edge when one can obtain the next tableau by adding a single cell.



Original source details: GELFAND-P000700

Diagram and typed description.





Original source details: GELFAND-P000700

Uncertain reading. The upper right row of cells is heavily retraced. There is a 1 above it.



Original source details: GELFAND-P000700

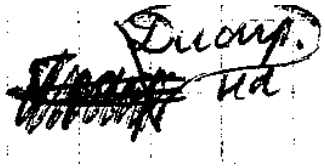
Undeciphered passage. The mark before the heading is heavily shaded over.

Branching thm. S_n, π — an irred. repr. of S_n

$S_{n-1} \subset S_n$

In $\pi|_{S_{n-1}}$ each irred. repr. occurs with multiplicity 1.

[crossed out] [inserted and circled: Diagr.] at step n decomposes into precisely the preceding ones.



Original source details: GELFAND-P000700

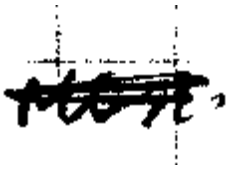
Undeciphered passage. The original word below “Diagr.” is crossed out.

— There are as many one-dimensional subspaces as there are paths in the graph leading to 1.

— Thus the count is a combinatorial problem that is unsolved (only via Frobenius’s formula and so on).

Gelfand–Tsetlin basis

Question: $S_m \subset S_n$. [crossed out] how many paths reach a fixed Young tableau?



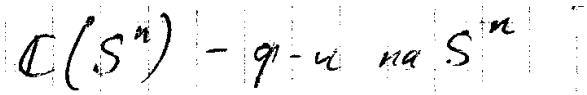
Original source details: GELFAND-P000700

Undeciphered passage. A short word after the inclusion is crossed out.

Length of the first row

$$\ell_1(\Lambda) = \max\{m : \ \pi|_{S_m} \supset \mathrm{id}\}$$

Regular repr. $C(S^n)$ — fns. on S^n.



Original source details: GELFAND-P000700

Uncertain reading. The manuscript writes n as a superscript in both instances of S^n.

$$(U_g f)(h) = f(g^{-1}h)$$

Plancherel measure on Young tableaux

It is known that

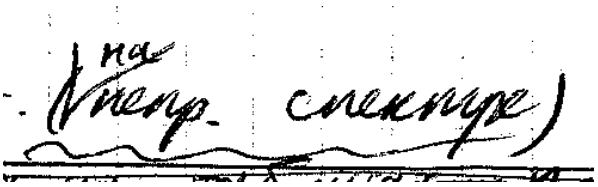
$$\underset{\text{dim. of the regular repr.}}{n!} = \sum_{\Lambda} (\dim \Lambda)^2$$

Plancherel measure

$$\mu(\Lambda) = \frac{(\dim \Lambda)^2}{n!}$$

Gelfand's problem.

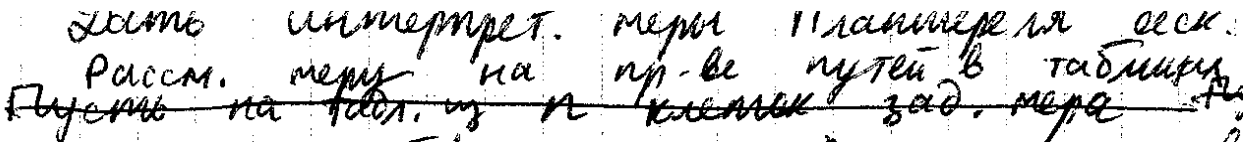
Give an interpretation of the Plancherel measure of an infinite-dimensional repr. ([“on” inserted above] the continuous spectrum)



Original source details: GELFAND-P000700

Uncertain reading. “On” is inserted above the start of the phrase in parentheses.

[written above the crossed-out line] Consider a measure on the space of paths in tableaux

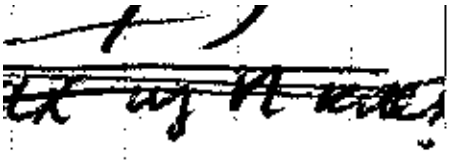


Original source details: GELFAND-P000700

Uncertain reading. The new phrase stands immediately above the horizontally crossed-out line.

[crossed out] Suppose that on tableaux with n cells a measure is given

[crossed out] Suppose there is a measure on tableaux with n cell...



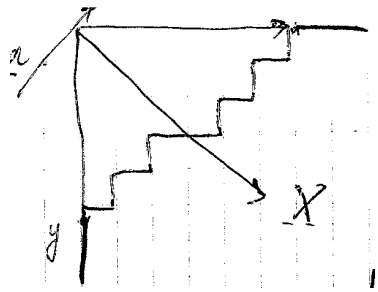
Original source details: GELFAND-P000700

Uncertain reading. The end of the crossed-out phrase is obscured by strokes at the right edge.

Suppose cells are added. Consider the conditional probability that a cell appears given the history

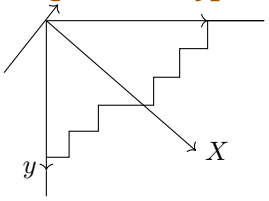
GELFAND-P000701

Theorem



Original source details: GELFAND-P000701

Diagram and typed description.

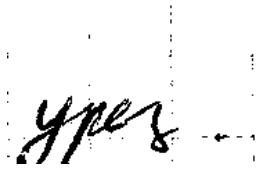


Stepped outline with the labels y and X. Strokes at the horizontal arrowhead may form x.

Consider the coord. system

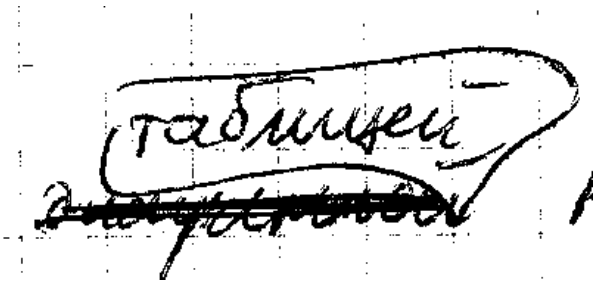
$$\left\| \begin{aligned} X &= \frac{x+y}{2}; \\ Y &= \frac{x-y}{2}; \end{aligned} \right.$$

Consider the graph of the fn. $|X|$, trunc. [crossed out: diagram] [inserted above: tableau] of Young.



Original source details: GELFAND-P000701

Uncertain reading. The abbreviation tentatively reads “ypez.” (trunc.).



Original source details: GELFAND-P000701

Uncertain reading. “Tableau” is circled above the crossed-out word, which tentatively reads “diagram.”

The tableau is identified with the fn. $Y = \widetilde{\Lambda}(X)$

$$\mu_n \left\{ \widetilde{\Lambda} : \sup \left| \frac{1}{\sqrt{n}} \widetilde{\Lambda}(\sqrt{n}\, X) - \Omega(X) \right| < \varepsilon \right\} \longrightarrow 1, \qquad \forall \varepsilon > 0$$

compress by a factor of $\sqrt{n}$. Then all are near the graph of the fn.

$$\Omega(X) = \frac{2}{\pi} \left(X \arcsin X + \sqrt{1 - X^2} \right) \qquad \left(\Omega'(X) = \frac{2}{\pi} \arcsin X \right)$$

This is conv. in measure, [crossed out]

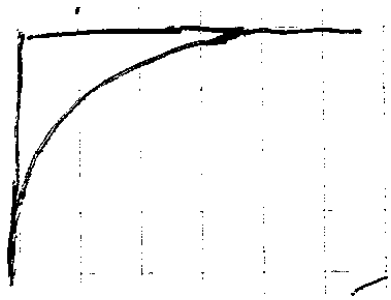


Original source details: GELFAND-P000701

Undeciphered passage. A short passage after the comma is crossed out.

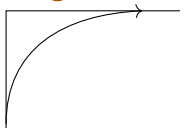
There is a thm. on conv. almost everywhere. (for a seq. of tableaux)

A diagram is an infinite path. It converges for almost all diagrams.



Original source details: GELFAND-P000701

Diagram and typed description.



Corollary:

$$\mu_n\left\{\Lambda : \left|\frac{\ell_1(\Lambda)}{2\sqrt{n}} - 1\right| < \varepsilon\right\}$$

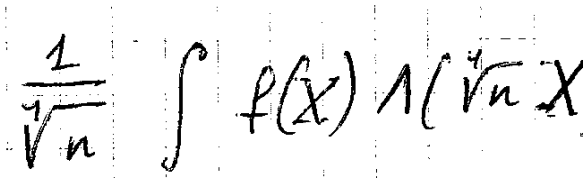
(with prob. close to 1 there is an inv. vector)

(soln. of Ulam's problem)

This is the law of large numbers. Deviations give white noise.

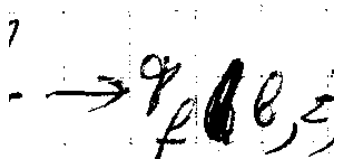
For the functional $\int f(X)\Lambda(X)\,dX$ there is a central limit thm.

$$\mu_n\left\{\Lambda : \frac{1}{\sqrt[4]{n}}\int f(X)\Lambda(\sqrt[4]{n}\,X)\,dX - \int f(X)\Omega(X)\,dX < \ell\right\}$$
$$\longrightarrow \varphi_f(\ell,\varepsilon)$$



Original source details: GELFAND-P000701

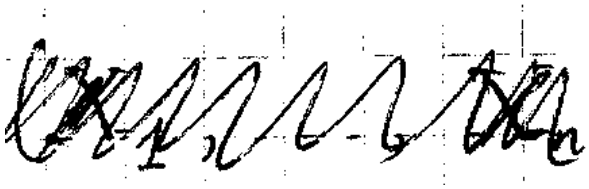
Uncertain reading. An index 4 is visible at both radicals.



Original source details: GELFAND-P000701

Uncertain reading. The right-hand side tentatively reads $\varphi_f(\ell,\varepsilon)$. The mark before ℓ is heavily overwritten.

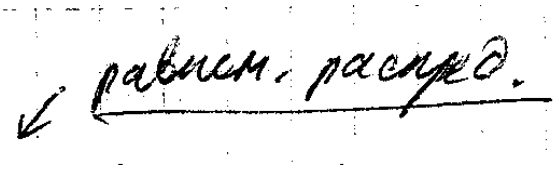
the Markov property and stationarity can be checked.



Original source details: GELFAND-P000701

Undeciphered passage. Crossed-out writing before $(x_1,\ldots,x_n)$. A parenthesis, x_1 and a final index n are visible.

$$(x_1,\ldots,x_n), \qquad \underset{\swarrow}{\underline{\text{uniformly distrib.}}}\ 0 \le x_i \le 1 \quad ; \quad i = 1,\ldots,n$$

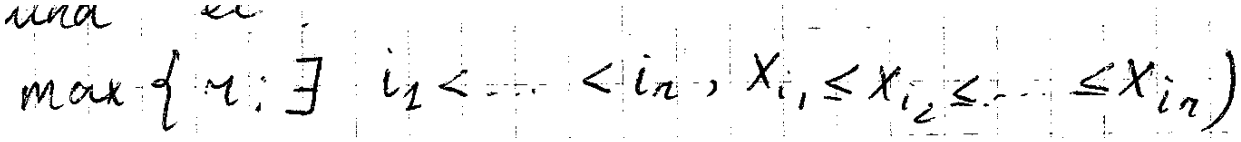


Original source details: GELFAND-P000701

Uncertain reading. Underlined annotation with an arrow: "uniformly distrib."

Consider a monot. incr. subseq. (not necessarily consecutive) What is its length?

$$\max\{r;\ \exists\, i_1 < \cdots < i_n, \quad X_{i_1} \le X_{i_2} \le \cdots \le X_{i_n})$$



Original source details: GELFAND-P000701

Uncertain reading. The letter after max reads r. The final indices read n. The last delimiter is a round parenthesis.

grows like $\underline{\underline{2\sqrt{n}}}$ (with respect to Lebesgue measure on the unit cube)

1970, Knuth, Pac. Journ.



Original source details: GELFAND-P000701

Uncertain reading. The journal abbreviation is written “Pac. Journ.”

1961, [crossed out] Shenstead



Original source details: GELFAND-P000701

Undeciphered passage. Several letter strokes before the surname are crossed out. The manuscript spells the name “Shenstead.”

There is a map into diagrams
(bumping)

GELFAND-P000702

4/IV-77 Gelfand Sem.

A. Tyurin. FG_g — a group with $2g$ gen. $a_1, \ldots, a_g, b_1, \ldots, b_g$ and rel.

$$\prod_i [a_i, b_i] = 1, \qquad [a_i, b_i] = a_i b_i a_i^{-1} b_i^{-1}$$

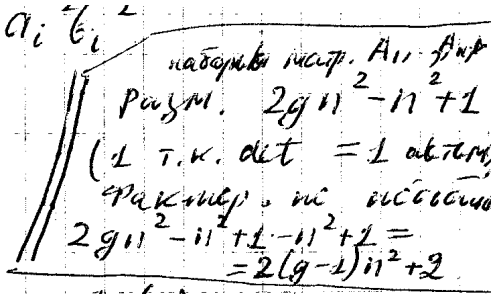


Original source details: GELFAND-P000702

Uncertain reading. The day of the month is heavily retraced. It reads as 4.

R_C — set of classes of reprs. in $GL(n, C)$

an alg. set with singularities.



Original source details: GELFAND-P000702

Diagram and typed description. [Right margin, in a frame with a double left rule:]

> tuples of matrices $A_1, \dots, A_{2g}$
> Dim. $2gn^2 - n^2 + 1$
> (1 since $\det = 1$ automatically,
> Quotient by similarity
> $2gn^2 - n^2 + 1 - n^2 + 1 =$
> $= 2(g-1)n^2 + 2$



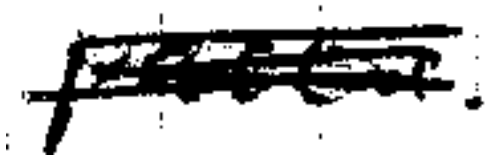
Original source details: GELFAND-P000702

Uncertain reading. The subscript on the last matrix is overwritten. It tentatively reads $2g$.

$$\pi_1(X) \longleftrightarrow FG_g.$$

Repr. $\varrho \rightsquigarrow$ bundle E_ϱ on X

Analyt. [crossed out] equiv. $\underset{\text{def}}{\Longleftrightarrow}$ $E_{\varrho_1} \cong E_{\varrho_2}$

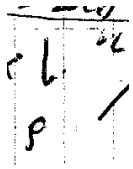


Original source details: GELFAND-P000702

Uncertain reading. The crossed-out abbreviation between “Analyt.” and “equiv.” tentatively reads “предст.” (repr.).

[At right, below the frame:]

$$\underset{\pi_1}{U(X)} \times \underset{\varrho}{V^n}/\pi_1 \longrightarrow X$$



Original source details: GELFAND-P000702

Uncertain reading. The letter with superscript n is nearly vertical in the right-hand formula. It is set as V^n following the formula below.

Packet of representations

$$H(X,E) \subset R_C$$
$$\uparrow$$
aff. space of dim. $(n^2-1)(g-1)$

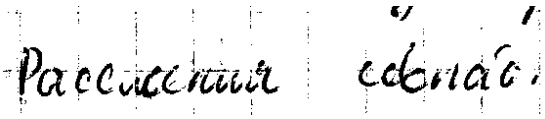
For a fixed curve X and a fixed bundle E.

The space of reprs. has dimension $2((g-1)n^2+1)$

A repr. in general position will be irreducible.

$$E_\varrho = \underset{\substack{\uparrow \\ \text{universal cover}}}{U(X)} \times V^n/\pi_1 \qquad x \in U(X)$$
$$(x,\xi) \sim (xg, \varrho(g)\xi)$$

Bundles are conn. $\Rightarrow$ what can one say about reprs.?



Original source details: GELFAND-P000702

Uncertain reading. The abbreviation after "Bundles" tentatively reads "связ." (conn.).

Any two irred. reprs. are equiv. with respect to some Riem. surface (belong to some packet).



Original source details: GELFAND-P000702

Uncertain reading. The middle of the abbreviation after "Any two" is overwritten. It tentatively reads "неприв." (irred.).



Original source details: GELFAND-P000702

Uncertain reading. The opening strokes of "эквивал." (equiv.) are written over earlier strokes.

Fix $X \Rightarrow$

$$R_C \xrightarrow{\text{affine bundle}} S_n(X) \qquad \text{(moduli of bundles)}$$

GELFAND-P000703

Typical problem: consider packets of reprs. corresp. to a one-dim. bundle L.

How a description of the whole set $H(X,L) \subset \mathbb{C}^{*2g}$ — space of characters.



Original source details: GELFAND-P000703

Uncertain reading. The word after "description" tentatively reads "всё" (whole).

L — a one-dimensional bundle.

L: given by transition fns. $\{\alpha_{ij}\} \in H^1(\{U_i\}, \mathcal{O}_X^*)$.

L: obt. from [crossed out] a repr. of $\pi_1 \Leftrightarrow \{\alpha_{ij}\} \in H^1(X, \mathbb{C}^*)$ (one can take $\alpha_{ij} = \text{const}$).



Undeciphered passage. Two rules obscure the word after “from.” Two lower strokes remain visible.

Original source details: GELFAND-P000703

Sections: f_i on U_j.

$$f_i = \alpha_{ij} f_j.$$



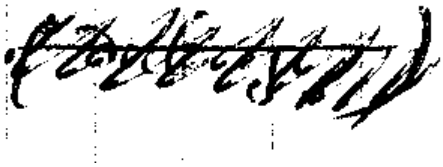
Uncertain reading. The lower stroke in the symbol after f_i resembles j. Readings i and ij are also possible.

Original source details: GELFAND-P000703

$$\alpha_{ij} = \text{const} \quad \Rightarrow \quad \frac{df_i}{f_i} = \omega\big|_{U_i}$$

— one differential (meromorphic).

Singularities of ω: poles of order ≤ 1 [crossed out: (divisor?)].



Uncertain reading. The crossed-out word in parentheses tentatively reads “divisor.”

Original source details: GELFAND-P000703

The residues are integers.

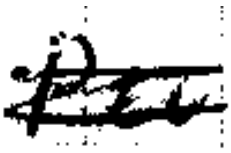
Principal parts — divisor

$$[\omega] = \sum \operatorname{Res}_x \omega \cdot x$$

(zeros and poles of the fns. f_i).

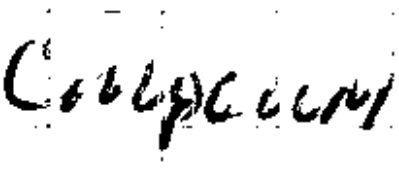
2 other representatives $\Rightarrow$
$\Rightarrow$ equiv. divisor.

Fix the divisor. [crossed out] Construct ω with the given set of principal parts.



Undeciphered passage. Two rules cross out a short passage before the verb.

Original source details: GELFAND-P000703



Uncertain reading. The verb tentatively reads “Соорудим” (construct). The middle letters run together.

Original source details: GELFAND-P000703

Solve the eq. $\frac{df}{f} = \omega + \omega_0$ globally on $U(X)$.

ω_0 — a holom. differential on X.

$$f(g(u)) = \underset{\text{character!}}{\chi_g} \quad f(u).$$

Here we obt.

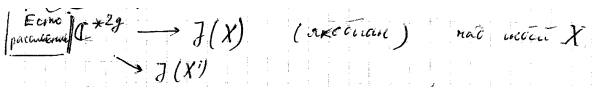
All characters corresp. to the packet.

Corollary: the packet of reprs. uniquely recovers X from Terelli's thm.



Original source details: GELFAND-P000703

Uncertain reading. The manuscript spells the name "Терелли" (Terelli), referring to Torelli's theorem.



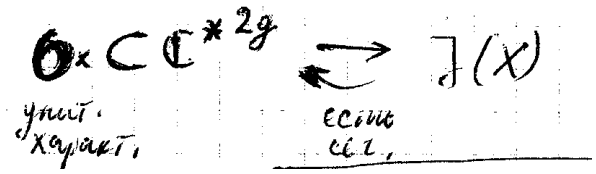
Original source details: GELFAND-P000703

Diagram and typed description.

There is a fibration $\mathbb{C}^{*2g} \longrightarrow J(X)$, $\mathbb{C}^{*2g} \searrow J(X')$

(Jacobian) over all X.

0th fibre — exponentials of primitives of holom. differentials.



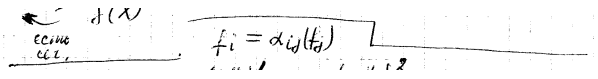
Original source details: GELFAND-P000703

Diagram and typed description.

$[O_x?] \subset \mathbb{C}^{*2g} \rightleftarrows J(X)$

unit. chars. — there is a sect.

The overwritten round symbol with subscript x may be O_x or Θ_x.



Original source details: GELFAND-P000703

Diagram and typed description.

$f_i = \alpha_{ij}(f_j)$

Schwarzian derivative:

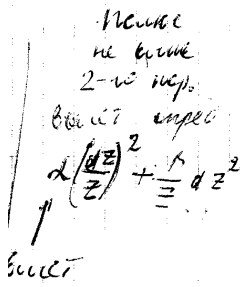
$$S_z^f = \left(\frac{f''}{f'}\right)' - \frac{1}{2}\left(\frac{f''}{f'}\right)^2$$

$$S_t^{f(g(t))}\,dt^2 = S_z^f\,dz^2 + \underbrace{S_t^g\,dt^2}_{\substack{\text{for fractional-linear}\\ \text{this is 0}}} .$$

We obt. general quadr. differentials.

[In the right margin, beyond the rule:] Pole of order at most 2. Residue is def.

$$\underset{\substack{\uparrow\\ \text{residue}}}{\alpha} \left(\frac{dz}{z}\right)^2 + \frac{r}{z}\,dz^2$$



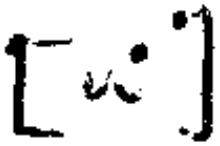
Uncertain reading. The first coefficient tentatively reads α, possibly λ. The numerator of the fraction reads r. The arrow marked "residue" points to the first coefficient.

Original source details: GELFAND-P000703

GELFAND-P000704

$$S_{z_i}^{f_i}\,dz_i^2$$

$[\omega]$ — principal part of the quadr. differential.



Uncertain reading. The sign in square brackets resembles ω. A dot or short stroke is visible above it at right, with two more dots by the right bracket.

Original source details: GELFAND-P000704

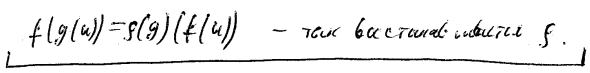


Diagram and typed description.

$$f(g(u)) = \varrho(g)(f(u))$$

— this is how one recovers ϱ.

Original source details: GELFAND-P000704